Draft version November 4, 2026
Typeset using LaTeX **twocolumn** style in AASTeX7.0.1

`galpy` with differentiable, accelerated backends: galactic dynamics in `jax` and `PyTorch`

Jo Bovy[1]

[1]*David A. Dunlap Department of Astronomy and Astrophysics, University of Toronto, 50 St. George Street, Toronto, ON M5S 3H4, Canada*

ABSTRACT

We describe an extension of `galpy`, the `Python` library for galactic dynamics, that adds `jax` and `PyTorch` as computational backends alongside its existing `numpy` and `C` implementation. With it, `galpy`'s gravitational potentials, orbit integration, action–angle transformations and their inverses, and distribution functions, including those for tidal streams, run on GPUs, can be compiled as part of a user's program, and can be differentiated with respect to their inputs and parameters, from the force at a point to the samples drawn from a distribution function. The user-facing design is kept simple: the active backend follows from the type of the arrays that are passed in, the `galpy` interface is unchanged, and the `numpy` path gives the same results as before. We describe the design of the backend framework and its implementation in each of `galpy`'s modules, with examples of use, and we discuss its limitations. On a GPU, batches of orbits, samples from distribution functions, and stream models run one to five orders of magnitude faster than on a single CPU core. To illustrate the new capabilities, we present simple examples of fitting the Galactic potential with a hypervelocity star and of learning the inverse action–angle transformation with a neural network trained through `galpy`. An AI coding agent wrote essentially all of the code under the detailed direction and review of the human author, and we describe how we organized this work and how we established the correctness of the result. This work is the start of `galpy`'s version 2, which is available at https://github.com/jobovy/galpy and documented at https://docs.galpy.org/en/v2/.

Keywords: Galaxy dynamics (591) — Stellar dynamics (1596) — Milky Way dynamics (1051) — Stellar streams (2166) — Astronomy software (1855) — Computational methods (1965)

1. INTRODUCTION

Software for galactic dynamics has become a standard part of the toolkit of Galactic astronomy. Codes such as the NEMO Stellar Dynamics Toolbox (P. J. Teuben 1995), `galpy` (J. Bovy 2015), `gala` (A. M. Price-Whelan 2017), and `AGAMA` (E. Vasiliev 2019) provide gravitational potentials, orbit integration, action–angle coordinates, and distribution functions (DFs) in a form that can be applied directly to observed stars, both to characterize the orbits of individual objects and collections of objects, and to build dynamical models that are fit to data. Such codes are useful because the motion of a star in a given gravitational potential is a solved problem that nevertheless requires careful numerical work, which a shared, well-tested implementation saves every user from repeating, and because they let developers build sophisticated models within a well-defined framework that users can apply without implementing them; `galpy`'s action–angle methods are an example, because their implementation is far from trivial and they have been used in many analyses that would otherwise not have been attempted. The kinematic data from the Gaia mission (Gaia Collaboration et al. 2016, 2018, 2023) have made such codes indispensable, because for the first time orbits can be computed for hundreds of millions of stars.

J. Bovy (2015) introduced `galpy` as a `Python` package, with `C` implementations of the computationally-expensive parts, for computing orbits, action–angle coordinates, and DFs in general galactic potentials. It has since been used in more than a thousand papers. Most of these use `galpy` as a tool: orbits are integrated, and orbital parameters and actions computed, in a model of the Milky Way to place objects in the Galaxy and to characterize their motion, for stars in the disk with measured abundances and ages (e.g., N. Frankel et al. 2020; M. Xiang & H.-W. Rix 2022), the substructure of the stellar halo (e.g., V. Belokurov et al. 2020; J. M. M. Lane et al. 2023), globular clusters and dwarf galaxies (e.g., T. K. Fritz et al. 2018; A. B. Pace et al. 2022), hy-

pervelocity and runaway stars and the natal kicks of pulsars and compact objects (e.g., D. Boubert et al. 2018; T. Marchetti et al. 2022; T. N. O'Doherty et al. 2023), open clusters and the star-forming regions near the Sun (e.g., C. Zucker et al. 2022; C. Swiggum et al. 2024), the hosts of exoplanets and interstellar objects (e.g., G. Stefánsson et al. 2023; J. C. Forbes et al. 2025), and galaxies other than our own (e.g., S. Tonnesen 2019). A smaller set of papers, perhaps one in six, uses `galpy` as a modeling framework in which a DF or a potential is fit to data: models of tidal streams and of their perturbations by dark-matter subhalos (e.g., J. Bovy et al. 2016; K. Malhan & R. A. Ibata 2019; N. Banik et al. 2021), action-based models of the disk (e.g., W. H. Trick et al. 2016; J. T. Mackereth et al. 2019), DF and Jeans models of the halo (e.g., J. M. M. Lane et al. 2023), models of the Galactic bar (e.g., J. A. S. Hunt & J. Bovy 2018; M. Lucey et al. 2023), and measurements of the Milky Way's mass and potential, from its rotation curve and halo tracers (e.g., M. Cautun et al. 2020), from the local escape velocity (e.g., L. Necib & T. Lin 2022), and from the clustering of stream stars in action space (e.g., T. Yang et al. 2020; S. Reino et al. 2021). It is for this second kind of use that derivatives matter, because fitting a model to data requires the likelihood to be evaluated many times over the model's parameters, and the efficiency of this search is set by whether its derivatives are available.

`galpy` has changed and grown a lot since 2015. The number of potentials grew from 22 to 72, with the growth shifting from specific density profiles to general solvers (basis-function, multipole, and interpolated potentials) and to wrappers that make any potential time-dependent, rotating, or tilted, and a library of Milky-Way models from the literature and velocity-dependent forces, such as dynamical friction and the fictitious forces in non-inertial reference frames, were added. The `Orbit` class was rewritten as an arbitrarily-shaped array of orbits that are integrated in parallel, such that millions of orbits can be analyzed at once; orbits can be initialized by name through a query to SIMBAD, and their eccentricities, peri- and apocenters, and maximum heights can be estimated from the actions at a small fraction of the cost of an integration (J. T. Mackereth & J. Bovy 2018). DFs for tidal streams were extended to stream gaps (J. L. Sanders et al. 2016) and to particle-spray models (M. A. Fardal et al. 2015; Y. Chen et al. 2025, implemented in `galpy` by Y. Qian et al. 2022), a general framework for isotropic and anisotropic spherical DFs was added (J. M. M. Lane et al. 2022), as were tools for Jeans modeling, inverse action–angle transformations, chaos diagnostics, and surfaces of section. Over the same period the code grew from 38,000 to 95,000 lines of `Python` and from 8,000 to 26,000 lines of `C`, and the test suite grew from 560 to 2,360 tests that cover 99.94 % of the code.

Over the same decade, numerical computing has been reshaped by frameworks built for machine learning, chiefly `jax` (J. Bradbury et al. 2018) and `PyTorch` (A. Paszke et al. 2019). These provide array programming that runs on CPUs and GPUs alike, just-in-time compilation of whole programs, and automatic differentiation of arbitrary code (see A. G. Baydin et al. 2018, for a review). A dynamics code written in such a framework gains three things: it runs on a GPU, where large batches of orbits or stars are processed at once; it can be compiled into a single program rather than executed one array operation at a time; and every quantity that it computes—a force, an orbit, an action, the value of a DF—can be differentiated with respect to every input, be it an initial condition or a parameter of the potential. Derivatives are what gradient-based inference requires, whether this is Hamiltonian Monte Carlo (S. Duane et al. 1987; M. D. Hoffman & A. Gelman 2014), variational inference, or the optimization of models with many parameters, such as neural networks that stand in for a potential or a DF. Codes built on these frameworks are now in use in cosmology (e.g., C. Modi et al. 2021; J.-E. Campagne et al. 2023; Y. Li et al. 2024) and, over the past two years, in galactic dynamics. `galax` (N. Starkman et al. 2026b), a re-implementation of `gala` in `jax`, provides gravitational potentials, orbit integration, and particle-spray stream models, all of them differentiable and able to run on a GPU; it supplied the halo potentials whose acceleration directions S. Wu et al. (2026) and Euclid Collaboration et al. (2026) compare with the curvature of stellar streams around other galaxies to constrain the shapes of their halos, the orbits with which N. Starkman et al. (2026a) forecast how well such curvature tests separate dark-matter models, and the GPU integration of streams perturbed by subhalos that N. Anau Montel & F. Schmidt (2026) differentiate to obtain Fisher forecasts. With other `jax` codes, G. Viterbo & T. Buck (2025) fit the progenitor and the host potential of a mock stream by gradient descent through an N-body simulation, D. Chemaly et al. (2026) generate the many particle-spray models needed to infer the distribution of halo flattenings from a population of streams, and J. Nibauer et al. (2025) compute the response of streams to subhalos by integrating linearized equations of motion. With gradient-based samplers in `jax`, J. Nibauer & A. Bonaca (2025) fit a spline model of the track of GD-1 to measure the accelerations along it, and A. M. Price-Whelan et al. (2025) fit a flexible model

of vertical orbits to the variation of stellar abundances. Normalizing flows for the DF, differentiated automatically, recover the potential from phase-space snapshots of simulated galaxies (G. M. Green et al. 2023; J. An et al. 2021; M. R. Buckley et al. 2023) and from 5.6 million *Gaia* stars (T. Kalda & G. M. Green 2025). Differentiating an entire simulation (G. Viterbo & T. Buck 2025; N. Anau Montel & F. Schmidt 2026), or fitting a neural-network potential to millions of stars (T. Kalda & G. M. Green 2025), is not practical with codes that only return values.

In this paper, we describe an extension of `galpy` that adds `jax` and `PyTorch` as *backends* to its existing `numpy` (and `scipy`) and `C` implementation. We support both frameworks deliberately: a solution that works for both cannot depend on the particulars of either, which keeps the implementation agnostic of the backend and should make it easy to add a third framework, or to replace one, as the ecosystem of array frameworks changes. The principle that guided this work is that nothing should change for the existing user: the `numpy` path produces results that are byte-for-byte identical to those of the previous version, with a handful of approved exceptions that we list, the interface is unchanged, and `galpy`'s existing test suite now runs under every backend, in eager and in compiled mode, with the same tolerances, again with very few exceptions. Backends are selected by the type of the arrays that are passed in or through a context manager; `galpy`'s code is written against a backend-agnostic array namespace; and `galpy` does not compile anything itself, but is written such that it can be compiled and differentiated as part of a user's own program, of which a `galpy` call is generally only a part. All of `galpy`'s core functionality is covered: the potentials and their forces and derivatives, orbit integration either inside the backend or through the `C` integrators with their variational equations, the action–angle transformations and their inverses, and the DFs, including those for tidal streams. All of these are differentiable with respect to their inputs and parameters, and they run on GPUs. We list the few things that are not covered where they come up; the main gap is the use of `astropy` units inside compiled programs, which a `jax`-aware units package such as `unxt` (N. Starkman et al. 2025) may make possible in the future.

Claude, an AI coding agent developed by Anthropic, wrote essentially all of the code described in this paper, working in the `Claude Code` environment under the detailed direction and review of the human author (JB) between June and October 2026. We discuss how this was done, how we organized the work, and how we established the correctness of the code in § 8; a statement on the use of AI, which also explains the use of "we" in a paper with a single human author, appears at the end of the paper.

The outline of this paper is as follows. § 2 describes the design of the backend framework: how a backend is chosen, how we made `galpy`'s code backend-agnostic, and how we made the test suite run under every backend. §s 3–7 go through `galpy`'s main modules in turn—potentials, orbit integration and its derivatives, action–angle coordinates, DFs, and models of tidal streams—each starting from the general approach and then giving the details that are specific to the module, with examples of use. § 8 describes the development process, § 9 presents fuller examples that combine several parts of the code in a single analysis, and § 10 concludes. An Appendix gives the variational equations behind the orbit derivatives, which are also useful outside of the backend frameworks, for computing the sensitivity of an orbit to its initial conditions and Lyapunov exponents.

Code to reproduce all of the figures in this paper, as well as the working notes of the development, is available in a `git` repository associated with this paper, separate from the `galpy` repository, at

2. DESIGN OF THE BACKEND FRAMEWORK

2.1. *Principles*

We set six rules before writing any code, and the rest of this paper is largely a description of how we kept them. First, the `numpy` path had to remain byte-identical to the previous version of `galpy`: every backend helper is a strict pass-through when the active array library is `numpy`, such that migrated code written as `xp.sqrt(x)`, where `xp` is the resolved namespace, runs the identical `numpy` function on the identical object. We enforce this rule with tests and, for each pull request (PR), by comparing checksums of outputs between a checkout with and one without the change; where the `numpy` result did change, we flagged and approved the change, and we point these places out where they arise. There are three deliberate changes: the King model now solves its Poisson equation to a relative tolerance of 10^{-10} rather than `scipy`'s default of 10^{-3}, which changes its potential and DF at the 10^{-5} to 10^{-3} level, and the velocity sampler of the quasi-isothermal disk DF, the samplers of the two-dimensional disk DFs, and the sampler of the stream DF draw by inverse transformation rather than by rejection, which changes the particular draws but not their distribution, except for the two-dimensional disk DFs, whose previous draws had a mean azimuthal velocity that was biased by a few per cent and whose new draws do not (§ 6.3). In addition, the forces of the Einasto profile and of the power law with an exponential cutoff were

rewritten in forms that are accurate and differentiable at small radii, such that their `numpy` values differ from the previous version by a few units in the last place (at most 5×10^{-15} relative), which propagates at the same level to everything computed from these potentials' `Python` forces; the formulas of a dozen other potentials were restructured for the same reasons, but evaluate in the same floating-point operation order as before and remain byte-identical, which a test checks. Accordingly, the tolerances of a handful of `numpy` tests were loosened slightly. Second, the `galpy` interface had to stay unchanged: the same classes, methods, and keywords, with the backend generally determined by the arrays that are passed in (see below). Third, `galpy`'s code was to be written against a backend-agnostic array namespace, with backend-specific code confined to a new `galpy.backend` subpackage and never written as `if backend == 'jax'` inside the physics modules. Fourth, `galpy` must use the basic array operations of `jax` and `PyTorch` and little else; we implemented the numerical methods that `scipy` supplies on the `numpy` path—special functions, root finding, quadrature, interpolation, sampling—in `galpy.backend` and tested them against `scipy` and against arbitrary-precision references, rather than taking them from the frameworks' own, uneven, `scipy`-like modules. Fifth, we decided that `galpy` should never compile anything internally itself: instead we required the code to be traceable by `jax.jit` and `torch.compile`, and the user decides what to compile. Sixth, the existing test suite, run under every backend with the same tolerances, had to pass in full; this was the main acceptance test, but not the only one, because the new functionality—differentiation, compilation, running on a GPU—has no existing tests, and we wrote new ones for it, as described in § 8.7. Supporting two frameworks from the start served the third rule (§ 1).

2.2. *Choosing a backend*

The general design approach for using the backends is that the backend follows from the data: a `galpy` function that receives a `jax` array or a `PyTorch` tensor computes with that framework and returns an array of the same kind, while a function that receives `numpy` arrays or `Python` scalars computes with `numpy` as before. Thus, the radial force of the standard Milky-Way model is evaluated on a `jax` array as

```
from galpy.potential import MWPotential2014
R= jnp.linspace(0.5,2.,1001)
FR= MWPotential2014.Rforce(R,0.) # jax array
```

without any further instruction, and with a `PyTorch` tensor the same call returns a tensor. The data include an object's parameters: a potential, distribution function, or action-angle object that is set up with a backend array as one of its parameters computes with that backend in every call, also at `Python` floats,

```
pot= NFWPotential(a=jnp.array(2.))
pot(1.1,0.2) # jax array
```

and an `Orbit` integrated in such a potential returns backend arrays as well. In short, backends are *greedy*: any backend array among the inputs of a calculation, be it a coordinate, an initial condition, or the parameter of an object, turns on the backend for that calculation and its output. We chose this rule because we assume that a user who passes a backend array wants the backend to perform the computation and to receive backend arrays as results, whether to differentiate, to compile, or to run on an accelerator. A parameter that is not differentiated keeps the fast paths that a differentiated one cannot take: an orbit in a potential with a concrete backend parameter is still integrated by the `C` integrators, with the `C`-computed sensitivities that make its result differentiable with respect to the initial conditions. When there is no array to dispatch on, as when a potential is evaluated at `Python` floats, the default backend is used; it is `numpy` unless a different one is set in the configuration file or through a context manager,

```
import galpy.backend
with galpy.backend.use('jax'):
    Phi= MWPotential2014(1.,0.) # jax
```

The configured or context default only applies in the absence of backend data; the data always take precedence over it. Similarly, an `Orbit` that is set up from a tensor on a GPU is integrated there by an in-backend solver (the default `C` integrators run on the CPU),

```
o= Orbit(torch.tensor([1.,0.1,1.1,0.,0.05,0.],
                      device='cuda'))
o.integrate(ts,MWPotential2014,method='torchode')
o.R(ts) # a CUDA tensor
```

Internally, `galpy`'s migrated functions also accept an explicit namespace as an `xp=` argument, which `galpy` uses to keep nested calls on one namespace; it is not meant for users. The full resolution order is thus: an explicit `xp=` argument, the array arguments and the objects' parameters, the context or configured default, and finally `numpy`. The context manager also accepts `force=True`, which makes the default take precedence over the data. Under a forced backend, `galpy` runs as much as possible on that backend, including the internals that otherwise use `numpy` and `scipy` for `numpy` inputs; the test suite depends on this option to run `galpy`'s existing tests, which are written with `numpy` inputs, on the backend code, and it is not meant for production code.

While a call with backend arrays returns backend arrays, a few conventions are needed at the edges of the

backend layer. `galpy` computes in double precision, and its internal constants are `Python` floats, such that under `PyTorch`, whose default is single precision, `galpy` enters a double-precision default for the duration of a call; a user's single-precision tensor stays single precision in and out. Outputs that are scalars in the `numpy` sense, such as a single potential value for scalar inputs under a forced backend, are returned as `numpy` values, because `astropy` units cannot wrap a backend array. Units are handled outside the backend layer: the existing unit-handling decorators convert a `Quantity` input to `galpy`'s internal units before it reaches the backend code and build a `Quantity` output from the result afterwards. A potential parameter that is given as a backend array carries no units and is taken to be in internal units already, which is what makes it possible to differentiate with respect to it. Because `jax.jit` cannot trace `astropy Quantity` objects, the use of units inside a compiled program is not supported; a user-defined density that uses units is evaluated outside the compiled program, through a callback, and cannot be differentiated. A `jax`-aware units package such as `unxt` (N. Starkman et al. 2025) could lift this restriction, and we may add support for it in the future.

2.3. *Backend-agnostic code*

The code in the physics modules is written in a backend-agnostic manner wherever possible. To achieve this, every migrated compute method starts by resolving the array namespace from its coordinate arguments and then uses only that namespace; for example, the potential of a Plummer sphere is computed as

```
def _evaluate(self,R,z,phi=0.,t=0.):
    xp= get_namespace(R,z,phi,t)
    return -1./xp.sqrt(R**2+z**2+self._b2)
```

The namespace is the plain `numpy` module for `numpy` data, `jax.numpy` for `jax` arrays, and the array-API-compatible view of `PyTorch` for tensors, obtained through the `array-api-compat` package. This is the entire change for the majority of `galpy`'s potentials, which are elementary functions of the coordinates; the work is in the remaining potentials, which use `scipy` or `numpy`-specific functions and which we describe in the module sections. The dispatch costs about 3 μs per call when run eagerly and nothing under compilation, because it is then resolved at trace time.

We placed the dispatch at `galpy`'s private compute layer (`_evaluate`, `_Rforce`, and so on), such that the public methods, their unit handling, and their documentation are untouched. The public entry points carry one new decorator that declares which of the arguments are coordinates, for example `@backend_input('R','z','phi','t')` for a method whose coordinate arguments are R, z, ϕ, and t. This decorator, which does nothing on `numpy` data, coerces the declared arguments, and only those, onto the active namespace, whether they are passed positionally or by keyword, and leaves derivative orders, flags, masses, option strings, and grids alone. It also resolves a single device for the call from the first backend array it finds, such that a `Python` scalar passed alongside a GPU array does not end up on the host CPU. The central coercion matters more than it might appear: `PyTorch`, unlike `jax`, rejects `Python` scalars in most tensor operations, and about two thousand of the tests that failed under `PyTorch` when we first ran the backends failed for this reason alone. Coordinate transformations in `galpy.util.coords` follow a similar pattern, where a marker decorator opts a migrated function into framework-preserving scalar promotion, such that we could migrate the functions one at a time.

2.4. *Numerical building blocks*

`galpy`'s `numpy` code uses `scipy` for special functions, root finding, quadrature, and interpolation, and `numpy`'s global random-number generator for sampling, none of which dispatch to `jax` or `PyTorch`, while the frameworks' own versions are incomplete and, in places, wrong. The `galpy.backend` subpackage therefore provides `scipy`-like replacements that delegate to `scipy` on `numpy` data and run natively otherwise.

For special functions, a router dispatches each function to `scipy.special`, `jax.scipy.special`, or `torch.special` unless the function is listed as missing or unreliable for that framework, in which case we implemented a fallback directly in `galpy`. The list is short but important: `jax`'s hypergeometric functions ${}_2F_1$ and ${}_1F_1$ currently return infinities with `nan` gradients for arguments $z < -1$, which is exactly the regime in which `galpy` uses them; `jax`'s incomplete gamma function iterates per element, which is about a hundred times slower than a vectorized series for large arrays on the CPU and slow on a GPU at every size; and `PyTorch`'s incomplete gamma function has no derivative with respect to its order and loses six digits near an order of 20, where its implementation switches between a series and a continued fraction. We gave the fallbacks fixed iteration counts, such that they trace and vectorize, and checked them against `mpmath` at 30 to 50 digits; the worst case among them agrees with the reference to 7×10^{-15}, which is at least as accurate as the `scipy` path. A test asserts that the list of unreliable functions still matches the installed frameworks, such that entries are removed as the frameworks improve.

For root finding, `galpy.backend` provides a drop-in replacement for `scipy`'s `brentq`, which on backend data runs a fixed number of sign-preserving bisection steps (a rolled rather than an unrolled loop under a `jax` trace), set by the bracket width and the tolerance or by the maximum number of iterations when the bracket is itself traced, followed by one Newton step that polishes the value and, through the implicit-function theorem, carries the derivative of the root with respect to whatever the function depends on. This one root finder now serves every action–angle and DF calculation, where `galpy` previously had several. Quadrature uses fixed-order Gauss–Legendre rules, with semi-infinite, boundary-layer, finite-part, and nested variants, and is therefore a differentiable approximation rather than an adaptive method; we set the order for each use by comparing with the adaptive `scipy` result, a comparison that in two cases found the `scipy` result to be the one in error (two of the fixes described in § 8.6). Interpolation comes in two forms: a fitted `scipy` spline is converted to a piecewise-polynomial table that is evaluated by namespace operations, which is differentiable in the evaluation point and is used for tables that do not depend on parameters; and cubic splines whose coefficients are solved for in the backend, which are differentiable in the tabulated values and are used where the table itself depends on a parameter, as for the velocity dispersion in the dynamical-friction force.

Sampling follows the same logic: every sampler in `galpy` that was a rejection sampler with `numpy` proposals has an inverse-cumulative counterpart on the backends, in which uniform variates are transformed through a tabulated cumulative distribution; this has static shapes, traces, and makes the samples differentiable with respect to the parameters of the distribution, because the transformation of a fixed set of uniforms is a smooth function of those parameters. Random keys are explicit: a `jax` key or its `PyTorch` equivalent is passed as `key=`, while `key=None` keeps the historical behavior of `numpy`'s global generator, such that existing scripts produce the same samples as before, except for the three samplers that were replaced on the `numpy` path as well (§ 2.1). For a Hernquist sphere, this reads as follows:

```
from galpy.backend import random as grandom
hp= HernquistPotential(amp=2.,a=1.3)
df= isotropicHernquistdf(pot=hp)
sam= df.sample(n=10**6)          # numpy, as before
key= grandom.key(42,backend='jax')
sam= df.sample(n=10**6,key=key)  # jax, differentiable
```

2.5. *GPUs*

Nothing in the above is specific to a device, and `galpy` runs on a GPU by being given arrays that live there. Throughout this paper we use two GPUs, which stand in for the two kinds that a user is likely to have: an NVIDIA TITAN Xp in the development machine, a consumer card from 2017 (about US$1,200 at launch) with 12 GB of memory that executes double-precision arithmetic at one thirty-second of its single-precision rate, and an NVIDIA H100 on a node of the Trillium cluster, a data-center accelerator (of order US$30,000 at the time of writing) with 80 GB of memory and a double-precision rate half of its single-precision one. The one thing the library must do is place the constants that it creates—rotation matrices, expansion coefficients, quadrature nodes, spline tables—on the device of the coordinates rather than on the host CPU, and we changed every site where a stored table meets a coordinate array to do so. Because these sites are invisible on a CPU, where host and device coincide, we found them by running the test suite on a GPU, which also served to find code that silently converted to `numpy`. Figure 1 displays what the GPU buys for a simple operation in `galpy`, the evaluation of a force at N points: on either GPU the computation alone is launch-bound up to $N \approx 10^4$ and then scales until it saturates the device's memory bandwidth, at about 10^9 points per second on a consumer card and 7×10^{10} on an H100; including the transfer of the input and the result over the PCIe bus, the gain over one CPU core is a more modest factor of five to six at $N = 10^7$. GPUs pay off when the data stay on the device across many operations, which is what a compiled program does.

2.6. *Compilation and differentiation from the outside*

`galpy` does not call `jax.jit` or `torch.compile` on its own code, because compilation is most effective at the level of a user's whole computation, in which a `galpy` call is one step, and a library that compiles its pieces separately defeats this. The user therefore writes code as they would without `galpy`, for example for a loss function that integrates an orbit in a potential and compares its position to an observed value,

```
def loss(q):
    pot= LogarithmicHaloPotential(normalize=1.,q=q)
    o= Orbit(jnp.asarray(ic))
    o.integrate(ts,pot,method='diffrax')
    return jnp.sum((o.x(ts)-x_obs)**2)
grad_loss= jax.jit(jax.grad(loss))
```

and it is `galpy`'s job to be traceable. Figure 2 displays what compilation costs and what it buys for a simplest operation, the radial force of a Plummer potential at 10^6 points, on one CPU core and on the two GPUs of § 2.5: compiling it takes about a tenth of a second on each. Under `jax`, a compiled call is four times faster than an eager one on the CPU, such that the compilation has

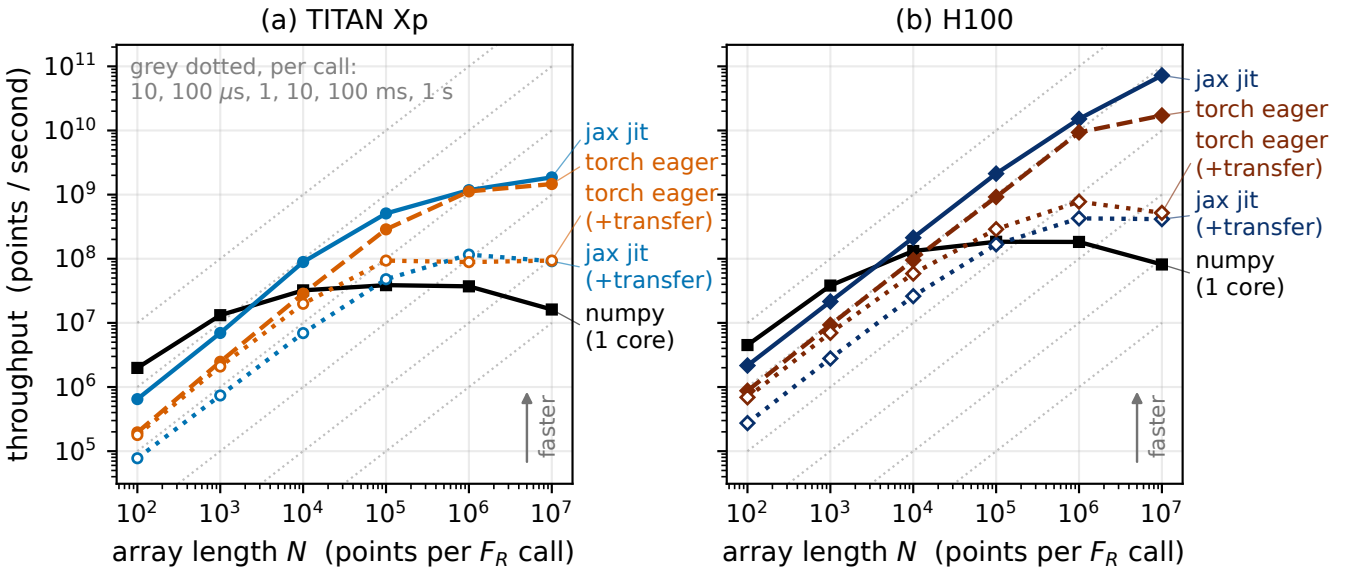


Figure 1. Throughput of the radial force of a Plummer potential as a function of the number of points per call. The left panel shows the development machine, with `numpy` on one CPU core and `jax` and `PyTorch` on an NVIDIA TITAN Xp; the right panel shows a node of the Trillium cluster, with an NVIDIA H100. Solid and dashed lines are compute-only (compiled `jax` and eager `PyTorch`); dotted lines with open markers include the transfer of inputs and outputs between host and device. Light grey dotted lines mark constant wall time per call, which the GPU curves follow while launch-bound. The GPU gives its largest gains when the inputs and outputs stay on the device.

paid for itself after two calls, and ten times faster on the GPUs, where it pays for itself after 60 to 90 calls. `PyTorch` tells a different story: `torch.compile` cannot be used on the TITAN Xp at all, because its code generator requires a GPU of compute capability 7.0 or higher, and on the H100 an eager `PyTorch` call, which is seven times faster than an eager `jax` call, is already within a factor of two of the compiled one, such that the compilation pays for itself only after about ten thousand calls. What `torch.compile` removes is the per-operation dispatch of a longer computation, such as an orbit integration. Figure 3 shows the cost of a derivative: under compiled `jax` the gradient of the potential with respect to the coordinates at N points costs between one and two times the value itself, on the CPU as on the GPUs, and under eager `PyTorch` two to four times, independent of N.

Making `galpy` traceable required a set of changes that recur throughout the code, which we list here once. We gave data-dependent loops fixed iteration counts: the bisection discussed above, the arithmetic–geometric mean in the elliptic integrals, the series and continued fractions of the special functions, and the refinement of a stream track, which is iterated a fixed number of times rather than to convergence. We replaced branches on data by `xp.where` with both sides evaluated, which requires clamping the unused side to a safe argument, because a `nan` produced on the discarded side still poisons the reverse-mode derivative; checks that raise on invalid input skip themselves when the input is traced. We made shapes static and, where an array shrinks as a loop proceeds, pad it to the next power of four, because eager `jax` recompiles for every new shape; one sampler spent 47 minutes compiling before this change and two after. We had to distinguish between a value that is traced and one that is merely a backend array, because `torch.compile` turns `float(x)` into a symbolic scalar, and a `float` probe silently took the `scipy` branch of a surface-density integral inside a compiled region. We wrap lazily built `numpy` tables such that their first construction inside a compiled region is not traced. A compiled function that receives a `galpy` object, such as a potential, as a static argument decides from a hash of that argument whether an existing compilation can be reused; `Python`'s default hash of an object is its identity, which would recompile for every newly created potential even when it is identical to one seen before and, worse, would reuse a compilation for a potential whose amplitude has been changed in place. We therefore key `galpy` objects by their type and attributes, with `galpy` sub-objects keyed recursively, such that a changed amplitude, which as a `Python` float is a constant of the compiled program, causes the re-trace that it must and a freshly created but identical wrapper does not cause a

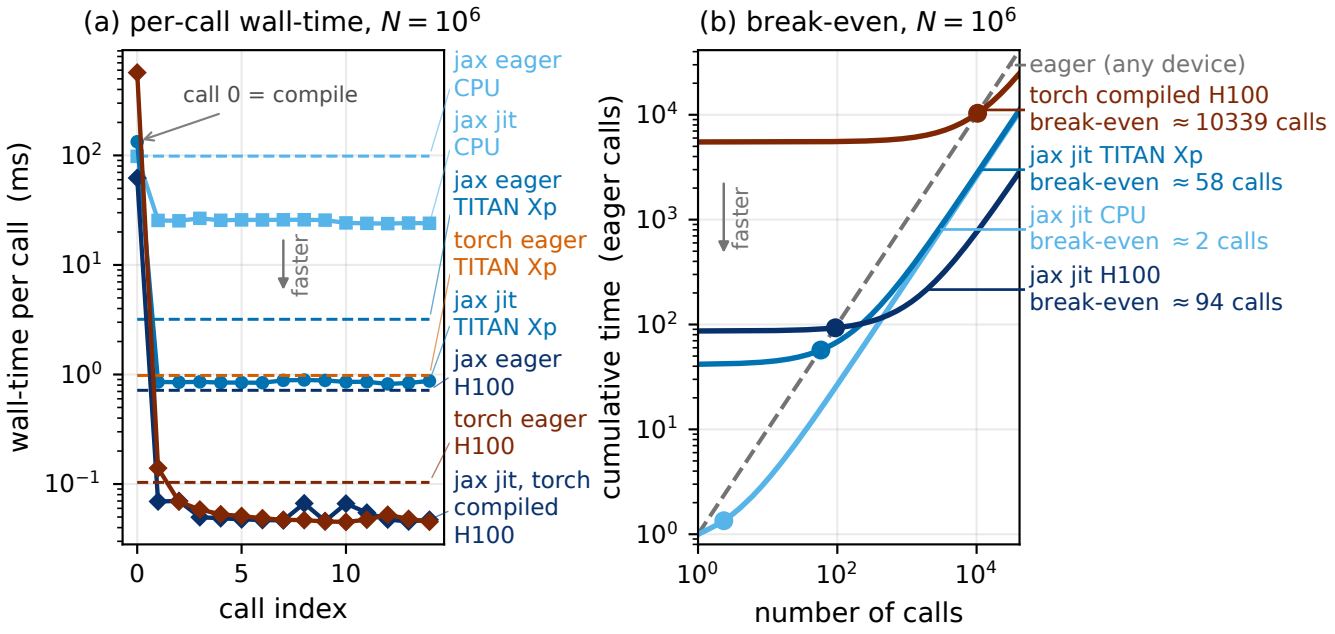


Figure 2. Cost and benefit of compilation for the radial force of a Plummer potential at $N = 10^6$ points under `jax` and `PyTorch`. Left: wall time of successive calls on one CPU core (`jax`), the TITAN Xp, and the H100; the first call compiles. Dashed lines are the eager cost per call; `torch.compile` does not support the TITAN Xp, so `PyTorch` appears there eagerly only. Right: the cumulative time of compiled and eager calls, in units of each device's eager call, and the number of calls after which compilation pays for itself.

needless one. The `PyTorch` compiler has its own difficulties: because a read of a `Python` context variable severs its graph, `galpy`'s backend selection, which is such a variable, reads a thread-local mirror while `PyTorch` is tracing. None of these changes affects what the `numpy` path computes; they are the price of the fifth rule, that the user, and not `galpy`, decides what to compile.

2.7. *The test suite under every backend*

We run `galpy`'s test suite unchanged under five modes: `numpy`, `jax` and `PyTorch` in eager mode, and `jax` and `PyTorch` in compiled mode. For the compiled modes, `galpy` has an optional global trace mode that wraps every call at the coercion boundary of § 2.3 in `jax.jit` or `torch.compile`; it exists for the test suite and is not needed in normal use, where the user compiles their own program. The mechanism is a fixture that wraps every test in the forced-backend context manager of § 2.2 and, for the compiled modes, in the global trace mode; it is selected by the `--backend` and `--jit` options of the test runner, and a `--device cuda` option additionally places every tensor on the GPU; § 8.5 describes what makes a pass in these modes meaningful and measures how much of the suite computes on the backend, and § 8.4 how we brought the suite to pass. Because a compiled-mode run can pass while every call falls back to eager execution, the test runner counts the fraction of calls that were traced and fails the run if this fraction drops below a floor, which is 99 % for a typical test file. The dedicated tests of the backend code, 18,807 tests in 100 files, are parametrized over the installed frameworks and additionally exercise the mixed case in which `numpy` and backend arrays meet in one call, which the forced mode cannot reach. In continuous integration, the two frameworks, the two modes, and 22 shards of the test suite form a matrix that runs on every PR; we run the GPU tests by hand, because the continuous integration framework available to us has no GPU.

3. POTENTIALS

The `Potential` classes are the foundation of `galpy` and were the first part of the code that we migrated. The approach is the one of § 2.3: every private compute method (`_evaluate`, `_Rforce`, `_zforce`, `_phitorque`, `_dens`, `_surfdens`, and the six second derivatives) resolves the array namespace from its coordinates and uses it throughout, and the public methods, through the coercion decorator, bring scalar and `numpy` inputs onto the namespace of any backend input. For the majority of `galpy`'s potentials this was the whole change, because they are elementary functions of the coordinates. The remainder use special functions or tabulated quantities: the basis-function and multipole expansions (`SCFPotential`, `MultipoleExpansionPotential`, and

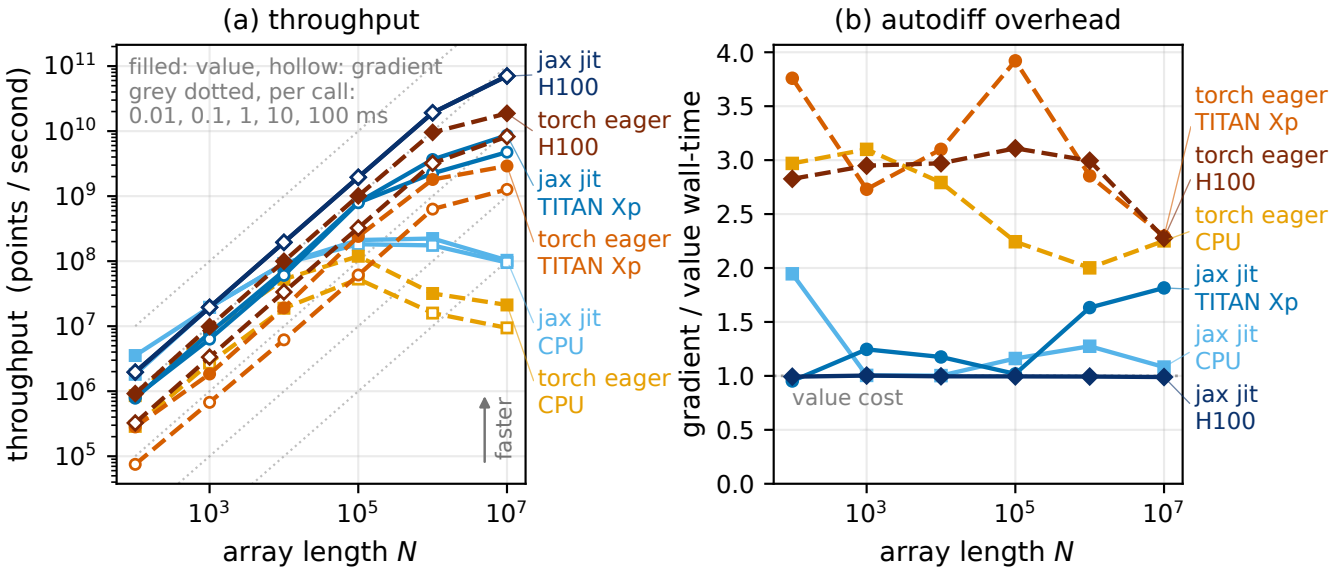


Figure 3. Cost of a derivative. Left: throughput of the Plummer potential (filled markers) and of its gradient with respect to the coordinates by reverse-mode automatic differentiation (open markers), under compiled `jax` and eager `PyTorch`, on one CPU core and on the two GPUs. Right: the ratio of the wall time of the gradient to that of the value. The gradient costs a small multiple of the value itself, independent of the number of points and of the device.

their disk variants) use the associated Legendre and Gegenbauer polynomials; the power-law potential with an exponential cutoff and the Einasto profile use incomplete gamma functions; the double-exponential disk uses Bessel functions in a Hankel-type integral; and the interpolated potentials and the King profile evaluate splines. These go through the special-function router and the interpolation layer of § 2.4. All of the potentials, forces, and wrappers in `galpy.potential` run on both backends, with the exception of the potentials that read N-body snapshots through `pynbody` (A. Pontzen et al. 2013) and the interface (J. J. Webb et al. 2020) to `AMUSE` (S. Portegies Zwart et al. 2009; F. I. Pelupessy et al. 2013; S. Portegies Zwart & S. McMillan 2018), which have no backend code. Even the wrapper that adiabatically contracts a dark-matter halo in response to the baryons, following M. Cautun et al. (2020), whose construction is a root find for the contracted profile, builds that profile on the backend when one of its parameters, or a parameter of one of the potentials that it contracts, is being differentiated. The convenience functions that act on potentials (the circular velocity and epicycle frequencies, the radius of a circular orbit of given angular momentum or energy, the enclosed mass, and the tidal radius and tidal tensor of J. J. Webb et al. 2019) are backend-aware as well; those that require a root are solved with the shared root finder and those that require an integral use Gauss–Legendre quadrature.

The derivatives of a potential with respect to the coordinates are, of course, its forces and second derivatives, which `galpy` implements by hand. Under a backend, these can also be obtained by differentiating the potential, and the agreement between the two is a strong test of both: for the basis-function expansions, the spiral arms, a rotating bar, and composite potentials, the force by automatic differentiation agrees with the hand-coded force to a few parts in 10^{15}, within twenty units in the last place (Figure 4), that is, the derivative that automatic differentiation produces is the analytic force up to rounding. While the test suite checks these identities for every potential, under both backends and in both modes, the derivatives that matter to the user are those with respect to the parameters: a potential constructed with a backend array as a parameter is differentiable with respect to that parameter through every method. For example, the derivative of the radial force of a Miyamoto–Nagai potential with respect to its scale length a reads with `jax` as

```
from galpy.potential import MiyamotoNagaiPotential
def Rforce_of_a(a):
    mp= MiyamotoNagaiPotential(amp=1.,a=a,b=0.3)
    return mp.Rforce(1.2,0.1)
dFR_da= jax.grad(Rforce_of_a)(0.5)
```

and the same with `PyTorch` is

```
a= torch.tensor(0.5,requires_grad=True)
mp= MiyamotoNagaiPotential(amp=1.,a=a,b=0.3)
mp.Rforce(torch.tensor(1.2),
```

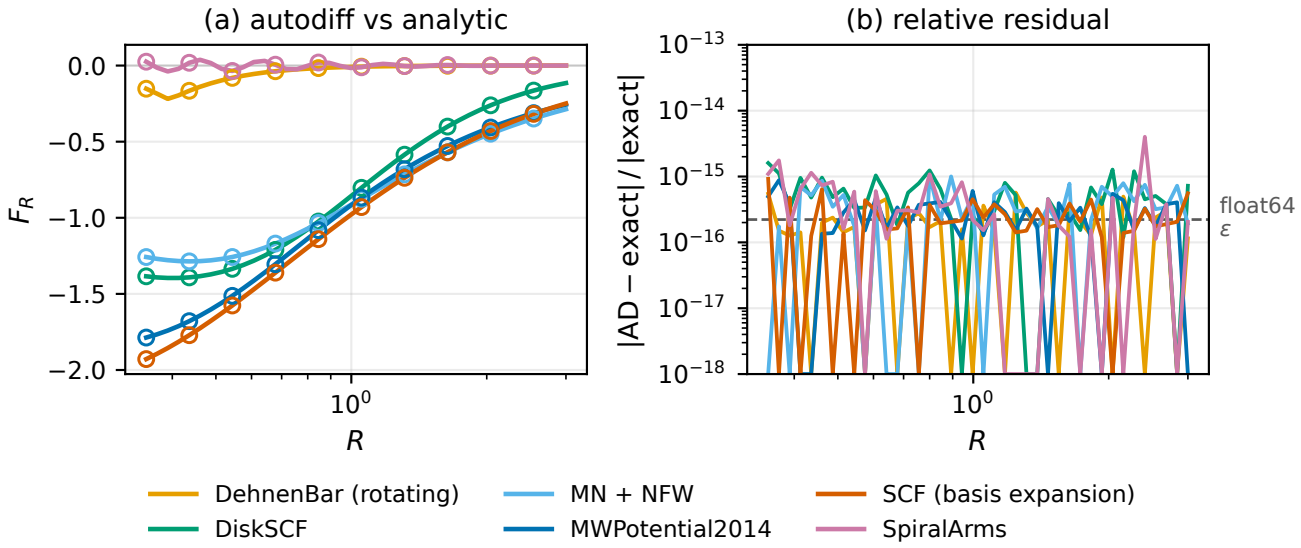


Figure 4. The radial force by automatic differentiation of the potential compared with the hand-coded force, for the Milky-Way model, a basis-function expansion and its disk variant, spiral arms, a rotating bar, and a disk-plus-halo composite. Left: the force along R in the mid-plane, lines from automatic differentiation and circles from the analytic implementation. Right: the relative difference between the two, which is at the level of the double-precision rounding error.

```
        torch.tensor(0.1)).backward()
a.grad # dFR/da
```

All 47 potentials that can be constructed from parameters alone are differentiable with respect to their amplitude (or, for the King model, its mass) in this way, both eagerly and under compilation. Figure 5 displays the accuracy of these parameter derivatives and a small example of their use, the recovery of the scale length of a Plummer potential from its force by gradient descent. The parameters of a potential constructed from a density, as the basis-function and multipole expansions and their disk variants can be, carry the derivative through the expansion coefficients.

The backend potentials differ from the `numpy` ones in a few respects. Where `galpy` computes an integral over a potential adaptively on the `numpy` path, as for the enclosed mass of a general density or the surface density, the backend computes a fixed-order Gauss–Legendre approximation; we set the order such that the two agree, but the backend result is an approximation of fixed cost rather than an adaptive one. Interpolated potentials evaluate the same spline as `scipy` does, to within one unit in the last place, by converting the fitted spline to a piecewise-polynomial table; they are differentiable in the evaluation point and, when they are built from a potential or function with a parameter that is being differentiated, also in the tabulated values, because the table and the spline coefficients are then computed on the backend and the derivative with respect to the parameter passes through them. A potential that moves along a precomputed orbit, as the `MovingObjectPotential`, interpolates that orbit on the backend when the orbit was integrated there, such that the field of the moving object is differentiable with respect to everything that the orbit depends on, and lives on the GPU when the orbit does; an orbit integrated with `numpy` stays on `numpy` and the field is computed at backend coordinates. And a `Python` branch on the value of a stored parameter, such as the special case of a vanishing scale length, is harmless when the parameter is differentiated eagerly but fails when the parameter is traced inside a compiled ODE solve; we replaced such branches where we found them, but authors of new potentials should keep this behavior in mind.

New potentials are written as before: a class that inherits from `Potential` and implements `_evaluate`, `_Rforce`, and `_zforce`. If these are written with the resolved namespace rather than with `numpy` directly, as illustrated in Figure 6, the new potential runs on every backend and can be differentiated with respect to its parameters without any further work, and the test suite's automatic checks of new potentials are run under every backend. A test checks that no migrated potential calls `numpy` directly in a compute method.

4. ORBIT INTEGRATION

4.1. *Three ways to integrate an orbit*

`Orbit` instances are set up and integrated as before, but an `Orbit` instance whose initial condition is a backend array, or whose potential carries a backend param-

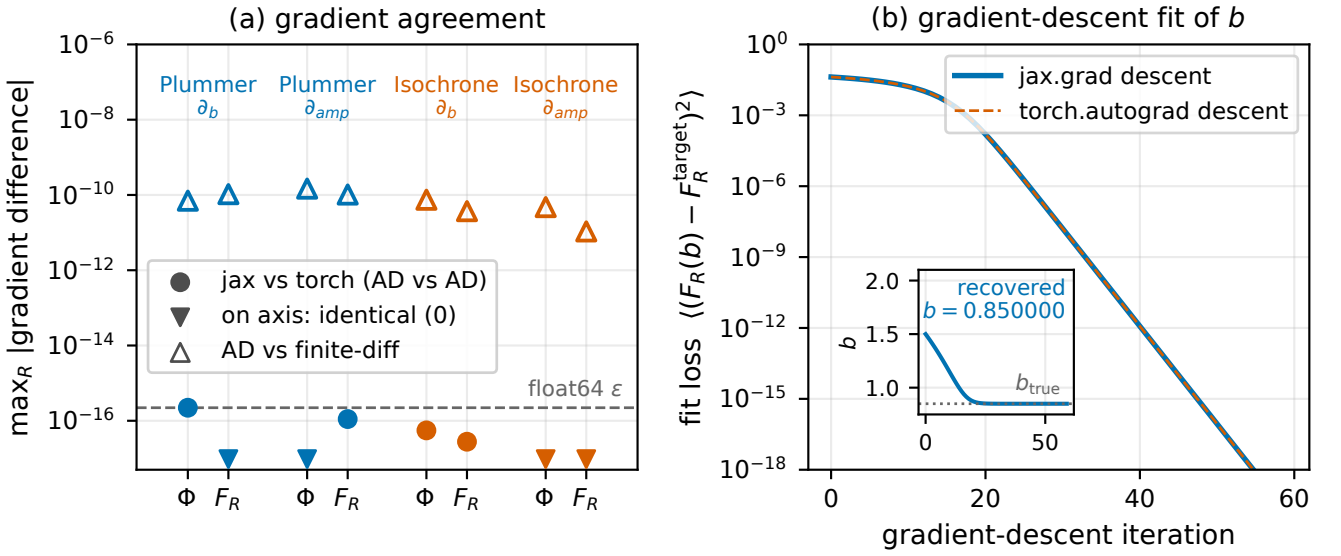


Figure 5. Derivatives of potentials with respect to their parameters. The left panel shows, for the Plummer and isochrone potentials, the largest difference between the derivatives of the potential and of the radial force with respect to the amplitude and scale parameters computed with `jax` and with `PyTorch` (circles, at the level of the double-precision rounding error; where the two are bit-identical, the difference is zero and cannot be shown on the logarithmic axis, which the filled triangles on the axis mark) and between either of these and a central finite difference (open triangles, at the level of the finite-difference error). The right panel displays the recovery of the scale length of a Plummer potential from the radial force at a set of points by gradient descent on the derivative, starting from $b = 1.5$ and converging to the true $b = 0.85$, with both backends.

eter, is integrated in one of two new ways, depending on what is to be differentiated. The first integrates the equations of motion inside the backend, with the backend's own ODE solver: `diffrax` (P. Kidger 2021) for `jax`, with an eighth-order Dormand–Prince method, and `torchdiffeq` (R. T. Q. Chen et al. 2018) or `torchode` (M. Lienen & S. Günnemann 2022) for `PyTorch`, with a fifth-order method. The equations of motion are those of `galpy`'s `C` integrators, evaluated in rectangular coordinates through the backend force layer, and the default tolerances are 10^{-12}. Because the whole integration is then a backend program, everything that enters it can be differentiated: the initial condition, the potential's parameters, the integration times. N orbits that share a time grid are integrated as a single solve along a batch axis, under `jax` with one step-size controller for the whole batch; given per-orbit time grids, an (N, n_t) array, `galpy` instead maps independent solves, each with its own controller, over the batch with `jax.vmap`, which is also what a user's own `jax.vmap` over single-orbit integrations gives and how `torchode` treats every batch; a user thus has both options. Either way, the solve runs on a GPU when the initial conditions do. This is the path to use for derivatives with respect to the parameters of the potential, as in the following example:

```
def final_R(b):
    o= Orbit(jnp.asarray(ic))
    pp= PlummerPotential(amp=1.,b=b)
    o.integrate(ts,pp,method='diffrax')
    return o.R(ts[-1])
dR_db= jax.grad(final_R)(0.6)
```

and under `PyTorch`:

```
b= torch.tensor(0.6,requires_grad=True)
o= Orbit(torch.as_tensor(ic))
pp= PlummerPotential(amp=1.,b=b)
o.integrate(torch.as_tensor(ts),pp,method='torchode')
o.R(ts[-1]).backward() # b.grad = dR/db
```

Second derivatives are available under `jax` by selecting the direct adjoint through the `inbackend_kwargs` keyword, and `jax.vmap` over initial conditions works. Figure 7 displays the simplest use of these derivatives: the sensitivity of an orbit to its initial conditions as a function of time, which the solver's derivative gives for free and which finite differences confirm, and a Gauss–Newton fit of the launch velocity of an orbit to a target end point using the Jacobian of the integration, which converges quadratically to machine precision in a handful of iterations.

The second way uses `galpy`'s compiled `C` integrators, which are much faster than a solver that runs inside `jax` or `PyTorch` on a CPU (for a single orbit in the Milky-Way model `MWPotential2014`, about ten times faster than the `diffrax` solve and a hundred times faster than `torchdiffeq`), but which are opaque to automatic differentiation. What they can provide is the derivative of

```python
from galpy.potential import Potential
from galpy.backend import get_namespace
class MNLikePotential(Potential):
    # a Miyamoto-Nagai disk written for every backend
    def __init__(self,amp=1.,a=1.,b=0.1,
                 normalize=False):
        Potential.__init__(self,amp=amp)
        self._a= a
        self._b2= b**2
        if normalize: self.normalize(normalize)
    def _evaluate(self,R,z,phi=0.,t=0.):
        xp= get_namespace(R,z)
        s= self._a+xp.sqrt(z**2+self._b2)
        return -1./xp.sqrt(R**2+s**2)
    def _Rforce(self,R,z,phi=0.,t=0.):
        xp= get_namespace(R,z)
        s= self._a+xp.sqrt(z**2+self._b2)
        return -R*(R**2+s**2)**-1.5
    def _zforce(self,R,z,phi=0.,t=0.):
        xp= get_namespace(R,z)
        sqb= xp.sqrt(z**2+self._b2)
        s= self._a+sqb
        return -z*s/sqb*(R**2+s**2)**-1.5
# differentiate an orbit in it w.r.t. the scale length a
def final_z(a):
    o= Orbit(jnp.asarray([1.,0.1,1.1,0.1,0.05,0.]))
    mp= MNLikePotential(a=a,b=0.3,normalize=1.)
    o.integrate(ts,mp,method='diffrax')
    return o.z(ts[-1])
jax.grad(final_z)(0.5)
```

Figure 6. A new `Potential` class written for every backend. The only difference from a class written for `numpy` is that the array functions are taken from the namespace resolved from the coordinates (`xp`) rather than from `numpy`. The class can then be used, and differentiated, like any other potential; the last lines take the derivative of the final height of an orbit with respect to the scale length of the new potential.

the orbit with respect to its initial condition, through the variational equations that `galpy` integrates alongside the orbit in `C` (Appendix A). When a `galpy` orbit with a backend initial condition is integrated with a `C` method, the `C` integrator returns both the orbit and the state-transition matrix $\mathbf{M}(t) = \partial \mathbf{w}(t)/\partial \mathbf{w}(0)$, and a custom derivative rule tells the backend that the derivative of the orbit with respect to the initial condition is $\mathbf{M}$; the reverse-mode derivative of any function of the orbit is then a single product of $\mathbf{M}^T$ with the cotangent, without a second call to `C`. For a six-dimensional orbit and a Runge–Kutta method, the six columns of $\mathbf{M}$ are integrated together with the orbit as a 42-component state; for the symplectic methods, the deviation vectors are propagated through the same drift and kick steps as the orbit. The derivative is exact to the precision of the solver: Figure 8 compares the state-transition matrix of a Milky-Way orbit over 3 Gyr from the `C` variational equations with reverse-mode automatic differentiation through `diffrax` and through `torchdiffeq` and with finite differences of independently integrated orbits, and the four agree to a few parts in 10^{10}. In code, nothing marks the difference between the two paths except the method:

```python
def end_point(ic):
    o= Orbit(ic)
    o.integrate(ts,MWPotential2014,
                method='dop853_c') # C + variational
    return jnp.stack([o.R(ts[-1]),o.z(ts[-1])])
M= jax.jacrev(end_point)(jnp.asarray(ic)) # STM rows
```

and a batch of orbits is differentiated with `jax.vmap`,

```python
Ms= jax.vmap(jax.jacrev(end_point))(ics) # ics: (N,6)
```

This path gives derivatives with respect to the initial conditions only, and to first order only; the variational equations do not carry derivatives with respect to the parameters of the potential, which must come from the in-backend solve.

`galpy` chooses between the two automatically. An `Orbit` whose initial condition is being differentiated—a traced `jax` array or a `PyTorch` tensor that requires a gradient—and that is integrated with a `C` method, in a potential that has `C` second derivatives, with non-traced times and non-traced potential parameters, uses the `C` integrator and its variational equations. In every other case in which something is differentiated, in particular whenever a potential parameter is traced or when the integration is to run on a GPU, the in-backend solver is used, under the name `method='diffrax'`, `'torchode'`, or `'torchdiffeq'`, which can also be requested directly. When nothing is being differentiated, that is, for a concrete backend initial condition, a `C` method integrates in `C` as before, without the variational equations, and only its result is a backend array; this is not a third way of integrating but the old one, and it means that a backend initial condition costs nothing until it is differentiated. The pure-`Python` integrators of `galpy` (`method='odeint'`, the `Python` leapfrog, and so on) are not routed at all: given a backend initial condition they integrate in `numpy` and return a backend array that is not differentiable, with a warning.

After integration, the orbit's methods work as before on backend arrays: positions and velocities at the integration times or, by interpolation with an in-backend spline, between them; energies and angular momenta; peri- and apocenter radii and maximum height, both from the integrated orbit and analytically through the action–angle code of § 5 (J. T. Mackereth & J. Bovy 2018); and the transformations to observed coordinates. Initializing an orbit from observed coordinates with

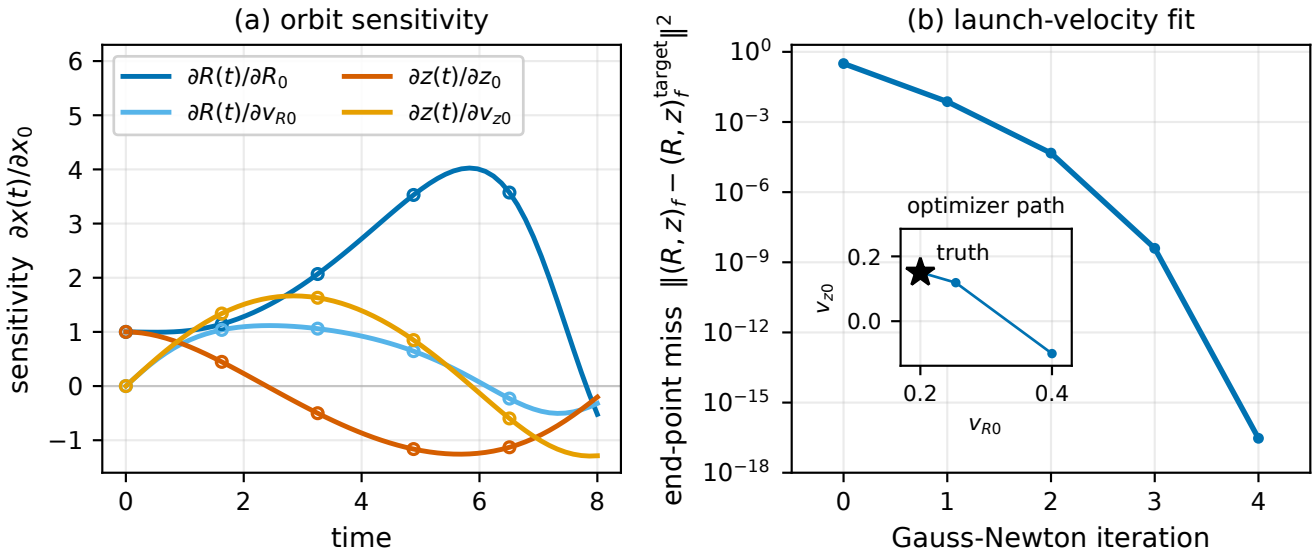


Figure 7. Derivatives of an in-backend orbit integration. Left: the sensitivities $\partial R(t)/\partial R_0$, $\partial R(t)/\partial v_{R,0}$, $\partial z(t)/\partial z_0$, and $\partial z(t)/\partial v_{z,0}$ of an orbit in a Plummer potential by reverse-mode differentiation through the `diffrax` solve (lines) and by central finite differences (circles). Right: the squared distance between the end point of the orbit and a target as the launch velocity $(v_{R,0}, v_{z,0})$ is fit by Gauss–Newton iterations on the Jacobian of the integration; the inset shows the path of the fit. The derivative through the solver agrees with the finite differences, and the fit that uses it converges in a handful of iterations.

backend arrays works as well, such that a derivative with respect to a star's distance or proper motion is available.

4.2. *Backend symplectic integrators for many orbits*

The derivative of an orbit with respect to its initial condition, in four ways. For a Sun-like orbit in `MWPotential2014` integrated for 3 Gyr, the left panel shows one element of the state-transition matrix, $\partial R(t)/\partial v_{R,0}$, from the `C` variational equations, from a finite difference of the `C` orbit, and from reverse-mode automatic differentiation through the `diffrax` and `torchdiffeq` solvers. The right panel shows the relative Frobenius norm of the difference between each of the latter three and the variational result; all agree to a few parts in 10^{10}.

The in-backend solvers of § 4.1 are adaptive Runge–Kutta methods: accurate, but expensive, because a step of `diffrax`'s eighth-order Dormand–Prince method evaluates the force thirteen times. For large batches of orbits in a smooth potential, where modest precision suffices and the cost is dominated by the force evaluations, a symplectic integrator with a fixed step is the better tool: it evaluates the force once per step, its energy error oscillates without a secular drift, it keeps little state per orbit, and its fixed steps make the derivative of a batch straightforward. `galpy` therefore also integrates inside the backends with a kick-drift-kick leapfrog with a constant step, `method='leapfrog_jax'` and `method='leapfrog_torch'`, as for example in the following:

```
o= Orbit(jnp.asarray(ics)) # ics: (N,6)
o.integrate(ts,MWPotential2014,method='leapfrog_jax',
            inbackend_kwargs={'nsteps':8000})
```

The same route provides backend versions of the fourth- and sixth-order symplectic integrators (H. Yoshida 1990; H. Kinoshita et al. 1991) of `galpy`'s `C` library, `method='symplec4_jax'` and `'symplec6_jax'` (and their `PyTorch` twins), which evaluate the force three and seven times per step and agree with their `C` counterparts at the same step to rounding; they are the better choice when more than a few digits are wanted: for the batch of disc orbits of § 4.5, 450 sixth-order steps over 2 Gyr (3150 force evaluations) reach the worst-case accuracy of the adaptive Dormand–Prince solve at tolerances of 10^{-8}, which takes about 250 steps (3250 force evaluations). The leapfrog is second order: its error falls by a factor of four per halving of the step, and in a static potential its energy error oscillates without the secular drift that a non-symplectic second-order method with the same steps shows; we verified both for a batch of 2000 orbits in `MWPotential2014` with a moving satellite and the acceleration of a non-inertial frame, against the `C` integrator. For reverse-mode derivatives, the steps are grouped into blocks under the backend's checkpointing (`jax.checkpoint` or `torch.utils.checkpoint`), such that the derivative stores the state at the start of each block and the states within one block, about $2\sqrt{N_{\rm steps}}$ states rather than $N_{\rm steps}$: for 10^5 orbits and 8000 steps in `MWPotential2014`, the compiled `jax` derivative needs

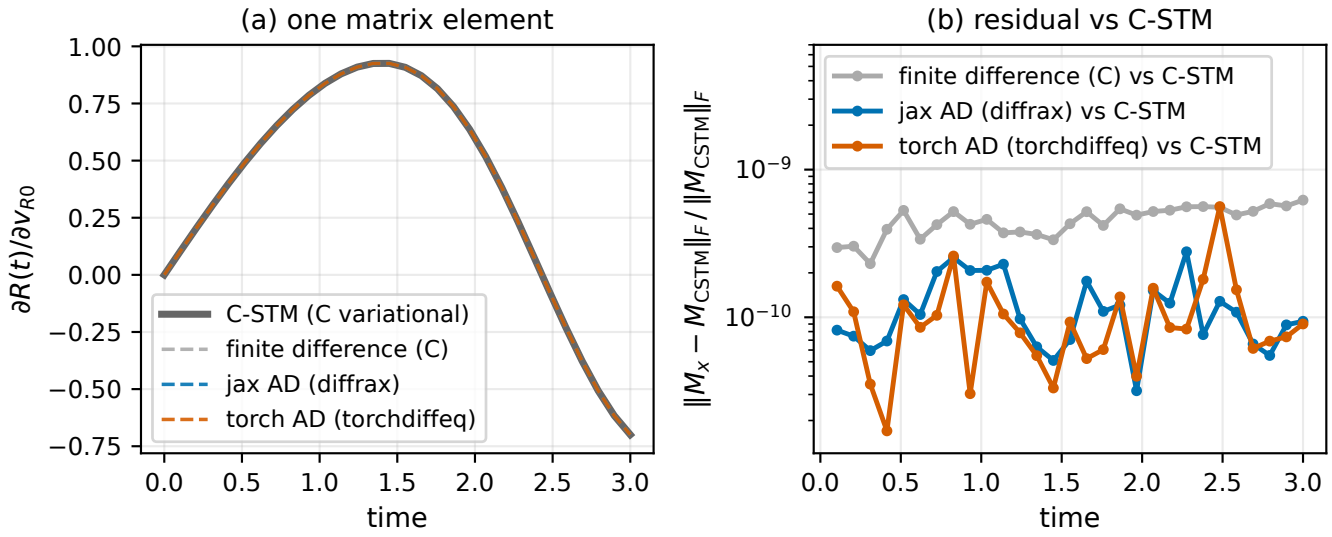


Figure 8. The derivative of an orbit with respect to its initial condition, in four ways. For a Sun-like orbit in `MWPotential2014` integrated for 3 Gyr, the left panel shows one element of the state-transition matrix, $\partial R(t)/\partial v_{R,0}$, from the `C` variational equations, from a finite difference of the `C` orbit, and from reverse-mode automatic differentiation through the `diffrax` and `torchdiffeq` solvers. The right panel shows the relative Frobenius norm of the difference between each of the latter three and the variational result; all agree to a few parts in 10^{10}.

1.4 GB of temporary memory instead of 39 GB. The derivatives agree with Richardson-extrapolated central differences of the same integration to 10^{-10}, and the `jax` and `PyTorch` versions agree with each other to 2×10^{-14}. An initial condition in single precision is integrated in single precision throughout, including the times and the forces, which matters on GPUs with a low double-precision rate (§ 4.5). Forces that do not derive from a potential, such as dynamical friction or the Coriolis force of a rotating frame, which `galpy` marks with the `isConservative` attribute, raise an error, because the closing kick would evaluate them at the half-step velocity and the method would then converge only at first order. As a second-order method, the leapfrog needs small steps to be precise: for 10^4 disc orbits integrated for 0.36 Gyr, 1000 leapfrog steps (1000 force evaluations) give a worst-case position error of 3×10^{-4} (in units of R_0), while 250 Dormand–Prince steps (3250 force evaluations) give 4×10^{-7}. The leapfrog is therefore the method of choice when the force is expensive to evaluate and modest precision suffices, the sixth-order integrator when the orbits share a timescale and precision matters, and the adaptive Runge–Kutta solvers remain the choice for batches that mix timescales.

4.3. *Variational equations and chaos*

The variational equations that the `C` state-transition matrix relies on were part of `galpy` for planar orbits since 2011; as part of this work, we extended them to three-dimensional orbits, to all of the `C` integrators including the symplectic ones, to one-dimensional orbits, and to dissipative, velocity-dependent forces, and we released them in `galpy v1.12`. Appendix A gives the equations and the checks that we applied to them. Their user-facing form is `Orbit.integrate_dxdv`, which integrates an orbit together with a deviation vector or with the full state-transition matrix, and `Orbit.lyapunov`, which computes the largest Lyapunov exponent, or the spectrum, by the renormalization method of G. Benettin et al. (1980a,b), for example as

```
o= Orbit([0.8,0.3,0.75,0.,0.2,0.])
o.integrate_dxdv([1.,0.,0.,0.,0.,0.],ts,
                 MWPotential2014,method='dop853_c')
o.getOrbit_dxdv() # the deviation along the orbit
o.lyapunov(ts,MWPotential2014,method='dop853_c',
           renorm_every=10)
```

For example, for the Hénon–Heiles potential at $E = 1/8$, the exponent of a chaotic orbit converges into the range quoted in the literature while that of a regular orbit at the same energy decays as $\ln t/t$ (Figure 9). Like the variational equations themselves, these are `numpy` and `C` features that do not require the backends: they accept backend initial conditions, but run on the CPU, return `numpy` arrays, and are not differentiable.

Largest Lyapunov exponent from `Orbit.lyapunov`, as a running estimate against time. Left: a chaotic and a regular orbit in the Hénon–Heiles potential at $E = 1/8$; the shaded band is the range of values in the literature for the chaotic orbit, and the dashed line is the $\ln t/t$ decay expected for a regular orbit. Right: a Sun-like orbit in `MWPotential2014`, which is regular. The estimate for the chaotic

Hénon–Heiles orbit converges into the range found in the literature, while that for the regular orbit decays as expected.

4.4. *Limitations*

In-backend orbit integration has a few limitations. `torchdiffeq` has no batching rule, such that a batch of orbits under `PyTorch` is a `Python` loop unless `torchode` is used. The eighth-order method of `torchdiffeq` carries a gradient error of about 10^{-5} in its backward pass, which is why `galpy` uses its fifth-order method. Additionally, the pure-`Python` integrators implemented in `galpy` are not routed to the backends (§ 4.1); they can run on a backend under the forced mode of § 2.2, but are slow there, because each step is a separate set of array operations, and we do not recommend it. Finally, the `C` integrators cannot be used on a GPU at all; their derivative rule runs the integration on the CPU.

4.5. *Speed*

Figure 10 compares the wall time of integrating N orbits for 2 Gyr in `MWPotential2014` with the `C` integrator, which parallelizes over orbits with OpenMP, on one core and on the 36 physical cores of the CPU node used in this work, and with the in-backend solvers on one CPU core and on two GPUs. The absolute times depend on the specifics of the problem, so we focus on the relative trends. On a single CPU core, the compiled `jax` solve and the eager `PyTorch` solve cost about the same per orbit as the `C` path for large batches, but they start from a floor of one to a few seconds, set by compilation and kernel launches, that the `C` path does not have, such that the `C` path is faster for up to a few thousand orbits; the compiled solve additionally takes seconds to compile, a cost that is paid once per shape of the problem. The `C` path is also the only one that makes use of more cores: its OpenMP loop speeds up almost linearly to the 36 cores of our node for 10^4 orbits or more, eager `PyTorch` gains at most a factor of 3.5 (on 16 threads), and compiled `jax`, whose runtime keeps no more than a few cores busy, gains nothing. The GPUs amortize a floor of half a second to a second over the batch: the compiled `jax` lines overtake one-core `C` at a few hundred orbits and the `PyTorch` lines at a few thousand. The consumer GPU, which executes double-precision arithmetic at one thirty-second of its single-precision rate, never beats the 36-thread `C` node with the adaptive solvers, while the H100 passes it at about 10^4 orbits and is 5 to 9 times cheaper per orbit for $N \geq 3 \times 10^5$. The sixth-order symplectic integrator of § 4.2, with the common step of that section, is the cheapest in-backend route for this batch of disc orbits, because its state per orbit is a fraction of the adaptive solvers' and its steps are the same for every orbit: on one CPU core it costs half of the adaptive `C` path, on the consumer GPU it beats the 36-thread `C` node for $N \geq 10^4$, which the adaptive solvers cannot there, and on the H100 it is 3 to 7 times cheaper than the adaptive solve on the same card and about 30 times cheaper than the 36-thread `C` node for $N \geq 10^5$; for fewer orbits, its compiled step loop is bound by its kernel launches, at 0.3–0.4 s for the 450 steps on either GPU. The derivative of an orbit with respect to its initial condition follows the same pattern (Figure 11): the `C` variational integrator computes it fastest for a single orbit, by one to three orders of magnitude over the backend solvers, and the GPU overtakes it at batches of a few hundred orbits.

To gauge what GPU-native code could add, we also compiled `galpy`'s `C` implementations of `MWPotential2014` and of its integrators as a CUDA kernel that integrates one orbit per GPU thread; this is an experiment and not part of `galpy`. After restructuring the kernel, such that the force is evaluated by a single out-of-line function and threads pick up a new orbit when theirs is done, `dop853` on the H100 is 4 to 11 times faster per orbit than `galpy`'s compiled `jax` integrator on the same GPU and 30 to 90 times faster than the `C` integrator on the 36 cores of our CPU node (2.4 μs per orbit with a direct port of `galpy`'s `C` code and 0.9 μs with special functions rewritten for the GPU), while agreeing with the latter to within its truncation error (Figure 10). The kernel is limited by per-thread state: an adaptive eighth-order step keeps ≈880 B per orbit live, so few threads fit on each multiprocessor. `galpy`'s sixth-order symplectic integrator with one common step for the whole batch reaches the same accuracy with 15% fewer force evaluations and a fraction of the state, and runs another 3 to 4 times faster. These numbers hold for a smooth, cheap potential and near-solar disc orbits with similar dynamical times, for which one step suits every orbit; a batch that mixes very different timescales, or has close encounters or a rapidly varying potential, has to step at the rate of its fastest orbit, and there adaptive integration wins. Such kernels are not differentiable, so they would complement the backend integrators rather than replace them (§ 10).

`galpy` is not the only galactic-dynamics code that integrates orbits with `diffrax`: `galax` (N. Starkman et al. 2026b), a re-implementation of `gala` in `jax`, does so by design, and comparing the two checks that the cost of the in-backend solve is in the solver and the potential rather than in the library around them. The green lines in Figure 10 show this comparison for the same orbits, potential, and tolerances. When we make `galax` control its step size per orbit, as the `galpy` lines do, the

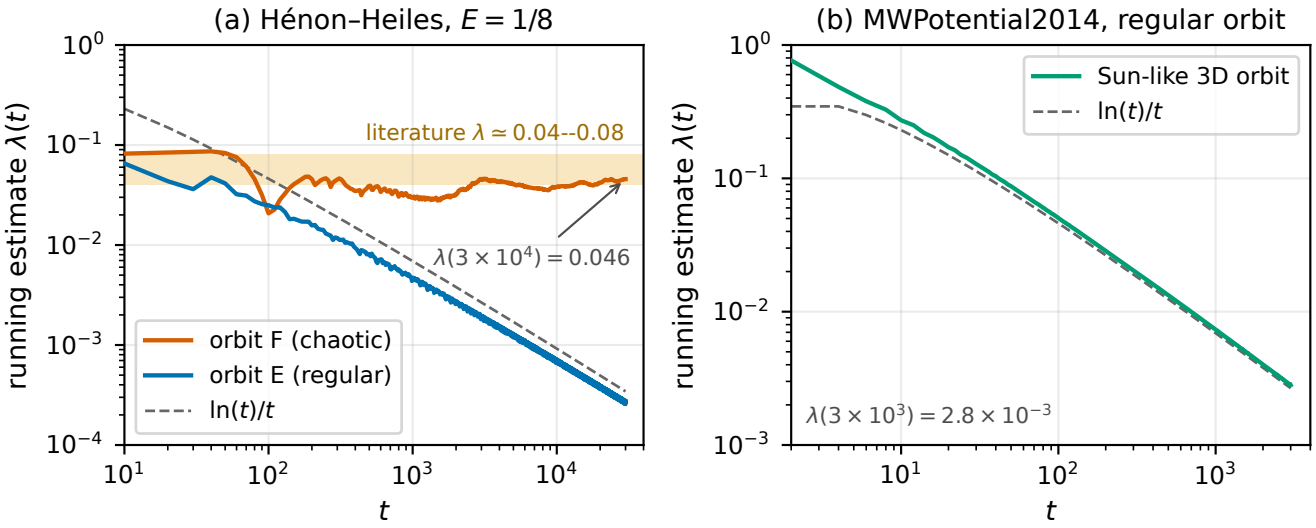


Figure 9. Largest Lyapunov exponent from `Orbit.lyapunov`, as a running estimate against time. Left: a chaotic and a regular orbit in the Hénon–Heiles potential at $E = 1/8$; the shaded band is the range of values in the literature for the chaotic orbit, and the dashed line is the $\ln t/t$ decay expected for a regular orbit. Right: a Sun-like orbit in `MWPotential2014`, which is regular. The estimate for the chaotic Hénon–Heiles orbit converges into the range found in the literature, while that for the regular orbit decays as expected.

two codes take the same time to within 13 per cent on both GPUs for batches of 10^5 orbits or more; for smaller batches, whose cost is dominated by fixed overheads, they differ by up to a factor of 1.6, in opposite directions on the two GPUs. In its default mode, `galax` integrates the whole batch as one system with a single step-size controller, whose error estimate is the root-mean-square over all orbits, such that one sequence of steps serves every orbit; this is cheaper per orbit for large batches, by a factor of two to three, at the price of an error control that holds only in the batch-averaged sense and that depends on which orbits happen to be in the batch. `galpy`'s `jax` solve does the same for a batch on a shared time grid (§ 4.1); the `galpy` lines in Figure 10 were measured with per-orbit control, as `jax.vmap` over single-orbit solves, which is the arrangement that the per-orbit `galax` line matches.

5. ACTION–ANGLE COORDINATES

5.1. *General approach*

The action–angle module is where `galpy` makes heavy use of numerical methods, all of which were discussed in § 2.4: root finding for the turning points of an orbit, quadrature of the action, frequency, and angle integrals between them, and, for the isochrone approximation of J. Bovy (2014), auxiliary orbit integrations with the in-backend solvers of § 4.1. All of `galpy.actionAngle` is migrated onto the backends except for `actionAngleTorus`, a convenience wrapper around the external torus-mapping code of J. Binney & P. J. McMillan (2016) that is not a core part of `galpy` and that we excluded from this work. The public interface is unchanged: an instance of one of the `actionAngle` classes, such as `actionAngleStaeckel`, is set up for a potential and called with phase-space positions, through `__call__` for the actions, `actionsFreqs` for actions and frequencies, and `actionsFreqsAngles` for all three; a call with backend arrays, or with a potential that carries a backend parameter, computes with that backend. Because these methods take a variable number of positional arguments, they resolve the namespace and promote scalars internally rather than through the coercion decorator of the potentials. Derivatives are available with respect to the phase-space position and, through the Python implementations, with respect to the parameters of the potential; these parameter derivatives agree with finite differences in which the potential and the `actionAngle` instance are rebuilt from the perturbed parameter to a few parts in 10^{10} for, e.g., the spherical, adiabatic, and Stäckel methods. Figure 12 shows the two kinds of derivative for the isochrone potential, where the actions are known in closed form: the derivatives of the radial and vertical actions with respect to the radial velocity, and the derivative of the radial action with respect to the scale radius of the potential, which is the derivative that an inference loop uses when its model is a function of the actions, as a DF of the actions is (§ 6.3).

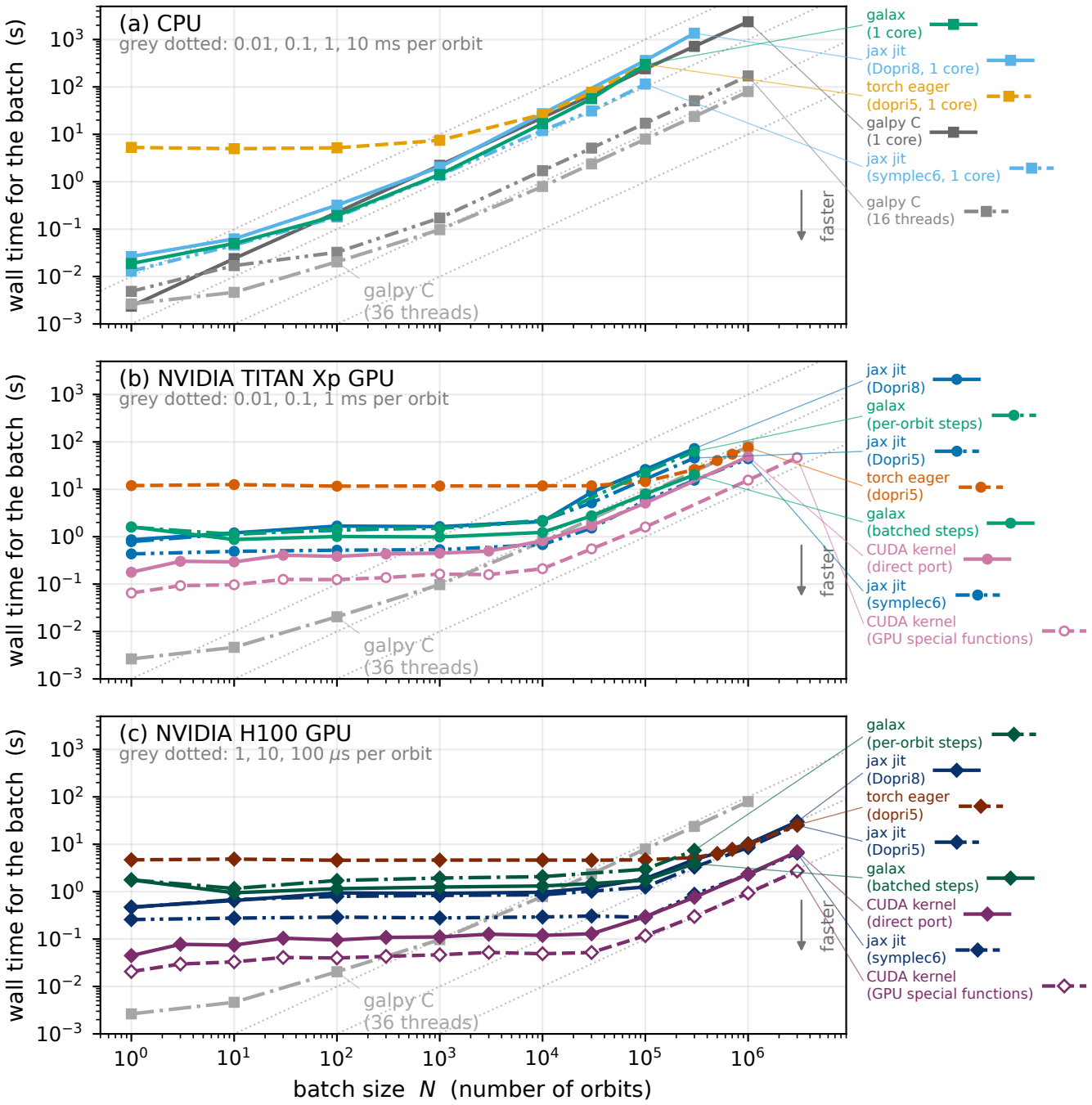


Figure 10. Wall time for integrating N orbits for 2 Gyr in `MWPotential2014` with relative and absolute tolerances of 10^{-8}, on (a) the CPU, (b) the TITAN Xp, and (c) the H100; each label repeats the style of its line. Grey: `galpy`'s `C` integrator `dop853_c`, which parallelizes over orbits with OpenMP, on one core, 16 threads, and the 36 physical cores of the node; the latter is repeated in the GPU panels for reference. Blue: the in-backend `diffrax` solve under `jax`, compiled, with per-orbit step control, with the eighth-order method that `galpy` uses by default and with a fifth-order method that performs the same number of force evaluations as the `PyTorch` solve, and the in-backend sixth-order symplectic integrator (`symplec6_jax`) with 450 common steps over the 2 Gyr, the step at which its worst-case position error over the batch equals that of the adaptive solve. Orange and red: the eager `torchdiffeq` solve. Green: `galax` with its default batched step-size control and with per-orbit control as in `galpy`. Purple: an experimental CUDA port of `galpy`'s `C` forces and `dop853` that integrates one orbit per GPU thread (not part of `galpy` and not differentiable), with a direct port of `galpy`'s `C` code and with special functions rewritten for the GPU, including the copies between host and device, which keep these lines flat up to $N \approx 10^4$. Light grey dotted lines mark a constant cost per orbit, and the arrows point towards lower wall time.

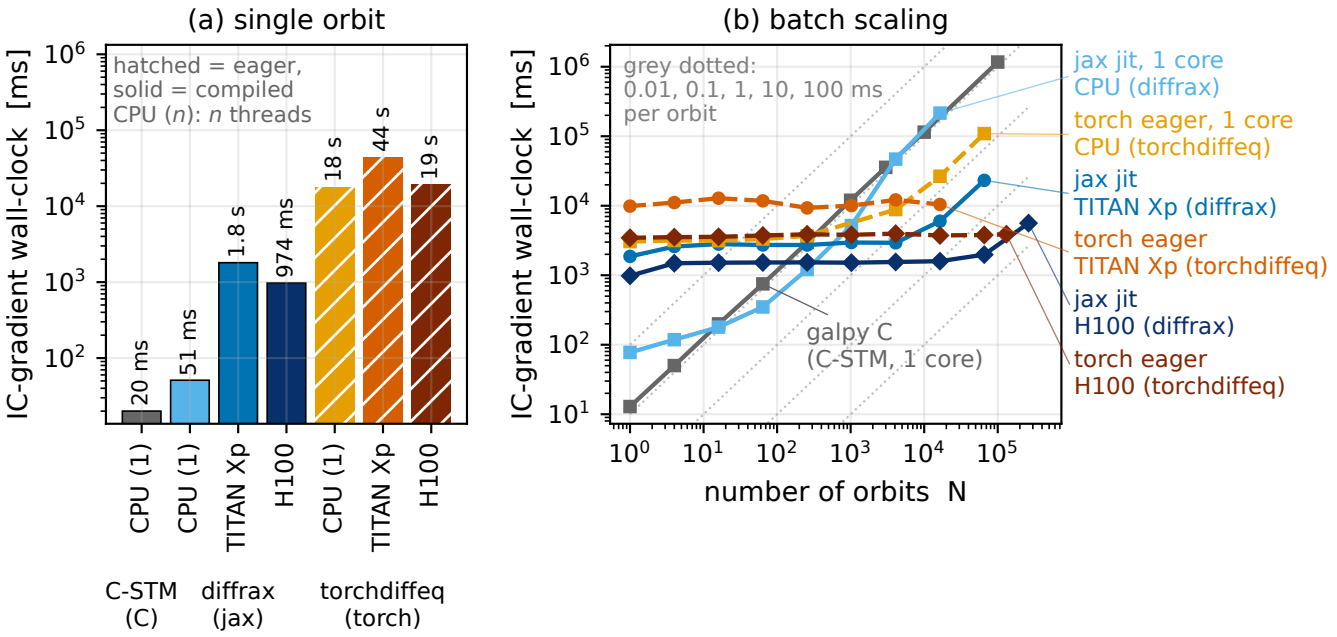


Figure 11. Cost of the derivative of an orbit with respect to its initial conditions, for a 2 Gyr orbit in `MWPotential2014`. Left: wall time for a single orbit with the `C` variational integrator and by reverse-mode differentiation through the `diffrax` and `torchdiffeq` solves on the CPU and the two GPUs; hatched bars are eager and solid bars compiled. Right: the same as a function of the number of orbits differentiated at once. The `C` variational integrator is the fastest way to obtain the derivative of a single orbit, while a GPU overtakes it for large batches. All CPU timings use one core.

5.2. *Spherical potentials and the isochrone approximation*

The isochrone and harmonic-oscillator actions are closed forms, which we migrated by swapping the namespace. `actionAngleSpherical` replaces its `scipy` root finds and adaptive quadratures with the shared root finder and a fixed-order Gauss–Legendre rule, with special handling of circular and near-circular orbits and of orbits that are unbound. The `actionAngleIsochroneApprox` method of J. Bovy (2014) integrates an auxiliary orbit, and does so with the in-backend solvers of § 4.1; it is therefore differentiable with respect to the potential's parameters, twice so under `jax` with the direct adjoint, and runs on a GPU.

5.3. *Inverse transformations*

The inverse transformation, from actions and angles to positions and velocities, exists in `galpy` for the isochrone and harmonic potentials in closed form and for general one-dimensional potentials through `actionAngleVerticalInverse`, which constructs a torus map by a momentum-matched canonical transformation and tabulates it as a function of the action. The construction—a root find for the maximum excursion, a spectral sweep over the angle, a batched Gauss–Newton solve for the anomaly, and Hermite-spline tables—and the evaluation run natively on the backends, trace under compilation, and are differentiable with respect to the action, the angle, and the parameters of the potential. Figure 13 demonstrates this for a self-gravitating isothermal disk: the derivative of the tori with respect to the velocity dispersion of the disk predicts the tori of a nearby disk, agrees with finite differences to 10^{-10}, and serves as the Jacobian of a Levenberg–Marquardt fit of the disk's parameters to stars scattered around three tori. In code, we differentiate the tori of a family of disks parametrized by their velocity dispersion as

```
def torus(sigma):
    disk= IsothermalDiskPotential(amp=1.,sigma=sigma)
    aAVI= actionAngleVerticalInverse(
        pot=disk,Es=Es,nta=128,setup_interp=True)
    return aAVI.xvFreqs(J,angles)[:2] # (z,v_z)
z,vz= torus(0.5)
dz_dsigma,dvz_dsigma= jax.jacfwd(torus)(0.5)
```

The older construction modes of this class, which use point transformations and are not recommended, are built on `numpy` and cannot take a potential with backend parameters.

Other `galpy`-native inverse action–angle transformations are under active development and will be added with full backend support soon.

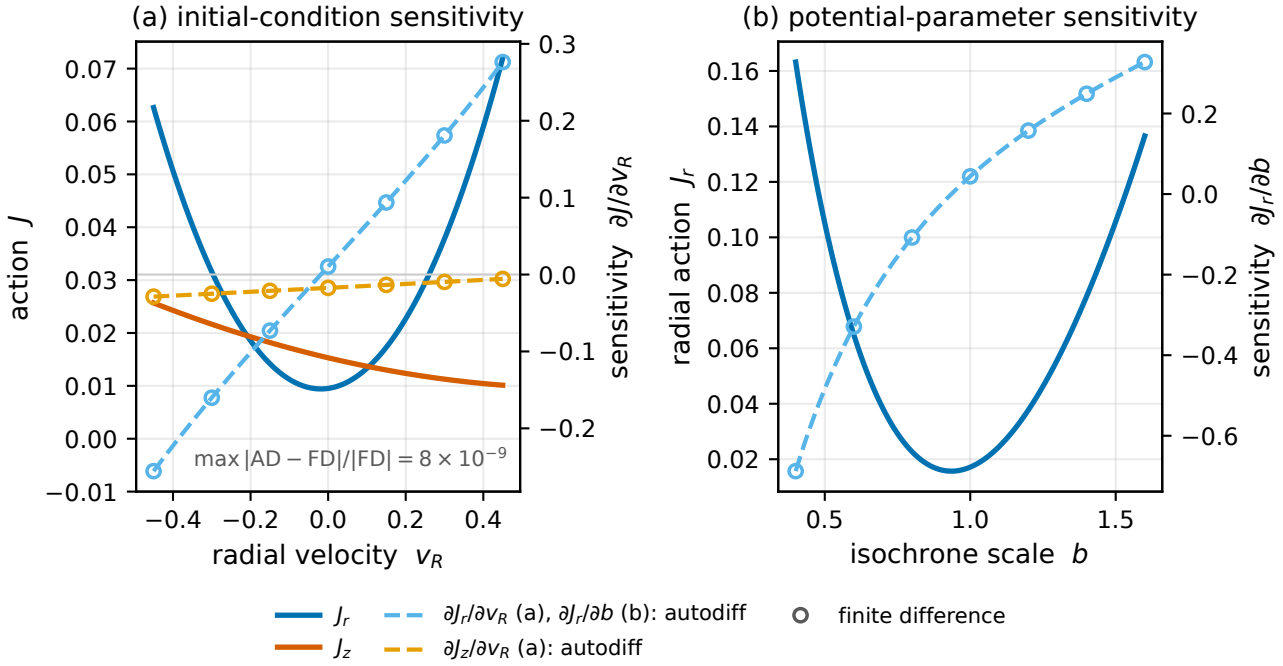


Figure 12. Derivatives of actions in the isochrone potential. Left: the radial and vertical actions of an orbit as a function of its radial velocity (solid, left axis) and their derivatives with respect to the radial velocity by automatic differentiation (dashed, right axis) with finite differences (circles). Right: the radial action as a function of the scale radius b of the potential and its derivative with respect to b.

5.4. *The Stäckel and adiabatic approximations*

`actionAngleStaeckel`, the workhorse for axisymmetric potentials (J. Binney 2012; J. Bovy & H.-W. Rix 2013), has two implementations that behave differently under a backend. As part of this work, we rewrote the pure-`Python` path (`c=False`) to be vectorized over phase-space points, where it previously looped over them, and the same code now runs under `numpy`, `jax`, and `PyTorch`: it finds the turning points with the fixed-count bisection and Newton polish of § 2.4 and evaluates the action, frequency, and angle integrals with the same Gauss–Legendre rule as the `C` implementation, in an anomaly variable that removes the square-root singularities at the turning points, such that the two agree to machine precision. Everything on this path is differentiable with respect to the phase-space coordinates and the potential's parameters, as for example in

```
def jr_of_a(a):
    mp= MiyamotoNagaiPotential(a=a,b=0.3)
    aAS= actionAngleStaeckel(pot=mp,delta=0.4,c=False)
    return aAS(R,vR,vT,z,vz)[0].sum()
djr_da= jax.grad(jr_of_a)(0.5)
```

with the restriction that the derivative of the action integral is of first order only. Figure 14 shows such a derivative, of the radial and vertical actions of a Sun-like orbit in `MWPotential2014` with respect to the flattening b/a of its Miyamoto–Nagai disk (its scale height over its scale length, at fixed scale length), with the potential, its normalization, and the Stäckel transformation rebuilt from b/a inside the differentiated function: the automatic derivative agrees with finite differences to 10^{-9}, at the bottom of the finite-difference error curve, and once compiled it costs about a millisecond.

The `C` path (`c=True`) is much faster for small numbers of points and, on a backend, does more than before: the `C` code computes, along with the actions, frequencies, and angles, their Jacobian with respect to the five phase-space coordinates, which `galpy` attaches to the result as a custom derivative rule, such that `jax.grad` or `PyTorch`'s `backward` through a `c=True` call costs one `C` call and a matrix product:

```
aAS= actionAngleStaeckel(pot=MWPotential2014,
                         delta=0.45,c=True)
def total_jr(R):
    return aAS(R,vR,vT,z,vz)[0].sum()
dJ= jax.jit(jax.grad(total_jr))
djr_dR= dJ(jnp.asarray([1.1,0.9]))
```

As for the orbit integrators, the `C` path gives derivatives with respect to the phase-space position only, to first order, and the focal length `delta` is a fixed number; derivatives with respect to the potential's parameters require `c=False`. The focal-length estimate `estimateDeltaStaeckel` runs on the backends and traces, such that a focal length estimated along an in-

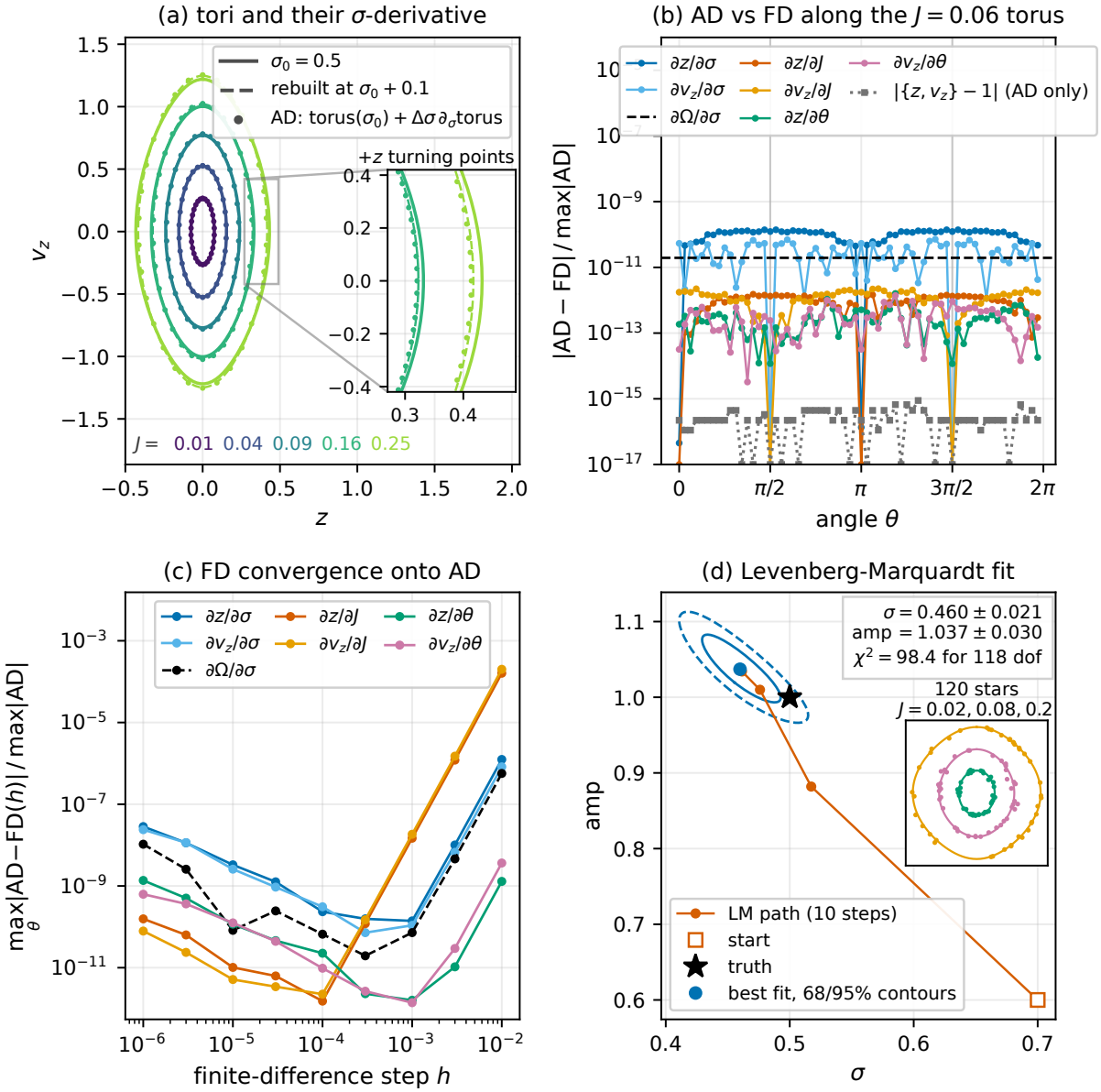


Figure 13. Differentiable inverse action–angle transformation in a self-gravitating isothermal disk with `actionAngleVerticalInverse`. (a) Tori at fixed actions for a disk with velocity dispersion σ_0 (solid), the same tori for a disk rebuilt with $\sigma_0+0.1$ (dashed), and the first-order prediction from the derivative of the tori with respect to σ by automatic differentiation (points). (b) The difference between the derivatives of z, v_z, and the frequency with respect to σ, J, and θ along the $J = 0.06$ torus and five-point finite differences with steps between 10^{-4} and 10^{-3} (the best step of the scan in panel c for each quantity), relative to the largest derivative. (c) The finite differences converge onto the automatic derivatives as h^4 until their round-off floor. (d) A Levenberg–Marquardt fit of (σ, amp) and 120 angles to 120 stars with noisy (z, v_z) on three tori, using the automatic Jacobian; the fit converges in ten steps and the truth lies inside the 68 % contour.

tegrated orbit can be part of a compiled program; for example, the following compiles

```
R,z= o.R(ts),o.z(ts)
aAS= actionAngleStaeckel(pot=MWPotential2014,c=False,
     delta=estimateDeltaStaeckel(MWPotential2014,R,z))
jr,lz,jz= aAS(R,o.vR(ts),o.vT(ts),z,o.vz(ts))
```

The grid-based interpolated version, `actionAngleStaeckelGrid`, builds its grid on the backend, such that the grid lives on the GPU when the potential does and the interpolated actions are differentiable with respect to the query point; when the potential carries a parameter that is being differentiated, the grid class delegates to the exact method, with a warning, because the grid cannot be differentiated through its construction. The adiabatic

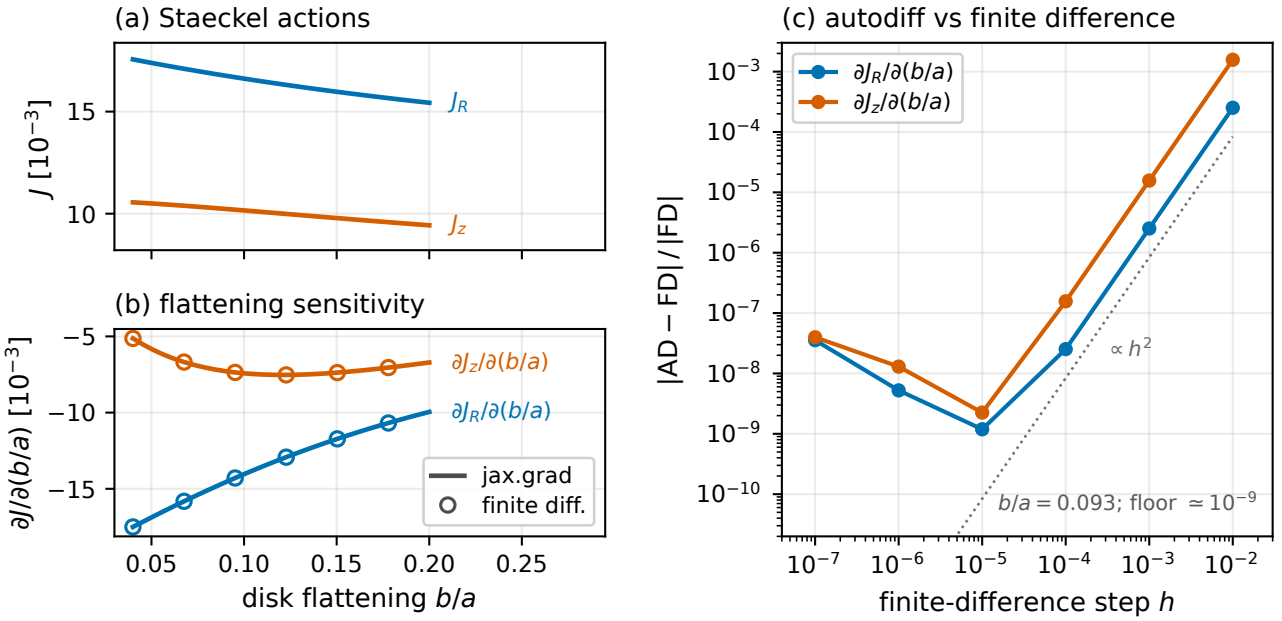


Figure 14. Derivatives of Stäckel actions with respect to a parameter of the potential. (a) The radial and vertical actions of a Sun-like orbit in `MWPotential2014` as a function of the flattening b/a of its Miyamoto–Nagai disk (`MWPotential2014` has $b/a = 0.093$), computed with `actionAngleStaeckel(c=False)` at fixed focal length. (b) Their derivatives with respect to b/a by `jax.grad` (lines) and by central finite differences in which the potential and the action–angle object are rebuilt (circles). (c) The relative difference between the two at $b/a = 0.093$ as a function of the finite-difference step: the finite difference converges as h^2 onto the automatic derivative until rounding error takes over below 10^{-9}.

approximation (J. Binney 2010) follows the same pattern, with a fused Jacobian in `C` for `c=True` and, for `c=False`, the migrated spherical and vertical machinery for its radial and vertical parts.

5.5. *Speed*

Figure 15 shows the speed of the full Stäckel transformation (actions, frequencies, and angles). On a single CPU core, the compiled `C` path remains the fastest. The new vectorized `Python` path costs 2–3 times more per point in plain `numpy`, but it is about 6 times faster than the per-object loop that it replaces for $N \gtrsim 10^3$. Eager `PyTorch` on the CPU runs at the speed of `numpy`, while eager `jax` is bound by dispatching hundreds of small operations per call and stays at a floor of several seconds. Compilation removes that floor, but for large N compiled `jax` is no faster than `numpy` on the CPU, because this quadrature-heavy code is bound by arithmetic that `numpy` already executes in compiled loops. (`PyTorch` is shown eagerly: `torch.compile` breaks this program into hundreds of graphs at the root finder's data-dependent loops, which `jax` stages as compiled loops, and the result is no faster than eager execution on the CPU and fails on the H100.) On a GPU, the same code overtakes the `C` path once N is large enough to amortize a launch floor. For eager `PyTorch` that floor is about 0.6 s on the H100, where it overtakes one `C` thread at $N \approx 2000$ and `C` on 16 OpenMP threads at $N \approx 3 \times 10^4$, and evaluates 10^6 points 78 times faster than one `C` thread. Compiled `jax` on the H100 starts from a floor of only 0.12 s and is five times faster than eager `PyTorch` up to $N = 10^5$ and 2.7 times at 10^6 (1.4 s, 200 times faster than one `C` thread and 19 times faster than 16 threads); it overtakes one `C` thread at $N \approx 400$ and 16 threads at $N \approx 5000$. The floor is set by the incomplete gamma function of the bulge of `MWPotential2014`: `jax`'s native version iterates per element, which is slow on a GPU at every array size, so compiled programs on a GPU use `galpy`'s vectorized series (§ 2.4) at every size; with the native function the floor was 0.7 s. On the TITAN Xp, with its slow double-precision arithmetic, eager `PyTorch` beats single-threaded `C` by a factor of 5–6 at large N, but not `C` on 16 threads; compiled `jax` starts from 0.2 s there and overtakes one `C` thread at $N \approx 10^3$, but not 16 threads. Thus the differentiable backends are also the fastest way to evaluate the transformation for large samples on a modern GPU.

6. DISTRIBUTION FUNCTIONS

6.1. *General approach*

`galpy` contains DF models to represent spherical and disky galactic components. These DF models are constructed from a potential and a set of parameters, they can be evaluated at phase-space points, integrated to obtain their moments (densities, mean velocities, and

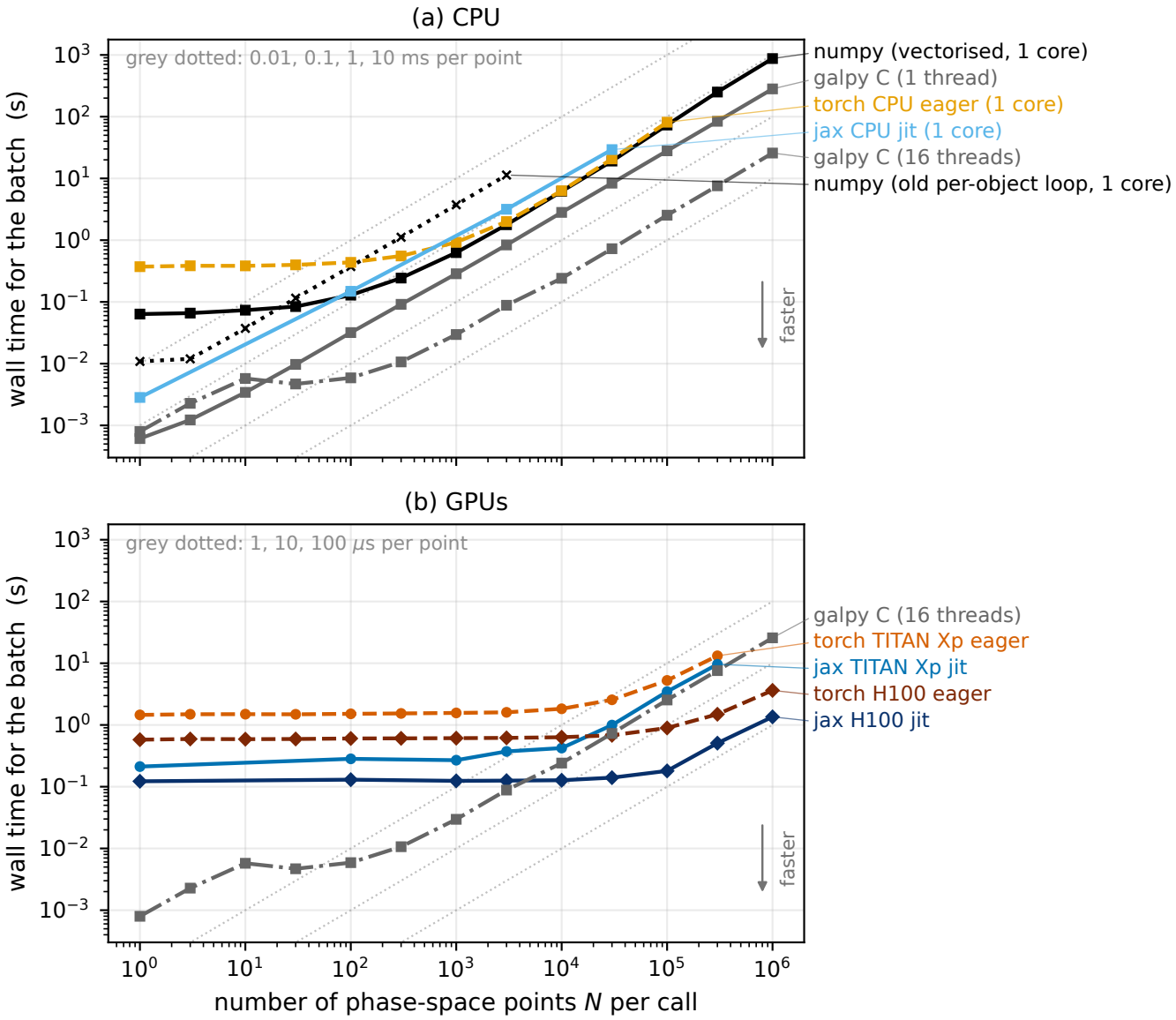


Figure 15. Speed of the Stäckel action–angle transformation (`actionAngleStaeckel.actionsFreqsAngles`) for `MWPotential2014` as a function of the number N of phase-space points per call: wall time on (a) the CPU and (b) the GPUs. CPU lines use one core, except for the dash-dotted `C` line, which uses 16 OpenMP threads and is repeated in (b) for reference; GPU lines are for eager `PyTorch` (dashed; `torch.compile` is no faster here, see the text) and for compiled `jax` (solid) on the TITAN Xp and the H100; host–device transfers add at most a few per cent to the eager `PyTorch` times and are not shown. The black dotted line with crosses is the per-object `numpy` loop that the vectorized path replaces; eager `jax`, which sits at a floor of several seconds on every device, is not shown. Points where the TITAN Xp runs out of memory are omitted. Light grey dotted lines mark a constant cost per point, and the arrows point towards lower wall time. After a launch floor, eager `PyTorch` on the H100 overtakes one `C` thread at $N \approx 2000$ and 16 threads at $N \approx 3 \times 10^4$; compiled `jax`, with a floor of 0.12 s, overtakes them at $N \approx 400$ and $N \approx 5000$.

velocity dispersions), and they can be sampled. On the backends, evaluation and the moments are differentiable for every DF; sampling is differentiable wherever it is done by inverse transformation, which is everywhere except for the many-position velocity sampler of the quasi-isothermal disk DF (§ 6.3); and construction is differentiable for every DF, including those whose construction is itself a numerical computation—the spherical DFs obtained by inverting a density, the King model, the quasi-isothermal disk DF, whose table of guiding-center radii is then built on the backend, and the stream models of § 7—with the one exception of the optional corrections

of the two-dimensional disk DFs for the difference between their input and output profiles, which we did not migrate (§ 6.3). The parameters of a DF, like those of a potential, can be given as backend arrays and are then differentiated through everything that follows.

Every sampler in `galpy` takes a `key=` argument: `key=None` draws from `numpy`'s global generator with the previous algorithm, and therefore reproduces the previous draws, for every sampler except the three of § 2.1 that were replaced on the `numpy` path too, while a backend key selects the inverse-cumulative samplers of § 2.4 and returns backend arrays. For a fixed key, the sample is then a smooth function of the DF's and the potential's parameters, and its derivative is the pathwise, or reparameterization, derivative: differentiating a statistic of the sample with respect to a parameter gives an unbiased estimate of the derivative of the expectation of that statistic. Figure 16 shows a first check that involves no derivative of the sampler itself: a kernel estimate of the velocity dispersion over a large, fixed sample is differentiated with respect to the radius at which it is evaluated, which enters only through the kernel weights, and compared with the derivative of the Jeans equation; the two share no code and agree within the Monte-Carlo noise. Figure 17 then shows the pathwise derivative itself, that of a sample through the construction of the DF, for an Osipkov–Merritt DF of a Hernquist sphere constructed and sampled inside a single compiled function of the scale radius: the derivative of the sampled velocity dispersion with respect to the scale radius agrees with the derivative of the Jeans-equation prediction to within the Monte-Carlo noise, and with a finite difference of the same sample to 10^{-7}. The same figure shows the other kind of derivative of a DF's construction, that of the King model's density profile with respect to the dimensionless central potential W_0 at fixed mass and tidal radius, which passes through the solution of the Poisson equation. In eager mode, `galpy` solves this ODE with `scipy` and attaches the derivative from the forward sensitivity equations, while under compilation it solves the same ODE in the backend.

6.2. *Spherical DFs*

The spherical DFs of `galpy`—the Eddington inversion for isotropic systems, the Osipkov–Merritt and constant-anisotropy inversions, the King model, and the closed-form Hernquist and NFW DFs—share one sampler, which draws radii by inverting the cumulative mass profile and velocities by inverting a tabulated $p(v|r)$, and which on the backends does both by interpolation in tables that are differentiable in their knots. The inversions themselves, which are integrals of a derivative of the density over the potential, use fixed-order Gauss–Legendre quadrature in substituted variables and are differentiable through the integrand, such that the DF can be differentiated with respect to the parameters of the density and potential that it is built from. The King model is the one DF whose `numpy` result changed on purpose: its Poisson equation is now solved to a relative tolerance of 10^{-10} rather than `scipy`'s default of 10^{-3}, which left errors of order 10^{-4} in its tables (§ 2.1). The constant-anisotropy DF is the one place where `jax` was already present in `galpy`, as an optional dependency for the fractional derivative in its inversion; its construction is now much faster than it was, by about a factor of four on `numpy` and by orders of magnitude under compilation, because the $f(E)$ inversion is a fixed-shape quadrature that compiles well. As an example, under `PyTorch` we obtain the derivative of a set of sampled radii with respect to the scale radius of a Hernquist potential as

```
a= torch.tensor(1.,dtype=torch.float64,
                requires_grad=True)
hp= HernquistPotential(amp=2.,a=a)
df= constantbetadf(pot=hp,beta=-0.2)
key= galpy.backend.random.key(7,backend='torch')
sam= df.sample(n=1000,key=key)
dr_da= torch.autograd.grad(sam.r().sum(),a)
```

and for a Hernquist sphere this derivative is r_i/a, which the sampler reproduces to 10^{-13}. Under `jax`, construction, evaluation, and sampling of these DFs run under `jax.jit`, and for the King model so does the derivative with respect to W_0, as in for example

```
def sigmar(W0):
    kdf= kingdf(W0=W0,M=2.3,rt=1.4)
    return kdf.sigmar(0.5)
dsig= jax.jit(jax.grad(sigmar)) # ODE by diffrax
dsig(3.)
```

A DF built from a density that is not the one generating the potential is differentiable with respect to the parameters of both the density and the potential; for example, with respect to those of the density:

```
def dispersion(a_tracer):
    hp= HernquistPotential(amp=0.1,a=a_tracer)
    df= eddingtondf(pot=NFWPotential(amp=1.,a=2.),
                    denspot=hp)
    return df.sigmar(1.)
jax.grad(dispersion)(0.7)
```

The closed-form DFs, including the analytic DFs of N. W. Evans et al. (1997) for power-law spheres, and their differential energy distributions $\mathrm{d}M/\mathrm{d}E$ are likewise differentiable.

6.3. *Disk DFs*

`galpy` contains models for two-dimensional and three-dimensional disk DFs. The two-dimensional axisymmetric disk DFs of W. Dehnen (1999), F. Shu (1969),

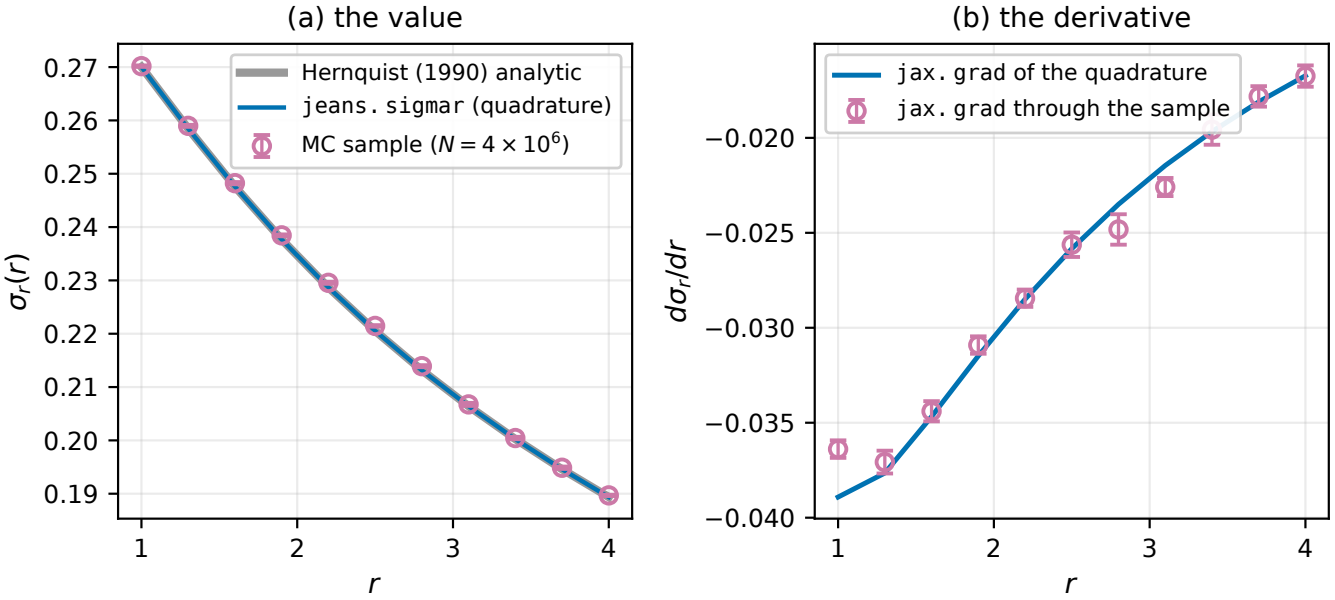


Figure 16. A spatial derivative of a kernel estimate over a fixed DF sample. Left: the radial velocity dispersion of the isotropic Hernquist DF from the analytic expression (L. Hernquist 1990), from the Jeans equation by quadrature, and from a sample of 4×10^6 particles. Right: its derivative with respect to radius, from `jax.grad` of the Jeans quadrature and from `jax.grad` of a kernel estimator over the fixed sample, with error bars from eight sub-samples; the two agree to a median of 0.5σ of the Monte-Carlo noise, with the innermost point at 5σ from the bias of the kernel.

and Schwarzschild and their moments are evaluated with nested Gauss–Legendre quadrature on the backends and are differentiable with respect to the parameters of their surface-density and velocity-dispersion profiles. Their samplers draw by inverse transformation on every backend: the guiding-center radius of a sample from the surface-density profile, its orbital phase from the relation between radius and radial angle that the action-angle quadrature of § 5.2 tabulates for every sampled orbit at once, its velocities at a given position from the DF on a velocity mesh, and its distance along a line of sight from the projected density; each orbit is then replicated according to the ratio of the epicycle frequency at its guiding radius to its radial frequency, which is the DF's measure in the sampled variables. With a key, the positions and velocities of the samples are differentiable with respect to the profile parameters, the number of samples, set by the replication, is not. This replaced adaptive rejection sampling and an orbit integration per sample on the `numpy` path too, and made sampling several times faster. It also corrected the samples: the old sampler weighted each orbit by the epicycle frequency at the sampled radius over the radial frequency of a different orbit, the one with zero radial velocity at the guiding radius, which biased the mean azimuthal velocity of its samples by a few per cent; the new samples reproduce the DF's surface density and velocity moments. We did not migrate the corrections of these DFs for the difference between their input and output profiles. `galpy` also contains a two-dimensional non-axisymmetric disk DF that is a perturbation of a Dehnen DF by a non-axisymmetric and/or time-dependent potential, which is evaluated by integrating orbits backwards in time to an initial axisymmetric DF (in `galpy.df.evolveddiskdf`). The moments of this evolved disk DF differentiate through that integration, which on the backends is one vectorized in-backend solve over the velocity grid rather than a loop.

As a three-dimensional disk DF, `galpy` implements the quasi-isothermal DF of J. Binney (2010); J. Binney & P. McMillan (2011), in the form used by J. Bovy & H.-W. Rix (2013), which is a function of the actions and is evaluated by first computing the actions with `actionAngleStaeckel` and then evaluating the DF. The quasi-isothermal DF is now evaluated and its moments computed on the backends, differentiably with respect to the parameters of the potential and the DF's scale lengths, and its velocity sampler at a given position draws by inverse transformation on a three-dimensional velocity mesh, such that a sample drawn with a fixed key is a differentiable function of the DF's parameters; this sampler replaced a rejection sampler on the `numpy` path as well, which is one of the three places where the `numpy` draws changed (the distribution is the same, the particular draws are not). In code this looks as follows:

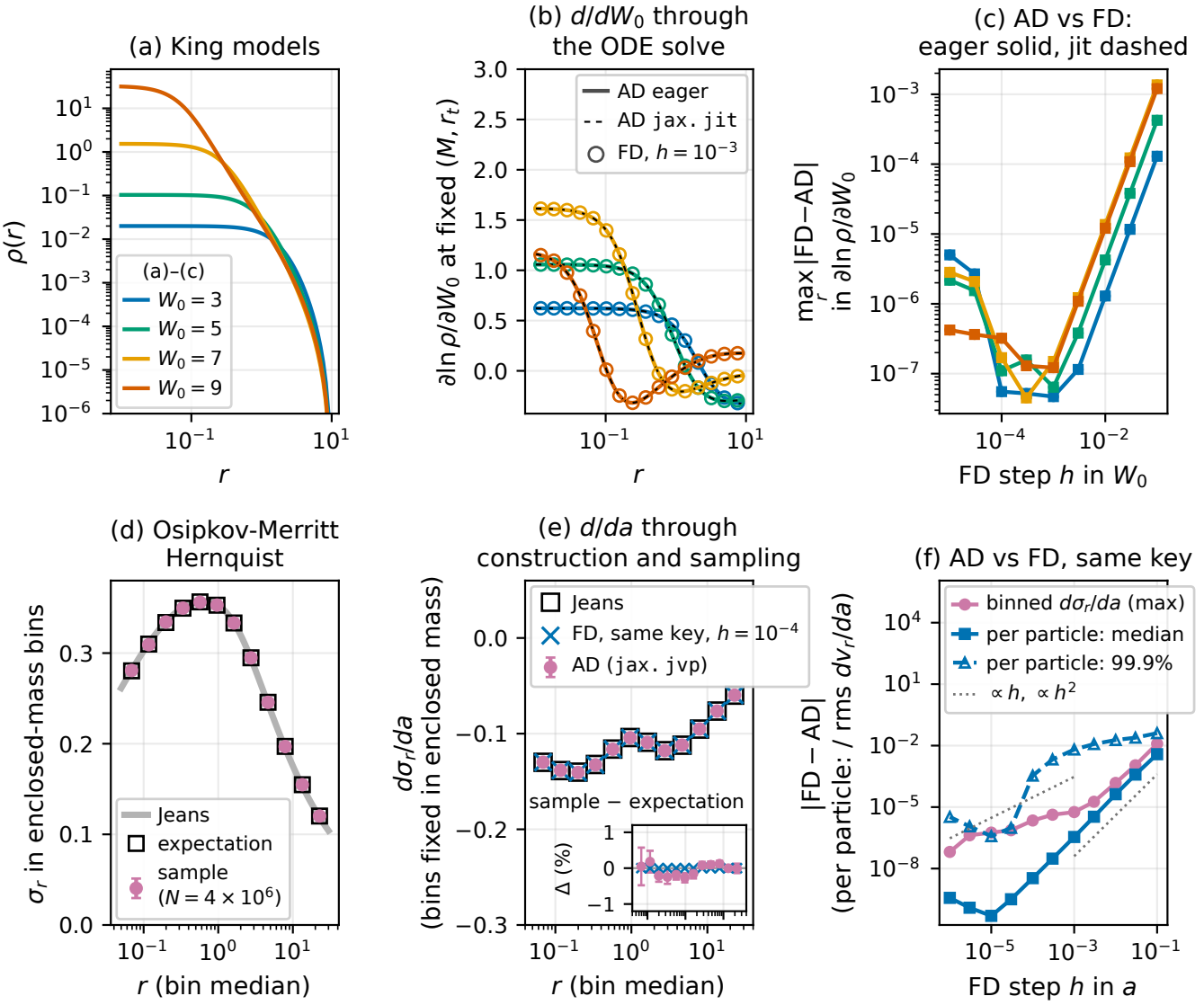


Figure 17. Derivatives through the construction and sampling of spherical DFs. Top row: (a) King models of unit mass and tidal radius $r_t = 10$ for four values of W_0; (b) the derivative $\partial \ln \rho / \partial W_0$ at fixed (M, r_t) through the solution of the Poisson equation, eagerly with `scipy` and under `jax.jit` with `diffrax`, compared with a finite difference in which the model is rebuilt, and (c) the convergence of that finite difference onto the automatic derivative. Bottom row: an Osipkov–Merritt DF of a Hernquist sphere with anisotropy radius $r_a = 1.5$, constructed and sampled (4×10^6 particles) in one compiled function of the scale radius a; panel (d) shows that the sampled velocity dispersion matches the Jeans equation, (e) demonstrates that its derivative with respect to a through construction and sampling matches both a finite difference of the same sample and the Jeans expectation within the Monte-Carlo noise, and panel (f) shows that the per-particle derivatives converge onto the automatic ones as h and h^2.

```
qdf= quasiisothermaldf(1./3.,0.2,0.1,1.,1.,
                       pot=MWPotential2014,aA=aAS)
vs= qdf.sampleV(0.9,0.05,n=10**4,
                key=grandom.key(0,backend='jax'))
sigma_R= jnp.std(vs[:,0])
```

Its construction tabulates the guiding-center radius as a function of angular momentum, which is built on the backend when the potential or a scale length carries a backend parameter, such that the DF and its moments are differentiable through the construction. Its interpolated velocity sampler, which samples many positions at once, is a rejection sampler with `numpy` proposals that runs on the backends, with its shrinking arrays padded to fixed sizes, but is not differentiable.

6.4. *Speed*

Sampling is where the backends change what is practical, as is clear from Figure 18, which displays the time

to draw N particles from four spherical DFs. On the CPU, eager `jax` and `PyTorch` are slower than `numpy` for small samples, because every one of the hundreds of array operations in a draw is dispatched separately and each costs a few to tens of microseconds, and they catch up only at $N \approx 10^4$; compiled `jax` is faster than `numpy` at every size, by a factor of 1.5 to 6 at $N = 10^7$ on one core, because the compiler fuses the hundreds of array operations of a draw into a few loops, where `numpy` executes each separately and writes its intermediate results to memory. On a GPU the sample of 10^7 particles takes tens of milliseconds on the consumer card and a few milliseconds on the H100, two to nearly four orders of magnitude faster than `numpy` on one core. The lower panel shows the cost of constructing a DF, which for the inversions is dominated by the $f(E)$ integral: more than a minute on `numpy` for the constant-anisotropy DF, a fraction of a second once compiled, with the compilation itself costing seconds and paid once.

7. TIDAL STREAMS

Tidal streams are the application in which derivatives have been most sorely missed, because a stream model is an expensive forward computation with a handful of parameters, the potential's among them, and fitting it to data has meant exploring that parameter space with a sampler that knows nothing of the model's smoothness, a cost that has limited such fits to a few parameters at a time. `galpy` contains three stream models, all of which now run on the backends: the action–angle model of J. Bovy (2014), `streamdf`, with its extension to stream gaps (J. L. Sanders et al. 2016); and the particle-spray models of M. A. Fardal et al. (2015) and Y. Chen et al. (2025), `streamspraydf` (see also Y. Qian et al. 2022, for the `galpy` implementation).

7.1. *The action–angle stream model*

`streamdf` models a stream as a distribution in frequency and angle space, computes the mean track of the stream in configuration space by iterating the transformation between the two, and samples and evaluates the DF around that track. On the backends the track computation is a pure function of the progenitor's orbit and the potential that is vectorized over the segments of the track and is differentiable from end to end: the auxiliary orbits are integrated in the backend, and the Jacobian of the action–angle transformation along the track, which was a finite-difference Jacobian in the `numpy` code, is the automatic Jacobian of the methods of § 5. The derivative of the track with respect to a parameter of the potential agrees with a finite difference of the same computation to a few parts in 10^4 (Figure 19), which is the level set by the fixed iteration count of the track refinement. Construction of a `streamdf` is differentiable with respect to the potential, and can be compiled, although the compilation of a construction, which contains tens of orbit integrations and the iteration of the track, takes a few minutes on one CPU core and is only worthwhile when many streams are to be built; the construction itself, once compiled, takes 7 to 30 s on one core, about twice as long as the `numpy` and `C` version. A `streamdf` is set up as before, with a progenitor orbit, an `actionAngle` instance, and a velocity dispersion, as in for example:

```
aAI= actionAngleIsochroneApprox(pot=lp,b=0.8,
        integrate_method='diffrax')
sdf= streamdf(sigv/220.,pot=lp,aA=aAI,
              progenitor=Orbit(jnp.asarray(prog)),
              tdisrupt=4.5/time_in_Gyr(220.,8.),
              leading=True,nTrackChunks=11)
sdf.sample(n=10**4,key=grandom.key(3,backend='jax'))
```

When `lp` in this code carries a backend parameter the track and the sample are differentiable with respect to it. Sampling and evaluating the DF, on the other hand, are 40–100 and 18–37 times faster under compilation than on `numpy` on one CPU core, and hundreds to thousands of times faster on a GPU; the sampler draws by inverse transformation on `numpy` too, which is the other place where the `numpy` draws changed, in the draws but not in their distribution. Figure 20 summarizes these costs.

7.2. *Particle-spray models*

The particle-spray models release particles from the progenitor at random times with random offsets in position and velocity and integrate them forward. On the backends, `sample()` and the track-fitting method `streamTrack()`, which fits a smooth track through the sampled particles, are pure functions of the progenitor's orbit, the potential, the progenitor mass (or a mass history), and a random key: the stripping times are drawn by inverse transformation of uniform variates, the offsets by reparameterized normal variates, the progenitor and the particles are integrated in the backend, and the track is fit with a penalized spline. Every step of that fit is a smooth function of the particles: each particle's closest point on the progenitor's orbit is found by Newton iterations on a spline of the orbit, the amount of smoothing by maximizing the restricted likelihood, and the extent of the track from a kernel-smoothed quantile, so that the fit, which is also the default for `numpy` inputs, gives the same track eagerly and compiled. The whole computation compiles and is differentiable with respect to every parameter; forward-mode derivatives, which are the efficient choice for a few parameters and

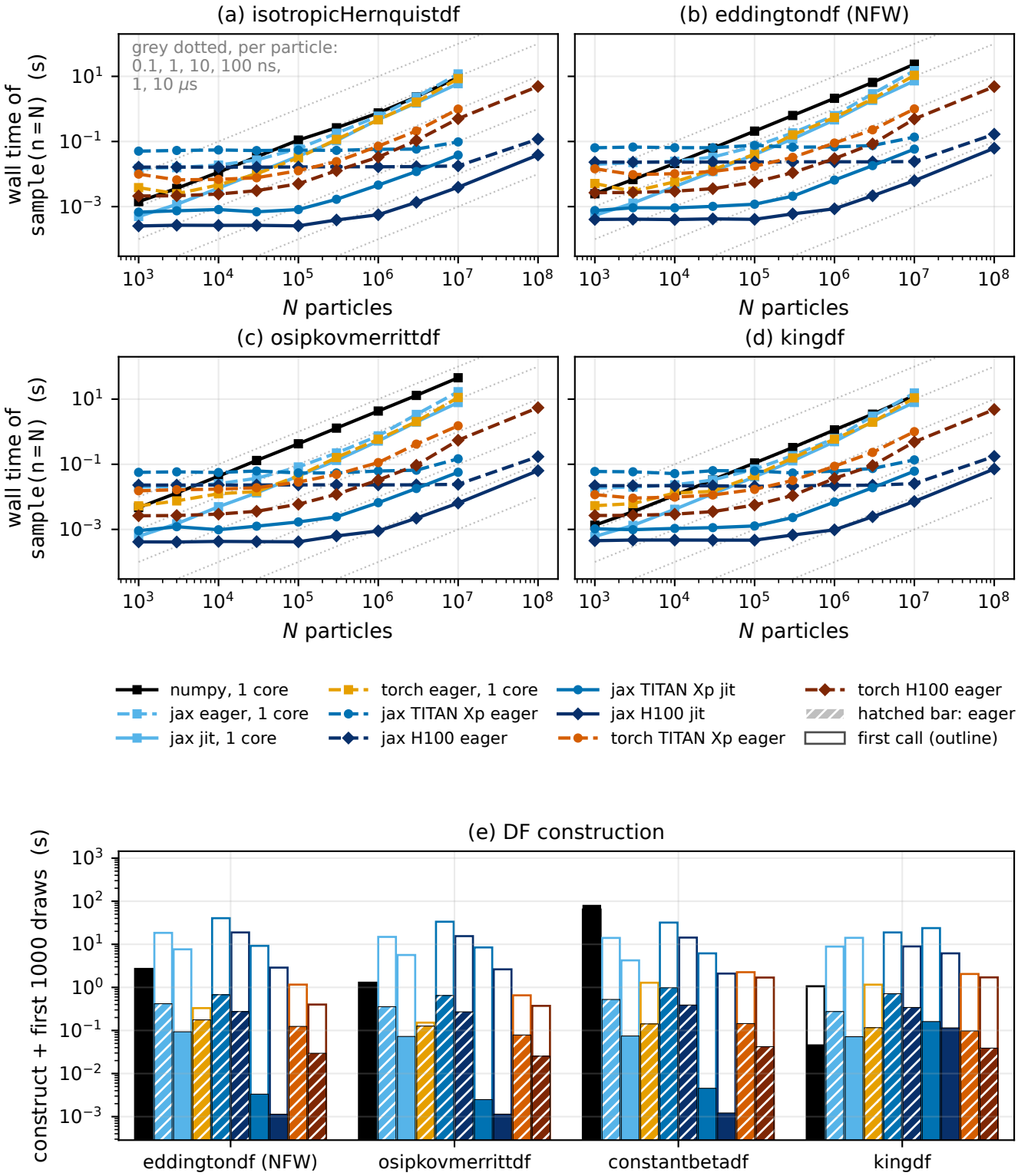


Figure 18. Speed of sampling and constructing spherical DFs. Top: wall time of `sample(n=N)` for the isotropic Hernquist DF, the Eddington inversion of an NFW profile, an Osipkov–Merritt DF, and a King model, on `numpy` and on the backends on the CPU, the TITAN Xp, and the H100; solid lines are compiled, dashed eager. Bottom: time to construct each DF and draw a first thousand particles, warm (filled) and including compilation on the first call in a fresh process (outline); hatched bars are eager. All CPU series use one core. Compiling with `jax.jit` or running on a GPU makes sampling much faster than with `numpy`, by two to nearly four orders of magnitude on a GPU for large samples.

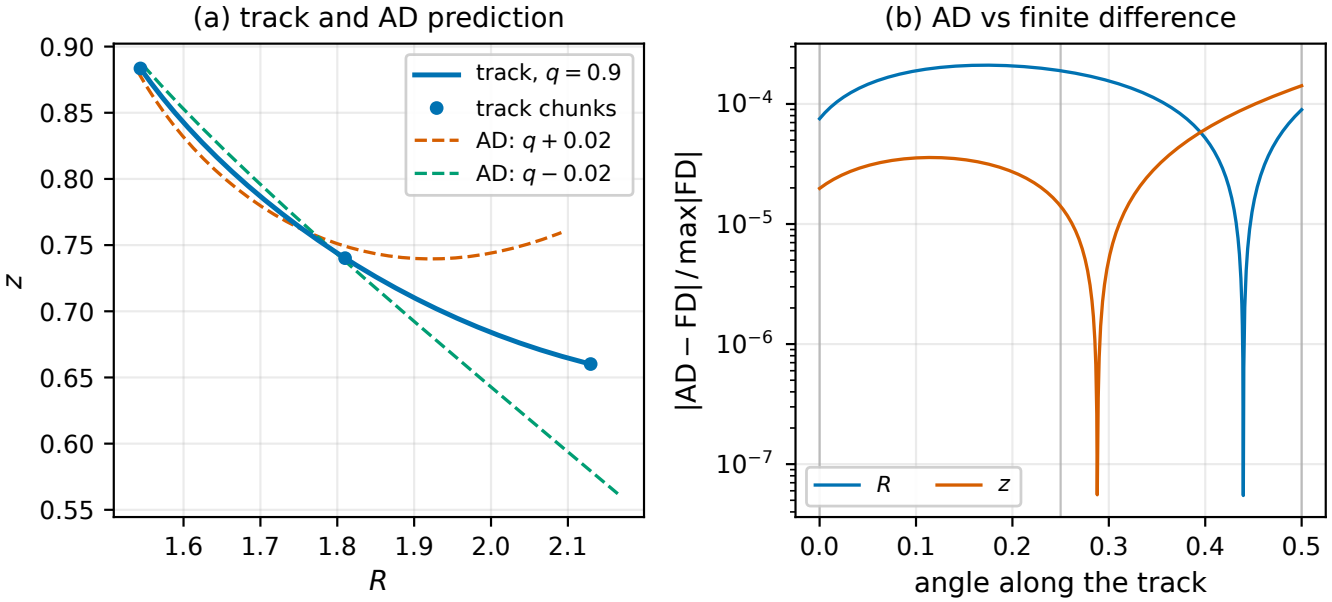


Figure 19. Derivative of a `streamdf` track with respect to the flattening q of a logarithmic halo. Left: the track in the meridional plane at $q = 0.9$ (the interpolated track through the three computed segments, which are marked) and its first-order extrapolations to $q \pm 0.02$ from the derivative by automatic differentiation through the whole track computation, including the interpolation. Right: the difference between this derivative and a central finite difference ($h = 10^{-3}$) of the same computation along the track, relative to the largest derivative of each coordinate; the vertical lines mark the computed segments. The two agree to a few parts in 10^4, the level set by the fixed iteration count of the track refinement.

many particles, require `diffrax`'s direct adjoint, like in the following example

```
key= galpy.backend.random.key(1,backend='jax')
def leading_arm(q):
    lp= LogarithmicHaloPotential(normalize=1.,q=q)
    spdf= fardal15spraydf(mass,progenitor=Orbit(ic),
        pot=lp,tdisrupt=td,
        integrate_kwargs={'adjoint':'direct'})
    return spdf.sample(n=1500,return_orbit=False,
                       tail='leading',key=key)
xv,dxv= jax.jvp(leading_arm,(q0,),(1.,))
```

where the forward-mode derivative gives the change of every particle's position for a unit change in the halo flattening, with the random numbers in common between the two. The fitted track can be differentiated in the same way,

```
def track(q):
    lp= LogarithmicHaloPotential(normalize=1.,q=q)
    spdf= fardal15spraydf(mass,progenitor=Orbit(ic),
        pot=lp,tdisrupt=td,
        integrate_kwargs={'adjoint':'direct'})
    tr= spdf.streamTrack(n=1500,tail='leading',key=key)
    return tr.x(tps)
track(0.9); jax.jacfwd(track)(0.9)
```

where the derivative includes the response of the fit itself: of each particle's closest point on the progenitor's orbit, through the implicit derivative of the Newton iterations, and of the amount of smoothing. This is the derivative that matters for a generative model like this one, in which the stream's stars are a sample from the model's DF: the derivative of the track with respect to a parameter can be taken either with the sample of stars held fixed, which is what one obtains by differentiating the track code alone, or by letting the sample move with the parameter, which is what a finite difference of the full model computes. The two differ, and the first is not a physical quantity; where `galpy` provides derivatives of stream models, they are of the second kind, and we test them against finite differences whose step is shown to be converged: central finite differences of the fitted track converge to the derivative above to 2×10^{-6}, and the derivative of the sampled stream with respect to the potential agrees with finite differences of the same sample to 10^{-6}. Figure 21 shows what such a derivative can do: a single forward-mode pass through the sampling, integration, and track fit of a 6000-particle stream after 4.5 Gyr in a logarithmic halo predicts the stream's track in a halo of different flattening as $\mathrm{track}(q_0) + \Delta q\, \mathrm{d\,track}/\mathrm{d}q$. This first-order prediction removes 80 % of the shift of the track for $|\Delta q| = 0.2$ and 99 % or more for $|\Delta q| \leq 0.006$, a range that covers the steps a fitter takes, with a residual that falls as $|\Delta q|^{1.9}$; moving the particles to first order and re-fitting the track instead leaves a residual that is 1.5 to 2.4 times larger at every step. Each spray particle's orbit is one more orbit in a batch, such that a sprayed stream costs

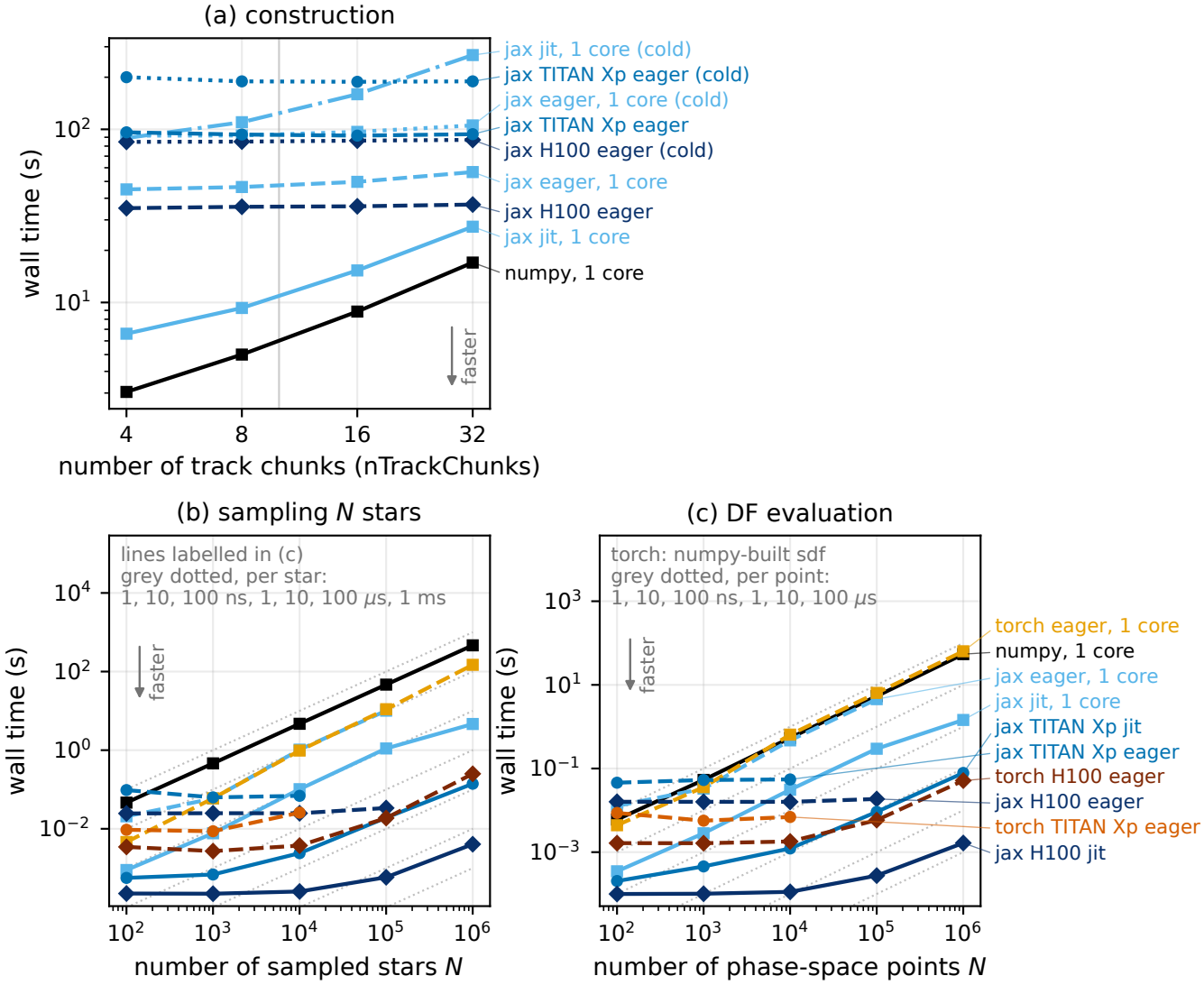


Figure 20. Speed of the action–angle stream model. (a) Wall time to construct a `streamdf`, including its track, as a function of the number of track segments, on `numpy` with the `C` integrators and on the backends, warm and including compilation (cold). (b) Sampling N stars and (c) evaluating the DF at N phase-space points from the constructed model. All CPU series use one core. The compiled `jax` series run to $N = 10^6$ on every device; the eager series end at the last N whose temporaries (arrays of N by the number of track points) fit in memory: 10^4 on the 12 GB TITAN Xp, and 10^5 for eager `jax` on the H100 (with `jax`'s default allocator settings) and on the CPU, where 10^6 needs more than 50 GB; eager `PyTorch` on the H100 reaches 10^6 using 50 GB.

about 0.6 ms per particle on one CPU core under compiled `jax` for batches of up to 3×10^4 particles (1.2–1.7 ms for 10^5–10^6), a fifth of the cost of the non-differentiable `C` version, and about 0.15 ms on the TITAN Xp and 10 μs on the H100 per particle (Figure 22), consistent with the cost per orbit of a large batch of in-backend orbit integrations (Figure 10), of which a spray particle's orbit is one.

7.3. *Stream gaps*

The stream-gap model of J. L. Sanders et al. (2016), `streamgapdf`, computes the velocity kick that a passing subhalo imparts along a stream and propagates it through the action–angle model of `streamdf`. The kick kernels for the Plummer and Hernquist subhalos, the interpolation of the kicks along the stream, and the transformation to the impact frame run on the backends and are differentiable with respect to the parameters of the impact, and the orbit integrations in the model go through the batched in-backend solver when given backend inputs. The construction of the model also runs on the backends and can be compiled: with a backend progenitor and an `actionAngleIsochroneApprox` that

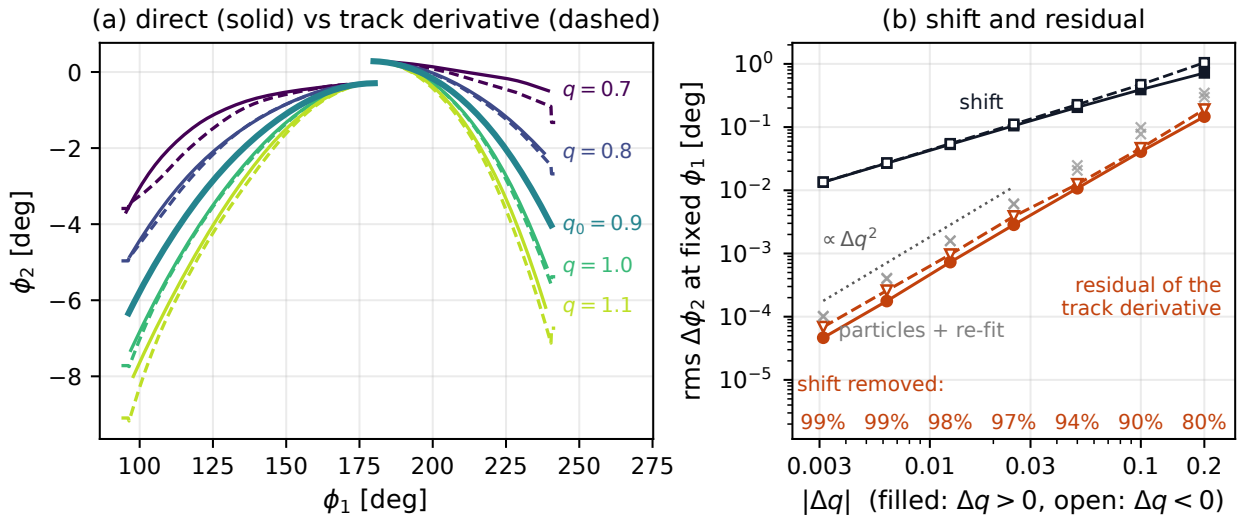


Figure 21. Predicting a stream's track in a different potential from one derivative. Left: the leading and trailing tracks (`streamTrack`) of a 6000-particle `fardal15spraydf` stream after 4.5 Gyr in a logarithmic halo with flattening $q_0 = 0.9$ (thick), the tracks in halos with $q = 0.7$ to 1.1 computed directly (solid), and their first-order prediction track$(q_0) + \Delta q\, \mathrm{d\,track}/\mathrm{d}q$ from the forward-mode derivative of the sampled and fitted track at q_0 (dashed), in a coordinate frame aligned with the progenitor's orbit. Right: the root-mean-square shift of the track at fixed ϕ_1 over the central 80 % of each tail and the residual of the first-order prediction, for $\Delta q > 0$ (filled) and $\Delta q < 0$ (open), with the same random numbers, frame, and track fit throughout; the numbers give the fraction of the shift that the prediction removes, and the residual falls nearly as Δq^2 (dotted). Crosses show the residual when instead the particles are moved to first order and the track is re-fitted.

integrates with `diffrax`, every stage of the set-up is a backend computation, and the construction compiled with `jax.jit` reproduces the eager one to 2×10^{-11}. The density along the stream, the mean frequency, and the frequency–angle kicks are differentiable through the construction with respect to the subhalo's mass, scale, impact parameter, and velocity, the time of the impact, and the parameters of the potential, and these derivatives agree with finite differences of rebuilt models to better than 10^{-6}. The same construction runs under `PyTorch`, with `torchode` for the orbit integrations, and gives the same density and mean frequency along the stream, but eagerly: the set-up of the model that takes a minute and a half under compiled `jax` takes half an hour under `PyTorch`, because its many small array operations are each dispatched separately, so that `jax` is the backend of choice for this model. Speed of the particle-spray stream model: wall time (left) and cost per particle (right) of `fardal15spraydf.sample(N)` with 4.5 Gyr of stripping, for the `numpy` version with the `C` integrator and for the differentiable backend version on the CPU, the TITAN Xp, and the H100, compiled (solid) and eager (dashed). All CPU series use one core; the `numpy` and eager `PyTorch` CPU series end where one call exceeds a minute, while the `jax` CPU series were continued to $N = 10^6$, where one compiled call takes 29 min. Once compiled, the differentiable version is faster than the `numpy` version with the `C` integrator at every N, and much faster on a GPU for large N.

8. HOW THE BACKENDS WERE BUILT

Because an AI agent wrote the code, the question of how one comes to trust such code is as much a part of this paper as the code itself. In this section, we describe the organization of the work, how we reviewed it, the stretch during which the agent worked without supervision, how we made the existing test suite pass under the new backends, and the checks through which we established that the code is correct. Because the division of labor between the human author and the agent is the subject here, this section names them separately, as JB and the agent.

8.1. *Organization*

Work started on 2026 June 1 with a design discussion between JB and Claude, Anthropic's AI coding agent, used through the `Claude Code` environment, which has access to a shell, to the `galpy` repository, and to `GitHub`. The discussion settled the scope (all of `galpy`, on both `jax` and `PyTorch`) and the rules of § 2.1: byte-identity of the `numpy` path, an unchanged interface, no compilation inside `galpy`, and the existing test suite as one of the main acceptance tests. The first code, which took a day to write, was a pilot that made two potentials run under all three array libraries and checked value parity, gradients against finite differences, and the identities between

the potential, its forces, and its second derivatives under automatic differentiation.

The work was organized in branches of the `galpy` repository. `galpy`'s main branch, from which version 1.x is released, continued to receive its usual maintenance and the fixes to the existing `numpy` and `C` code that the backend work turned up (§ 8.6). The backend code was developed on a long-lived *backend branch*, to which every backend change was proposed as a pull request (PR) and which was periodically rebased onto the main branch, without merge commits, so that it always contained the main branch's fixes; it has become `galpy`'s version 2, released as pre-releases while the main branch continues for now as version 1.x. Between 2026 June 1 and October 30, 723 PRs were opened for this work, of which 696 were merged: 602 on the backend branch, 29 for the three-dimensional variational equations, which had a branch of their own, and 65 against `galpy`'s main branch, of which 40 fixed defects in the existing `numpy` and `C` code that the backend work had found (§ 8.6), 16 added capabilities or speed-ups to that code, and 9 concerned the tests and the continuous integration. The merged PRs have a median size of 182 changed lines (quartiles 74 and 372 lines). We kept them small on purpose, because a PR is reviewable when it does one thing, and we split larger pieces of work into stacks of dependent PRs that are reviewed in order (101 PRs are members of such stacks); the migration of the spherical DFs, for example, is a base PR and four PRs that each port one family. Each PR carries a description of what was changed, the measurements that support it (timings, parity counts, gradient errors), and what was explicitly left out, such that the description can be reviewed without re-deriving it from the diff. The rate of PRs followed the kind of work: 88 backend PRs in June, when the agent migrated the potentials family by family from a fixed template; 176 in July, when it ported the orbit, action–angle, and DF modules and the autonomous stretch started; 114 and 112 in August and September, fewer and larger, when the work turned to the compiled modes, the test ledger, and the harder DFs, whose remaining problems were coupled; and 135 in October.

In round numbers, and in US dollars, the work cost the following. The agent ran on a fixed-price subscription to Anthropic's Claude: until 2026 June 25 the "Max" plan at its 20× tier ($200 per month), which Anthropic provided free of charge, afterwards the 5× tier at $100 per month, and for the final two weeks the 20× tier again, this time paid. Successive versions of the Claude Opus models did almost all of the coding (Claude Fable did it on six days, about 4 % of the agent's output), and Claude Fable was otherwise used only to draft this paper and its major revisions. Had the agent's usage (§ 8.2) been paid for through the Claude API instead, it would have cost about $31,000 at the list prices of the time, 61 % of it for re-reading the cached context at every step, 29 % for writing it to the cache, and 9 % for output. The larger cost was the continuous integration, because every backend PR runs `galpy`'s test suite in five modes over 22 shards, about 150 jobs of 20 to 90 minutes each; the GitHub Actions bill for the repository at list price was $13,100 in June, $16,500 in July, $15,800 in August, $10,200 in September, and $8,600 in October 2026, of which about $59,000 is attributable to the backend project (in August and September, the repository also received 58 PRs for another `galpy` project, and a PR to the main branch runs only the `numpy` suite). GitHub's Education program covered this cost, which would otherwise have been prohibitive. The continuous integration was crucial to establishing that the PRs were correct: it ran every test in every mode on every PR, and it often turned up errors that the local runs before pushing had not, in modes, shards, or versions of the dependencies that the agent had not tried. It was also the bottleneck of the project, because a PR could not be merged before its matrix had run, the shards queued behind each other when several PRs were open, and a flaky job meant a two-hour wait; several of our engineering decisions—cancelling the runs of superseded commits, running the compiled modes only on demand until they were stable, running the touched tests under forced backends locally before pushing—served to spend less time in that queue.

JB fixed a few rules early and enforced them throughout, beyond those of § 2.1. A tolerance could be loosened only for one test, only slightly, and with a justification in the PR. Test coverage was not allowed to decrease, not by a single line; in the end, the backend branch leaves 14 of its 56,950 executable statements uncovered, against 24 of 41,704 on the main branch. Tests had to make meaningful assertions at tight tolerances: values had to match their reference to a relative 10^{-13} or better, often 10^{-14}, and derivatives a finite difference that is shown to converge; a test could never merely check that a result is finite, and only tests of sampling had looser tolerances, set by their Monte Carlo noise. A test that passes under a backend only because the code silently fell back to `numpy` does not count (§ 8.5). Before pushing any PR, the agent ran its tests locally under forced `jax` and `PyTorch` backends, which the `numpy`-only checks cannot do, and measured the `numpy` byte-identity by comparing checksums of outputs between two checkouts that differ only in the files of the PR.

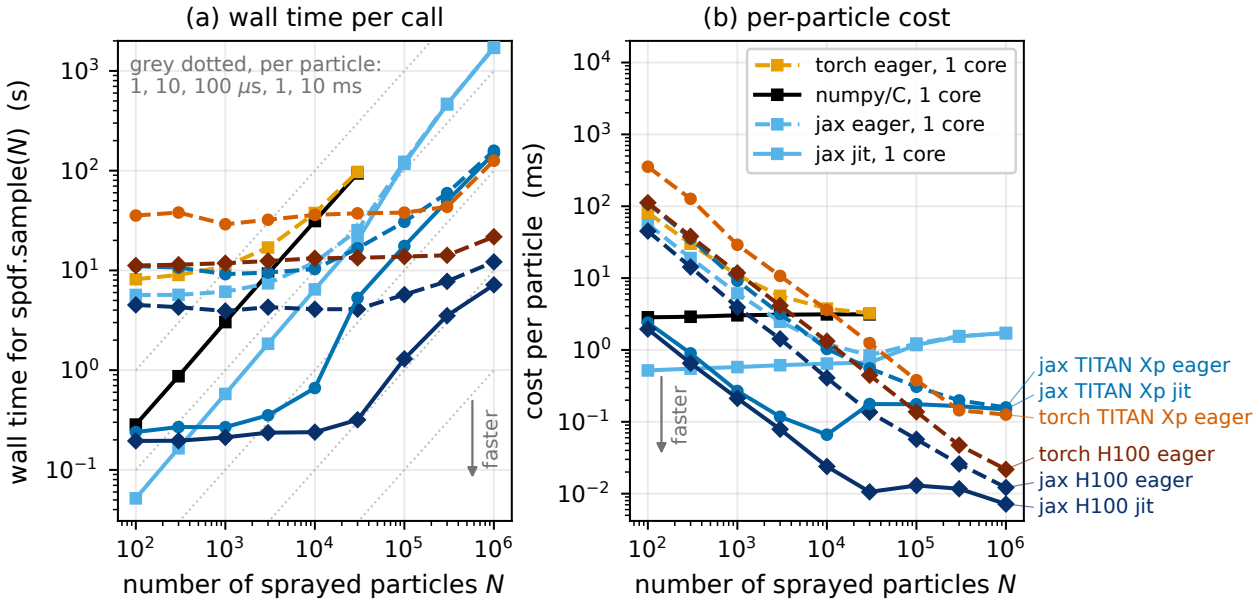


Figure 22. Speed of the particle-spray stream model: wall time (left) and cost per particle (right) of `fardal15spraydf.sample(N)` with 4.5 Gyr of stripping, for the `numpy` version with the `C` integrator and for the differentiable backend version on the CPU, the TITAN Xp, and the H100, compiled (solid) and eager (dashed). All CPU series use one core; the `numpy` and eager `PyTorch` CPU series end where one call exceeds a minute, while the `jax` CPU series were continued to $N = 10^6$, where one compiled call takes 29 min. Once compiled, the differentiable version is faster than the `numpy` version with the `C` integrator at every N, and much faster on a GPU for large N.

8.2. *Review*

Every PR was reviewed before it was merged into the backend branch, and JB merged all of them except during the stretch described in § 8.3, whose work JB reviewed before it was merged into the backend branch. Much of the review was conversational, in the session in which the agent worked, and concerned design as much as code: that a hand-rolled root finder belongs in a utility module or should be replaced by `scipy`'s, that speed-ups are to be demonstrated by timing both versions rather than asserted, that tests should go through public methods rather than internal functions, and so on. Reviews on `GitHub` itself caught the kinds of problems that reviews of human code catch: a variable omitted from an OpenMP `private` clause, a coverage drop of one line that the automated check had passed.

Claude Code records every message of a session, and the token counts of every response, in a transcript, from which we can estimate how much input the work took, if only roughly, because separating JB's own text from the material that he passed on (reports from other agents, reviews, logs) relies on simple heuristics. Between 2026 June 4 and October 30, JB sent about 2,500 messages, on 141 of the 149 days, almost a quarter of them while the agent was working. Most were short, with a median length of 14 words, and in total they contain about 57,000 words of JB's own text, or roughly 110 single-spaced pages, plus about 33,000 words that he passed on. Over the same period, the agent produced about 114 million output tokens (including its reasoning and its tool calls, a third of them in subagents that it started) and read about 44 billion input tokens, almost all of them its own context re-read from a cache at every step.

JB also used two other forms of review. First, at several points he ran an independent AI code reviewer (OpenAI's Codex) over the agent's work and passed its findings to the agent to address (18 reviews in all). These reviews found real defects: a gradient of the incomplete gamma function that was `nan` at the endpoint $x = 0$ although its value was correct, a surface-density routine that used the backend quadrature only when the height above the plane was traced, such that a derivative with respect to the radius at fixed height fell back to `scipy` and failed, and a branch-selection rule in the compiled path of the hypergeometric function that returned infinity where the eager path was correct. The agent reproduced each finding before fixing it and dropped the findings that it could not reproduce with a note saying why. Second, before merging complex PRs, JB instructed the agent to run adversarial reviews of its own work by independent agents whose only task was to break the PR against the real code; these found, for example, that a stream sampler handed the same random key to the position- and velocity-angle samplers,

and that the parity tests of an early PR compared at a tolerance four orders of magnitude too loose to catch the 10^{-16} differences from re-associated arithmetic. The latter finding set the pattern for all subsequent work: the original `numpy` expression is kept byte-for-byte and only the branching is rewritten, never an algebraically equivalent form.

8.3. *The autonomous stretch*

From 2026 July 26 to August 9, JB was away and the agent worked on its own, with the goal of making the whole existing test suite pass under compilation and of emptying the ledger of expected failures (§ 8.4). For this, the agent opened PRs against a separate development branch, which it merged itself once the continuous integration was green, under the understanding that JB would review the accumulated branch on his return. It could not merge into the backend branch or into `galpy`'s main branch, and it could not change the continuous-integration configuration. These were hard guardrails rather than instructions: the agent worked through its own `GitHub` account, whose access token lacks the permission to modify the workflow files that define the continuous integration, and branch rulesets on `GitHub` blocked all updates of the main and backend branches for the duration of the stretch. A scheduler woke the agent at regular intervals to keep it going, and the agent kept a dated log of what it did and why; this log, which runs to 60,000 lines for the summer, is part of the repository that accompanies this paper.

In the two unsupervised weeks, the agent merged 74 PRs. JB was pleased enough with the progress on his return that he let the unsupervised procedure continue for a few more weeks, alongside his supervised work, and 145 PRs in total went through the development branch before it was merged into the backend branch on 2026 September 2, after JB had reviewed it as a whole, with the help of a Codex review whose verdict, that "the work is coherent, the ledger is honest," matched JB's own. The stretch made clear both what such an agent can and cannot do. It burned the compiled-mode ledger down from 100 to 15 failing tests in three days, found and fixed a performance problem in the coercion of inputs that had cost 178 μs per call, resolved the interaction between `astropy` units and compilation, and found two pre-existing bugs in `galpy`'s `numpy` code that it validated against a 30-digit reference before filing them as issues. It also introduced regressions, all of which it found itself within days, using the tools it had built for the purpose, and fixed before JB returned; it misread a stale rule; it once left conflict markers in a working tree, the type of error that a human review would have spotted easily; and its summary table of PRs fell out of date and had to be rebuilt from `GitHub`. The scorecard that it wrote for itself judged the stretch against its goal honestly: the ledger went from 392 to 140 entries against a target of zero. The autonomous stretch succeeded in keeping the agent working productively while JB was away, but it did so because the agent had a clear, limited, and verifiable goal; work towards the broader aims of the project, which requires judgment about what to build and how, would have been much harder to leave unsupervised.

8.4. *Making the test suite pass under every backend*

One of the main acceptance tests for the backends was `galpy`'s own test suite in the five modes of § 2.7. Leaving aside the 27 tests of the out-of-scope torus-mapping code (the 23 tests of its own module and four tests of the stream DF that build the stream's track with it), the suite grew from 2,462 to 4,462 tests over the course of this work, mainly because we split tests that looped over all of `galpy`'s potentials into one test per potential, such that each failure could be traced to its potential. We did not delete or modify tests that failed under a backend, but listed them in a *ledger*, one line per test and mode, that marked them as expected failures, such that the size of the ledger was the measure of what remained to be done; a second ledger listed tests that passed but exceeded the time limit of 300 s per test in the continuous integration. We regenerated the ledgers from the actual test results on demand, such that they could not drift from reality, and the continuous integration posted a report of each PR's results per file and mode, ledgered failures included, to the PR, such that a reviewer saw at a glance what a PR fixed or broke. Figure 23 displays how we brought the suite to pass. On 2026 June 14, when the all-backend continuous integration was first run, 1,042 tests failed under `jax` and, a week later, 2,080 under `PyTorch`; the difference between the two is almost entirely due to `PyTorch`'s rejection of `Python` scalars where `jax` accepts them, and it disappeared when we centralized the coercion of inputs (§ 2.3). The eager ledgers were essentially empty by 2026 September 12, and we emptied the compiled-mode ledgers, which started at 100 and 146 tests when we introduced those modes, over the following month. At the time of writing, the ledgers are empty in every mode.

8.5. *How much of the suite runs on the backend*

A test that passes under a forced backend shows that the code returned the right answer, not that the backend computed it: a code path that converts its input

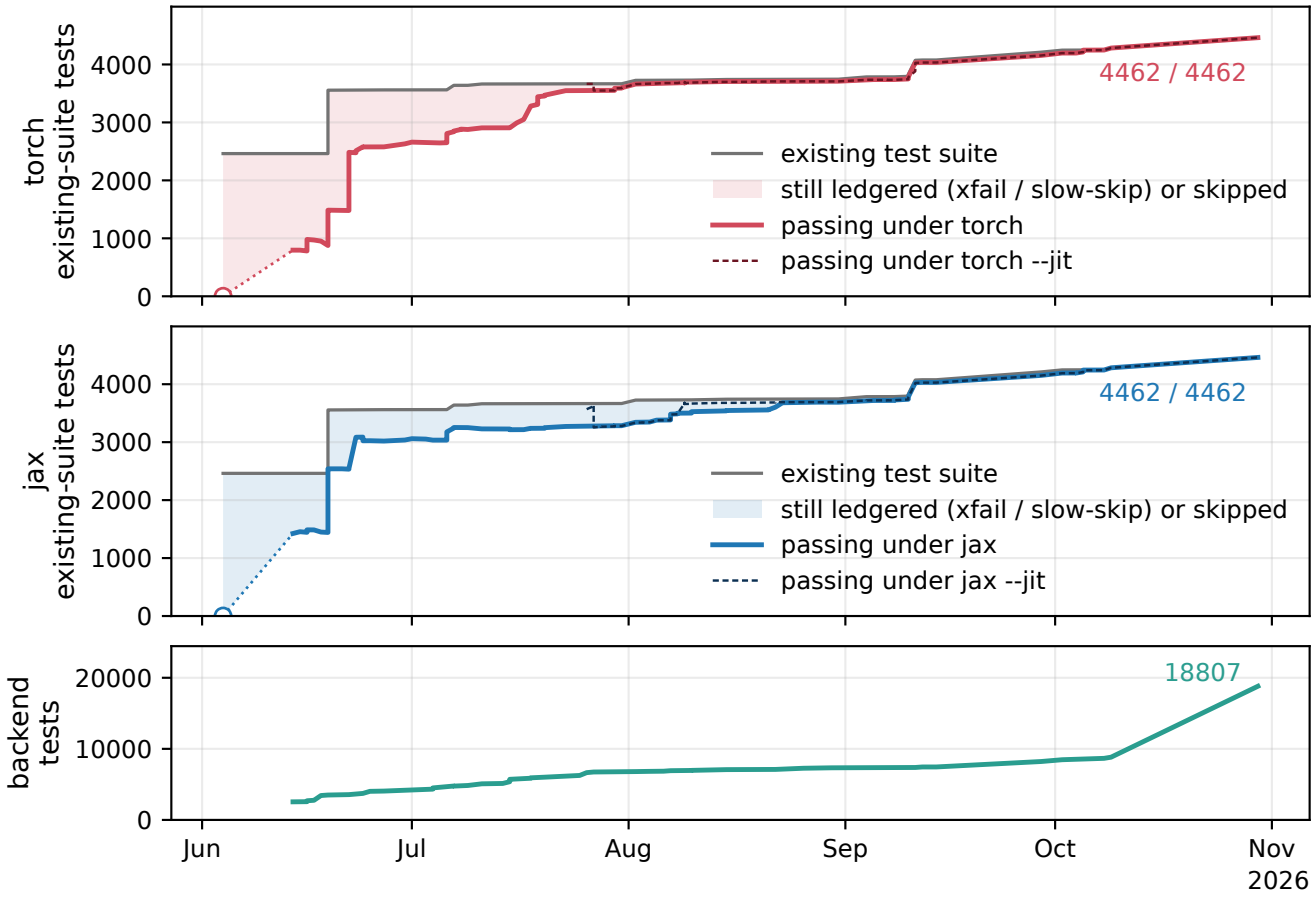


Figure 23. Tests of `galpy`'s existing test suite that pass under each backend, as a function of time, leaving aside the 27 tests of the out-of-scope torus-mapping code. The grey line is the size of the suite, which grows from 2,462 to 4,462 tests (most of the jump on June 19 comes from splitting the potential tests into one test per potential); the colored curves show the number of these tests that pass under `PyTorch` (top) and `jax` (middle) in eager mode (solid) and under compilation (dashed), that is, the size of the suite minus the tests that are ledgered as expected failures or as too slow, or skipped, under that mode (the shaded band; a ledger entry that covers a whole file counts as the tests that it matched at the time). In August, `jax` passes more tests under compilation than in eager mode because slow orbit tests that are skipped in eager mode run in reasonable time when compiled. The bottom panel shows the number of tests in the dedicated backend test modules; its jump in October is the two sweeps of § 8.7. At the end, all 4,462 tests pass in every mode.

to `numpy`, computes with `numpy` or `scipy`, and converts the result back passes every check of a value. In the first months, such *numpy islands* were common, because the natural way to add a backend path to a function, a branch on whether its input is a backend array, took the `numpy` branch for the `numpy` inputs of the existing tests even under a forced backend; a first backend version of the action–angle code passed its tests under `jax` in this way and was rejected in review. The forced mode of § 2.2 is meaningful because functions branch on the resolved namespace rather than on the type of their input, because objects that are constructed under a forced backend hold their parameters as backend arrays, such that their methods compute on the backend also when they are called from code that still uses `numpy`, and because on a GPU an implicit conversion of a device array to `numpy` raises an error rather than copying silently (§ 2.5).

To measure what remains, we instrumented a run of the existing suite under forced `PyTorch` to record every place where `galpy` hands backend data to `numpy`: implicit conversions by `numpy` and `scipy`, explicit copies to the host, and regions in which `galpy` deliberately computes with `numpy`.[2] We classified every such place by the reason for the hand-off and every test by the least benign of its reasons (Table 1). Four reasons concern

[2] We use `PyTorch` for this measurement, because `numpy` reads a `jax` array on the CPU through the buffer protocol, which bypasses the conversion method that the instrument intercepts.

Table 1. Tests of the existing suite by how they hand backend data to `numpy` under a forced `PyTorch` backend (4,353 tests that ran; cumulative).

hand-offs to `numpy` at most	tests	share
none	2,011	46.2 %
+ `galpy`'s `C` extensions	2,321	53.3 %
+ decisions on the host	3,153	72.4 %
+ output (units, plots)	3,322	76.3 %
+ `numpy` functions of the test	3,389	77.9 %
+ constant tables	3,949	90.7 %
+ `numpy`'s random numbers	4,098	94.1 %
+ the `Python` integrators	4,349	99.9 %
+ the documented exception (four tests)	4,353	100.0 %

work that has to happen on the host: `galpy`'s `C` extensions, for orbit integration and the Stäckel actions, whose derivatives come from the state-transition matrix and from custom derivative rules (§ 4.1); decisions that need a value on the host, such as a tolerance, a root bracket, the number of integration steps, or a cache key; output to `astropy` quantities or plots; and `numpy` functions that the test itself supplies, such as the density of an expansion. Three are `numpy` computations that could run on the backend but have no reason to: constant tables built at construction from parameters that are not differentiated, such as the spline of the guiding radius or the coefficients of an expansion (a differentiated parameter takes a backend path, as the sweep of § 8.7 verifies); sampling with `numpy`'s global random-number generator, which the tests seed (a backend random key runs the backend sampler); and the `Python` integrators of the `numpy` code, which warn that their result is not differentiable. The last is one documented exception that keeps `numpy` for `numpy` inputs under a forced backend, while a backend input or a differentiated parameter takes the backend path: a stream that the tests build with `numpy` before the backend is forced, whose lazily computed track stays in `numpy` like the stream itself; four tests reach it.

When we first ran the instrument, 220 tests (5.3 %) passed through a genuine island, `numpy` computation that could and should run on the backend: the transformation of an integrated orbit in the isochrone approximation (55 tests), the `numpy` track tables of `streamdf` (53), the profiles of the disk DFs (39), `scipy` root finding for orbit quantities such as the guiding radius and for the Lindblad resonances (49), and smaller sites in about 25 other functions. We moved all of these to the backend, in eight PRs, mostly with backend code that already existed; the root finders, which now run on the backend also for plain scalars under a forced backend, needed a bracketing that evaluates the function at many points of the bracket at once, because the 45 sequential evaluations of a bisection each cost a full eager dispatch. The same instrumented run now finds no test that passes through a genuine island (Table 1): 46 % of the tests never hand backend data to `numpy`, 78 % do so only for work that has to happen on the host, and the remaining 22 % use `numpy` only for constant tables, for sampling with `numpy`'s random-number generator, in the `Python` integrators, or in the documented exception.

8.6. *Fixes to `galpy`*

Running the same calculations through a second implementation, and differentiating them, is an effective way to find bugs in the first, and the backend work led to the 40 PRs against the main branch that fixed `galpy`'s `numpy` and `C` code (§ 8.1). The three-dimensional variational equations, which require correct second derivatives of the potential, found some of these: four potentials had incorrect planar second derivatives in `C`, and the `C` second derivatives of the basis-function potentials had the wrong sign in one component, which the usual check that the determinant of the state-transition matrix is one cannot catch. The comparison of the backend quadratures with the existing ones found others: the plain Gauss–Legendre quadrature of the action integrands in the adiabatic and Stäckel approximations loses about four digits at the turning points, because the integrands have square-root endpoints, and we regularized it; the default absolute tolerance of `scipy`'s adaptive quadrature truncated several small integrals (the outer tail of the Jeans velocity dispersion, the surface density far from the plane); and the `C` and `Python` implementations of the Chandrasekhar dynamical-friction force interpolated the velocity dispersion with different splines, which differed by a few parts in a thousand at small radii. The figures of this paper, which we regenerate from scratch against the current code, found yet others: the inverse-cumulative sampling of the spherical DFs biased the sampled velocity dispersion low by about 0.7 % at the default grid resolution, and the forces of the two-power density profiles were off by up to 10^{-3} near the inner power-law index of three. We fixed preexisting defects of this kind in their own PRs against the main branch, never quietly in the backend branch, such that each is documented and their count is known.

8.7. *Establishing correctness*

JB does not take the code to be correct because an agent reports that its tests pass; the reasons to believe that it is correct are the following, roughly in order of

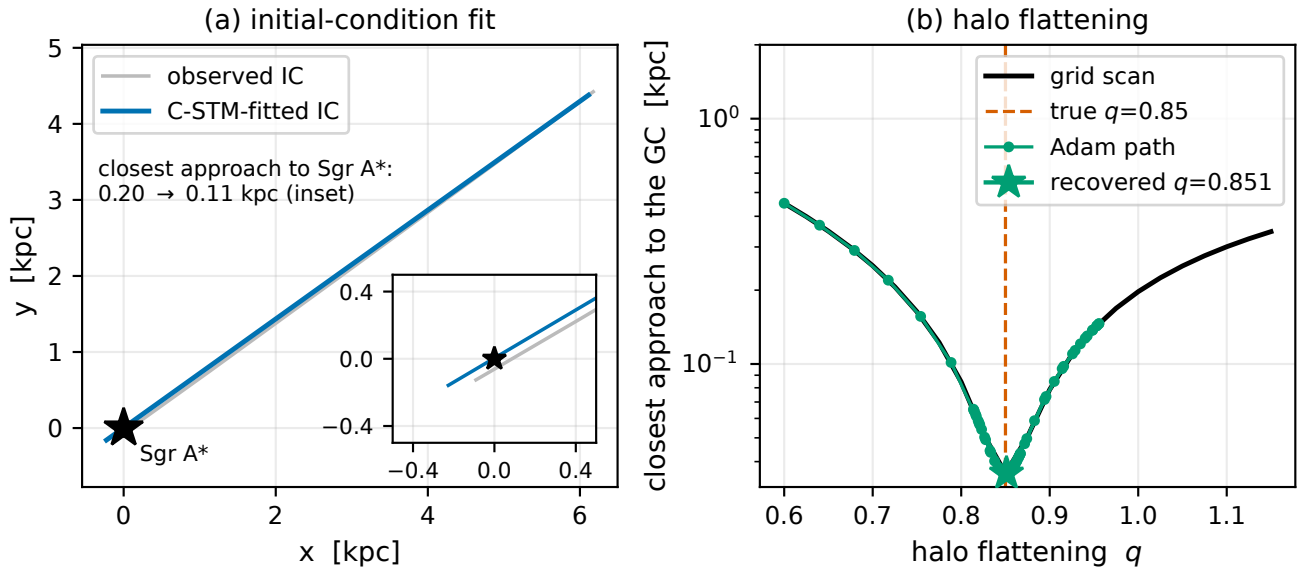


Figure 24. A mock hypervelocity star as a constraint on the Galactic potential. Left: the orbit of a star resembling S5-HVS1, integrated backwards from its observed phase-space position (grey) and from the position within the observational uncertainties that brings the orbit closest to Sgr A* (blue), found by Gauss–Newton iterations on the derivative of the orbit with respect to its initial condition from the `C` variational integrator. Right: the closest approach to the Galactic center of the backward orbit as a function of the flattening q of the halo (black, from a grid of integrations) and the path of a gradient-descent fit of q using the derivative of the closest approach through the in-backend orbit integration (points), which recovers the value from which the star was generated (dashed line).

weight. The `numpy` path is byte-identical to the previous version, measured by checksums of tens of thousands of output values per PR, such that the entire body of validation that `galpy` has accumulated over more than a decade carries over. The backend paths pass the same tests at the same tolerances, in five modes (§ 8.4). We compared every derivative that the code can compute with a finite difference whose convergence with step size is demonstrated rather than assumed, or with an independent reference where one exists: the derivatives of orbits with respect to their initial conditions, for example, are computed by the variational equations in `C` and by reverse-mode automatic differentiation through two different ODE solvers, and the three agree with each other and with finite differences of independently integrated orbits to a few parts in 10^{10} (Figure 8). That no derivative was missed is checked by a sweep: for every public potential, action–angle, and distribution-function class and the stream-impulse functions, it differentiates each numeric parameter and one input coordinate, 761 derivatives in 199 configurations, under compiled `jax` and eager `PyTorch`, with the inputs given as backend arrays, as `Python` floats, and under a forced backend, and compares them with converged finite differences of the `numpy` code. When we first ran it, about 60 combinations of class and parameter raised an error or worked only eagerly, and nine families of derivatives were silently wrong, zero, or disconnected from the parameter; for example, `PyTorch` orbit integration with `galpy`'s `C` integrators dropped the derivatives with respect to the potential's parameters, which made those of a stream model wrong by up to a factor of 45. We fixed all of them, and the sweep, 2,791 cases, now runs in the continuous integration; the one remaining exception is by design: under compilation, the anisotropy of a constant-anisotropy DF cannot be traced, because it selects the order of a derivative in the inversion. A second sweep gives each parameter and input coordinate in turn as a concrete backend array that is not differentiated, and checks that every output comes back on that backend with the `numpy` values (4,249 cases); when we first ran it, about 1,200 cases returned `numpy` arrays, raised an error, or returned an array of the wrong shape, and all of them now pass. Where the backend code implements a numerical method of its own—a special function, a quadrature, a root finder—we checked it against `scipy` and against `mpmath` at 30 to 50 digits and validated the reference itself before use, because the analytic formula one reaches for first is sometimes the one that is wrong. JB, a second AI system, and adversarial agents reviewed the code; running it on two GPUs, which the continuous integration cannot do, found the

device-placement bugs that are invisible on a CPU; and the figures of this paper, each of which is produced by a script that is re-run against the current code and whose output is checked against a stored hash, found several of the bugs listed above. None of these checks is sufficient on its own, and JB would not have accepted the code on the basis of any one of them, only on their combination together with his own review of the code itself.

9. EXAMPLES

While the sections above illustrate each part of the code on its own, here we give two examples that combine several of them in a single analysis. Both are deliberately small: they use mock data and run in minutes, and their point is to illustrate the computation rather than the result.

9.1. *A hypervelocity star as a probe of the potential*

A hypervelocity star ejected from the Galactic center must, when its orbit is integrated backwards, return to the Galactic center, and the closest approach of the backward orbit to the center is therefore a function of the Galactic potential that a measurement of the star's phase-space position constrains (see, e.g., O. Y. Gnedin et al. 2005; O. Contigiani et al. 2019). This is a natural problem for differentiable orbits: the objective is a function of an integrated orbit, and both the star's initial condition and the potential's parameters are to be varied. Figure 24 displays the two derivatives of § 4.1 at work on a mock star resembling S5-HVS1 (S. E. Koposov et al. 2020) in a cored, flattened logarithmic halo. In the left panel, we hold the potential fixed and adjust the star's phase-space position within its observational uncertainties to minimize the closest approach, using the derivative of the orbit with respect to its initial condition from the `C` variational integrator; seven Gauss–Newton iterations bring the closest approach from 0.2 to 0.1 kpc. In the right panel, we hold the star fixed and fit the flattening q of the halo by descending the closest approach with the derivative obtained by reverse-mode differentiation through the in-backend orbit integration, which the `C` path cannot provide; the recovered $q = 0.851$ matches the true 0.85 from which the mock star was generated. In code, the first fit uses the derivative of the backward orbit with respect to the star's observed coordinates,

```
def closest_approach_ic(vxvv):
    # vxvv: (ra,dec,d,pmra,pmdec,vlos)
    o= Orbit(vxvv,radec=True,ro=8.,vo=220.)
    o.integrate(-ts,pot,method='dop853_c') # C + STM
    return jnp.min(o.r(-ts))
J= jax.jacrev(closest_approach_ic)(
    jnp.asarray(vxvv_obs))
```

which a Gauss–Newton step then combines with the covariance of the observations; the second fit is

```
def closest_approach(q):
    pot= LogarithmicHaloPotential(normalize=1.,
                                  core=0.2,q=q)
    o= Orbit(jnp.asarray(vxvv),radec=True,ro=8.,vo=220.)
    o.integrate(-ts,pot,method='diffrax')
    return jnp.min(o.r(-ts))
objective= jax.jit(jax.value_and_grad(closest_approach))
q= adam_descent(objective,q0=1.1) # a few lines of optax
```

With real data, one would combine the two fits into one likelihood over the star's parameters and those of the potential, and the derivative of that likelihood with respect to all of them would be available for a gradient-based sampler.

9.2. *Learning the inverse action–angle transformation*

The transformation from actions and angles to positions and velocities has no closed form in a general potential, and `galpy`'s implementations of it are currently limited to one-dimensional potentials and to the torus machinery of J. Binney & P. J. McMillan (2016). A differentiable forward transformation makes a different approach possible: a neural network $G(\mathbf{J}, \boldsymbol{\theta})$ that takes actions and angles to phase-space positions can be trained by requiring that `galpy`'s forward transformation, applied to the network's output, return the actions and angles that went in. The loss then contains a cycle-consistency term, $|F(G(\mathbf{J}, \boldsymbol{\theta})) - (\mathbf{J}, \boldsymbol{\theta})|^2$, with F the `actionAngle` method, whose gradient with respect to the network's weights flows through F. In both demonstrations below, we add to it a supervised term, the mean-square phase-space error on training pairs obtained by applying F to phase-space points sampled in a box around the origin, which keeps the network on bound orbits early in training (in the second demonstration, the weight of the cycle term is ramped up from zero over the first 400 steps). Because the pairs come from the forward transformation, no reference inverse is needed anywhere in the training. We show this for the vertical motion in `MWPotential2014` at the solar radius, where the exact inverse of § 5.2 exists and serves as a check, but is not used in the training (Figure 25), and for the motion in the meridional plane with the Stäckel approximation, where no reference exists (Figure 26). In the first case a small network trained in a few minutes, identically under `jax` and `PyTorch`, reproduces the exact tori to a phase-space error of 0.5 to 2 %. In the second, we keep the problem small by fixing the angular momentum at that of a circular orbit at the solar radius, $L_z = 1$ in `galpy`'s natural units, such that the network maps the two actions J_R and J_z and the two meridional angles to (R, z, v_R, v_z); a four-layer network trained for half an hour on a GPU returns actions on the round trip with a scatter of 7×10^{-3} and reconstructs meridional orbits that are not in its training set. A network for the

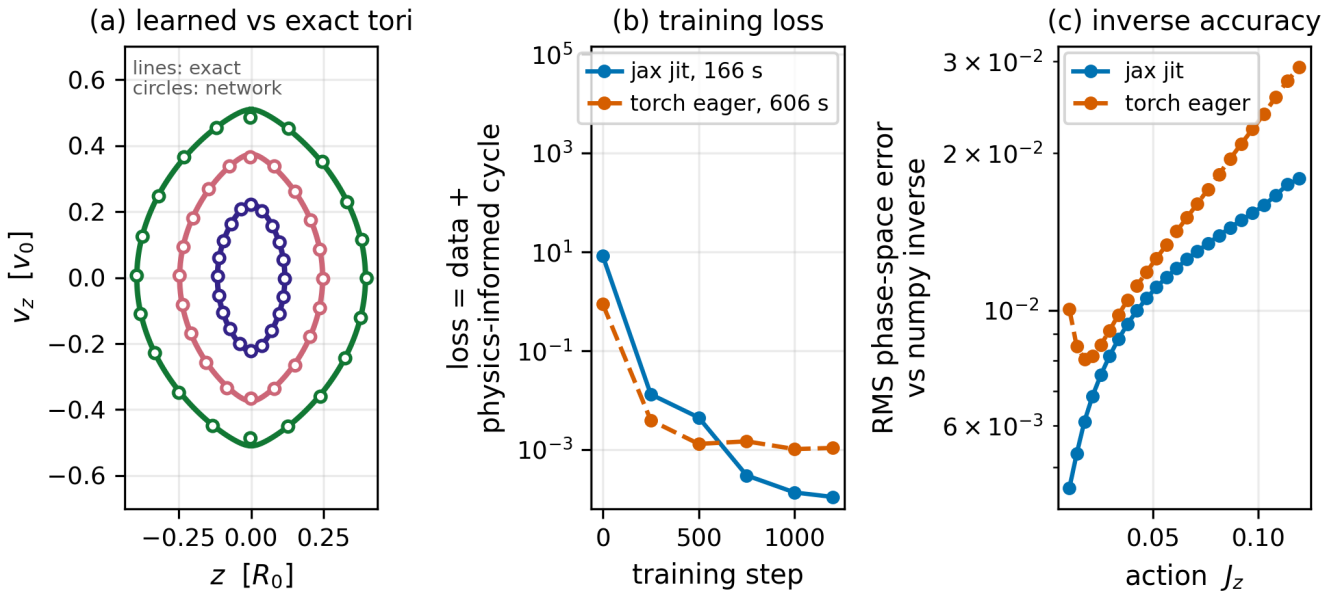


Figure 25. A neural network learns the inverse action–angle transformation of the vertical motion in `MWPotential2014` at the solar radius by differentiating through `galpy`'s forward transformation, in `jax` and in `PyTorch`. Left: the exact tori at three vertical actions (lines) and the network's output for the same actions and a grid of angles (circles). Middle: the training loss, the sum of the supervised and the cycle-consistency terms, in the two frameworks, with the training time in the legend: `jax` (`flax` and `optax`) with a compiled training step and `PyTorch` (`torch.nn`) eagerly, to show both options. Right: the root-mean-square phase-space error of the network against the exact inverse as a function of the action. The network reproduces the exact tori in both frameworks.

full three-dimensional transformation would take L_z and the azimuthal angle as further inputs. The cycle term of the training step is a few lines, whatever the framework,

```
def cycle_loss(params,J,theta):
    z,vz= net.apply(params,J,theta) # the network
    Jp,_,thetap= aAV.actionsFreqsAngles(z,vz) # galpy
    return (jnp.mean((Jp-J)**2/J**2)
            +jnp.mean(wrap(thetap-theta)**2))
grads= jax.jit(jax.grad(cycle_loss))(params,J,theta)
```

where `net` is a small multi-layer perceptron and `wrap` reduces an angle difference to $[-\pi,\pi)$; the supervised term is an ordinary mean-square error on the network's output. Neither network is a replacement for a proper inverse, and the point is not the accuracy of these particular fits but that `galpy`'s action–angle code sits inside a training loop, as a layer whose gradient the framework computes, and that the same code does so in both frameworks.

10. CONCLUSION

We have described an extension of `galpy` that makes it run, compile, and differentiate under `jax` and `PyTorch`, on CPUs and GPUs, while leaving its existing `numpy` and `C` implementation byte-for-byte unchanged, up to the handful of deliberate exceptions of § 2.1, and its interface as it was. Potentials, orbit integration, the action–angle transformations and their inverses, and the distribution functions, including those of tidal streams, accept the arrays of either framework and return derivatives with respect to their inputs and parameters. The existing test suite of `galpy`, now 4,462 tests apart from the 27 of the out-of-scope torus-mapping code, passes under `numpy`, under both frameworks in eager mode, and under both with compilation, and the dedicated tests of the backend code add 18,807 more. We tested derivatives against finite differences whose convergence is demonstrated, against the `C` variational equations, and against each other across the two frameworks, and where `galpy` implements a numerical method of its own, we tested it against `scipy` and against arbitrary-precision references. The work also found defects in the `numpy` and `C` code that had been there before, which 40 PRs to the main branch fixed.

What this makes possible is the gradient-based fitting of dynamical models that `galpy` already contains, with a likelihood whose derivative with respect to every parameter—of the potential, of the DF, of the orbits of the objects being modeled—costs a small multiple of the likelihood itself, and the use of these models as components of larger differentiable programs, such as neural networks that represent a potential or a DF, or a simulation whose output is compared with data. The speed-ups that a GPU provides over a single CPU core,

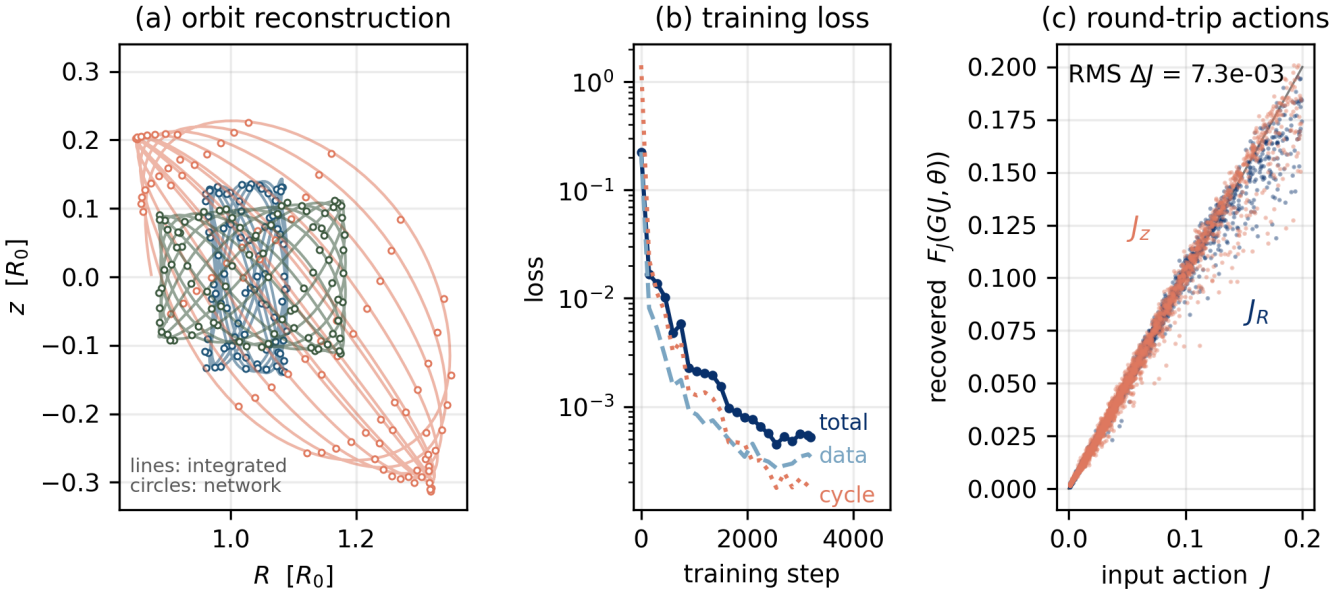


Figure 26. Neural inverse of the Stäckel action–angle transformation in `MWPotential2014` at fixed L_z, learned by differentiating through `galpy`'s forward transformation. Left: three orbits integrated in the potential (lines) and the positions returned by the network for the orbits' actions and angles (circles). Middle: the training loss, with its supervised ("data") and cycle-consistency terms. Right: the actions recovered by applying the forward transformation to the network's output against the input actions for a test set. The network recovers the input actions on the round trip and reconstructs orbits that are not in its training set.

of one to five orders of magnitude for large batches, are a second benefit, and one that depends on keeping the computation on the device by compiling it as a whole.

The GPU speed-ups are not uniform. Evaluating forces and drawing samples are limited by the memory bandwidth of the GPU and come close to it, but the in-backend orbit integration reaches only a few percent of the H100's double-precision peak: the batched array programs that `jax` and `PyTorch` generate step all orbits in lockstep, with a loop overhead per step and branch-heavy special functions, which maps poorly onto a GPU. GPU-native kernels that compile `galpy`'s existing `C` potentials and integrators as CUDA device functions, with one thread per orbit, are a natural complement. A prototype for `MWPotential2014` agrees with `galpy`'s `C` integrators to within their truncation error and, on the H100, integrates 10^6 disc orbits for 2 Gyr with `dop853` in 2.4 μs per orbit with `galpy`'s arithmetic and in 0.9 μs with GPU-tuned special functions, compared to 10 μs for compiled `jax` on the same GPU. Adaptive high-order integrators are limited by their per-thread state, and a fixed-step sixth-order symplectic integrator reaches the same accuracy another 3 to 3.7 times faster when, as here, the orbits share a dynamical timescale; adaptive stepping remains necessary when they do not. This speed came from restructuring the computation rather than from a direct translation, porting all of `galpy`'s potentials would be a substantial effort, and such kernels are not differentiable unless their variational or adjoint equations are written by hand as well. They would therefore complement the differentiable backends rather than replace them.

Some things are not done: the torus-mapping code, a wrapper of the external code of J. Binney & P. J. McMillan (2016), is not covered and will not be, `astropy` units cannot be used inside compiled programs, the interpolated velocity sampler of the quasi-isothermal disk DF runs on the backend but is a rejection sampler and therefore not differentiable, and the variational equations and Lyapunov exponents are features of the `C` integrators. Derivatives through the `C` integrators and the `C` action–angle methods are with respect to the initial conditions only and of first order. A backend is also a moving target: the special-function fallbacks exist because the frameworks' own versions are incomplete today, and the test that checks this will say when they are no longer needed. It already did once: `PyTorch` 2.15, released while we revised this paper, made its modified Bessel functions of the second kind differentiable, the test failed on the day of the release, and `galpy` now uses `PyTorch`'s own functions from that version on.

The code described here is the start of `galpy`'s version 2 series, which for now is distributed as pre-releases, starting with `v2.0.0b1`, that are installed with `pip install --pre galpy[jax]` or `galpy[torch]`; a plain `pip install galpy` remains at version 1.x; the docu-

mentation of version 2, which describes the backends, is at

https://docs.galpy.org/en/v2/ .

All code used to produce the figures in this paper, together with the working notes and logs of the development, is available at

https://github.com/jobovy/galpy-backends .

ACKNOWLEDGMENTS

It is a pleasure to thank Anthropic for providing a Claude Max subscription free of charge for the first part of this work, GitHub for covering the cost of the continuous integration through its Education program, and NVIDIA for the TITAN Xp GPU provided through its Academic Grant Program. We thank the contributors to `galpy`'s code, listed here in order of their number of contributions: Aladdin Seaifan, Ted Mackereth, James Lane, Mathew Bub, Nathaniel Starkman, Henry Leung, Wilma Trick, Samuel Wong, Jack Hong, Semyeong Oh, Rok Roškar, Turner Garrow, Jeremy Webb, Jason Sanders, Marius Cautun, Anna Juranova, Morgan Bennett, James Grimmett, Anya Dovgal, Gus Beane, Andrew Pontzen, Anthony Mercer, Mark Fardal, John Weatherall, Yingtian Chen, Hood Chatham, Mike German, Rhitik Bhatt, Junyang Lu, Gabriel Pfaffman, Yansong Qian, Jeffrey Simpson, Adrian Szpilfidel, Jovan Veljanoski, Carlos Vera-Ciro, and Jenny Wagner. We also thank the many users of `galpy` whose bug reports and questions over more than a decade shaped the code that is extended here. JB acknowledges financial support from NSERC (funding reference number RGPIN-2026-04837). This work was made possible by the open-source codes `numpy` (C. R. Harris et al. 2020), `scipy` (P. Virtanen et al. 2020), `matplotlib` (J. D. Hunter 2007), `astropy` (The Astropy Collaboration et al. 2022), `jax` (J. Bradbury et al. 2018), `PyTorch` (A. Paszke et al. 2019), `diffrax` and `equinox` (P. Kidger 2021; P. Kidger & C. Garcia 2021), `torchdiffeq` (R. T. Q. Chen et al. 2018), `torchode` (M. Lienen & S. Günnemann 2022), and the GNU Scientific Library (M. Galassi et al. 2009). Computations on the H100 GPU were performed on the Trillium supercomputer at the SciNet HPC Consortium (C. Loken et al. 2010; M. Ponce et al. 2019). SciNet is funded by Innovation, Science and Economic Development Canada; the Digital Research Alliance of Canada; the Ontario Research Fund: Research Excellence; and the University of Toronto. This research was enabled in part by support provided by Compute Ontario (https://www.computeontario.ca) and the Digital Research Alliance of Canada (https://alliancecan.ca). Version-control, continuous-integration, coverage, and documentation services are provided by GitHub, Codecov, and Read the Docs and are gratefully acknowledged.

STATEMENT ON THE USE OF AI

This paper has a single human author, JB, but is written in the first person plural, because the work that it describes was a collaboration between JB and Claude, an AI coding agent developed by Anthropic and used through the `Claude Code` environment. "We" refers to that collaboration; where the distinction matters, as in § 8, JB and the agent are named separately. The agent wrote the code described in this paper under JB's direction and review between 2026 June 1 and October 30; successive versions of the Claude models were used over this period, and each commit records the model that produced it. JB set the goals, the constraints, and the acceptance criteria, reviewed the pull requests, directed the design where the agent's choices were not what he wanted, and merged the code; the agent wrote the code, the tests, the documentation, and the working notes, and ran the measurements. For about two weeks in the summer the agent worked without supervision, as described in § 8.3, and JB reviewed its work from that period afterwards. JB also used OpenAI's Codex to review the agent's code at several points. The agent produced the figures of this paper and the analysis behind them from scripts that are part of the accompanying repository. The text of this paper was drafted by the agent from the working notes of the project and from JB's instructions and was then edited by JB, who is responsible for its content.

APPENDIX

A. VARIATIONAL EQUATIONS AND THE STATE-TRANSITION MATRIX

This appendix collects the equations behind `Orbit.integrate_dxdv`, `Orbit.lyapunov`, and the derivative rule of the `C` integrators of § 4.1. None of it is specific to a backend; it is implemented in `C` and in `numpy`.

Write the phase-space coordinates as $\mathbf{w} = (\mathbf{x}, \mathbf{v})$ in rectangular Galactocentric coordinates, in which `galpy`'s `C` integrators work, and the equations of motion as $\dot{\mathbf{w}} = \mathbf{f}(\mathbf{w}, t)$ with $\dot{\mathbf{x}} = \mathbf{v}$ and $\dot{\mathbf{v}} = \mathbf{a}(\mathbf{x}, \mathbf{v}, t)$. A small deviation $\delta\mathbf{w}$

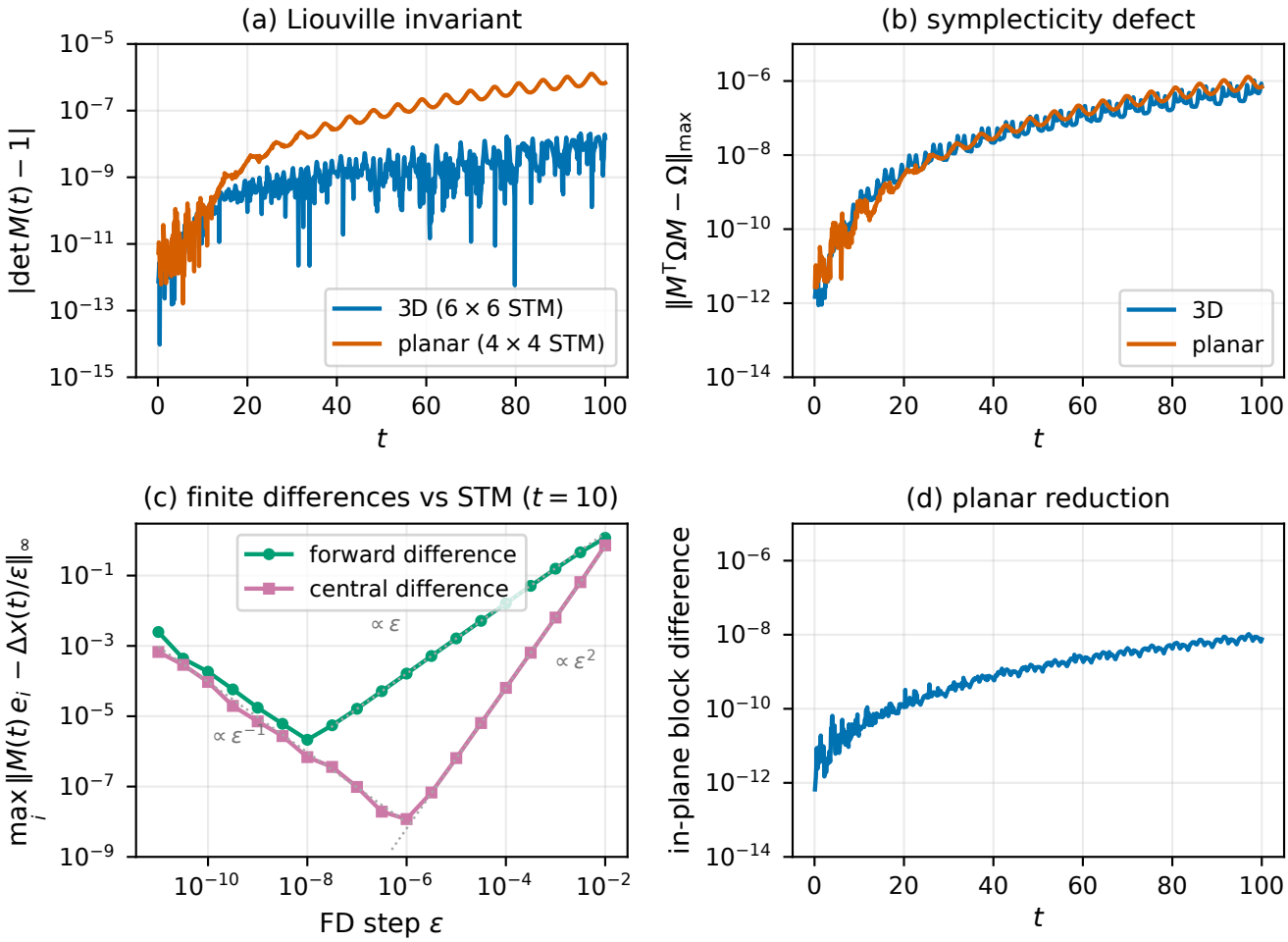


Figure 27. Checks of the `C` variational equations for an orbit in `MWPotential2014`, integrated with `dop853_c` at tolerances of 10^{-12}. (a) The Liouville defect $|\det \mathbf{M} - 1|$ of the state-transition matrix $\mathbf{M}$ and (b) the symplecticity defect for the three-dimensional and the planar state-transition matrix. (c) The difference between the columns of $\mathbf{M}$ at $t = 10$ and forward and central finite differences of independently integrated orbits as a function of the finite-difference step; the variational result agrees with the best central difference at its floor. (d) The in-plane block of the three-dimensional matrix for a planar initial condition agrees with the planar matrix, with no leakage out of the plane.

from an orbit evolves according to the linearized equations

$$\frac{\mathrm{d}\,\delta\mathbf{w}}{\mathrm{d}t} = \mathbf{J}(t)\,\delta\mathbf{w}\,, \qquad \mathbf{J} = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \partial\mathbf{a}/\partial\mathbf{x} & \partial\mathbf{a}/\partial\mathbf{v} \end{pmatrix}, \tag{A1}$$

evaluated along the orbit. For a conservative force, $\partial\mathbf{a}/\partial\mathbf{v} = \mathbf{0}$ and $\mathbf{K} \equiv \partial\mathbf{a}/\partial\mathbf{x} = -\nabla\nabla\Phi$ is the symmetric Hessian of the potential. `galpy`'s potentials provide their second derivatives in cylindrical coordinates, and $\mathbf{K}$ is assembled from them: with $\cos\phi = x/R$ and $\sin\phi = y/R$, the in-plane block is the one of the planar variational equations that `galpy` has had since 2011, built from $\partial^2\Phi/\partial R^2$, $\partial^2\Phi/\partial\phi^2$, and $\partial^2\Phi/\partial R\partial\phi$, and the vertical couplings are

$$\begin{aligned} K_{xz} &= -\cos\phi\,\frac{\partial^2\Phi}{\partial R\partial z} + \frac{\sin\phi}{R}\,\frac{\partial^2\Phi}{\partial z\partial\phi}\,, \\ K_{yz} &= -\sin\phi\,\frac{\partial^2\Phi}{\partial R\partial z} - \frac{\cos\phi}{R}\,\frac{\partial^2\Phi}{\partial z\partial\phi}\,, \\ K_{zz} &= -\frac{\partial^2\Phi}{\partial z^2}\,. \end{aligned} \tag{A2}$$

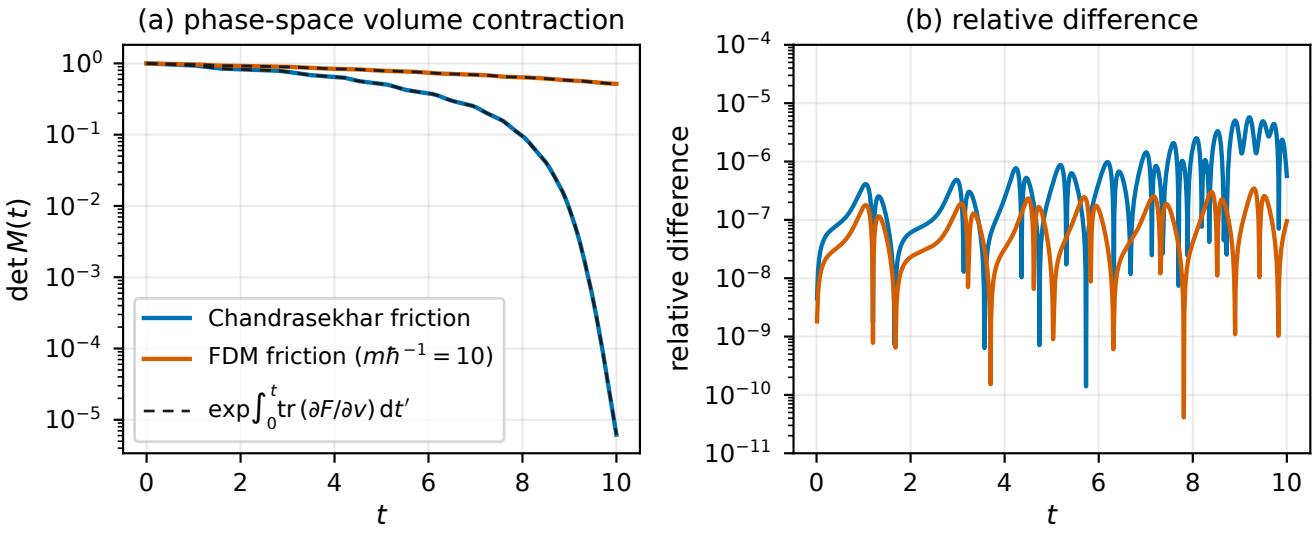


Figure 28. Phase-space volume under dissipative forces. Left: $\det \mathbf{M}(t)$ for an orbit in `MWPotential2014` with Chandrasekhar dynamical friction and with fuzzy-dark-matter friction (A. Szpilfidel et al. 2026), from the `C` variational equations (solid) and from the integral of the velocity divergence of the force along the orbit (dashed). Right: the relative difference between the two. The variational equations reproduce the contraction of phase-space volume by dissipative forces.

We added the mixed derivatives $\partial^2\Phi/\partial R\partial z$ and $\partial^2\Phi/\partial z\partial\phi$ to the `C` interface of every potential for this purpose, and the test suite checks the symmetry of $\mathbf{K}$. For a velocity-dependent force $\mathbf{F}(\mathbf{x}, \mathbf{v})$, such as dynamical friction or the Coriolis force in a rotating frame, the lower-right block $\partial\mathbf{F}/\partial\mathbf{v}$ is non-zero and the lower-left block need not be symmetric; both are implemented for all three of `galpy`'s velocity-dependent forces: the Chandrasekhar dynamical friction, the dynamical friction in fuzzy dark matter (A. Szpilfidel et al. 2026), and the fictitious forces of a non-inertial frame.

The state-transition matrix $\mathbf{M}(t) = \partial\mathbf{w}(t)/\partial\mathbf{w}(0)$ is the solution of equation (A1) for six independent deviation vectors, $\dot{\mathbf{M}} = \mathbf{J}\mathbf{M}$ with $\mathbf{M}(0) = \mathbf{I}$. The Runge–Kutta integrators integrate the orbit and all six columns as one 42-component state, such that the deviations see the same steps as the orbit; the symplectic integrators apply to the deviations the drift ($\delta\mathbf{x} \leftarrow \delta\mathbf{x} + h\,\delta\mathbf{v}$) and kick ($\delta\mathbf{v} \leftarrow \delta\mathbf{v} + h\,\mathbf{K}\,\delta\mathbf{x}$) sub-steps with the same coefficients as the orbit, which is the tangent map of the integrator and is itself symplectic. For a conservative force, $\mathbf{M}$ is symplectic, $\mathbf{M}^T\mathbf{\Omega}\mathbf{M} = \mathbf{\Omega}$ with $\mathbf{\Omega}$ the standard symplectic form, and in particular $\det\mathbf{M} = 1$, which is Liouville's theorem; for a dissipative force, $\det\mathbf{M}(t) = \exp\int_0^t \mathrm{tr}\,(\partial\mathbf{F}/\partial\mathbf{v})\,\mathrm{d}t'$, which measures the contraction of phase-space volume. These identities are necessary but not sufficient checks—a sign error in one block of $\mathbf{K}$ can leave the determinant equal to one—and the test that settles the correctness of $\mathbf{M}$ is the comparison of its columns with finite differences of independently integrated orbits, in which the finite-difference error follows its characteristic V shape with step size and the variational result sits at the bottom of the V. Figure 27 shows these checks for a Milky-Way orbit: the Liouville and symplecticity defects stay below 10^{-6} over a hundred time units, and the state-transition matrix agrees with the best central difference at the finite-difference floor of 10^{-8}. Figure 28 shows the dissipative case: for an orbit that sinks under Chandrasekhar dynamical friction, the phase-space volume contracts by five orders of magnitude over ten time units, and $\det\mathbf{M}$ follows the integral of the velocity divergence of the force to a few parts in 10^6; the same orbit under the fuzzy-dark-matter friction of `galpy` (A. Szpilfidel et al. 2026) contracts far less, because that force is suppressed at the orbit's velocities.

The derivative rule of § 4.1 follows from the chain rule: for a scalar function L of the orbit sampled at times t_k, $\partial L/\partial\mathbf{w}(0) = \sum_k \mathbf{M}(t_k)^T\,\partial L/\partial\mathbf{w}(t_k)$, which is a single matrix–vector product per output time once $\mathbf{M}$ is known. Lyapunov exponents are computed by integrating a deviation vector with equation (A1), renormalizing it to unit length every fixed number of steps, and accumulating the logarithms of the normalization factors (G. Benettin et al. 1980a,b); the spectrum follows by orthonormalizing a set of deviation vectors in the same way.

REFERENCES

An, J., Naik, A. P., Evans, N. W., & Burrage, C. 2021, MNRAS, 506, 5721, doi: 10.1093/mnras/stab2049

Anau Montel, N., & Schmidt, F. 2026, arXiv e-prints, arXiv:2606.09629. https://arxiv.org/abs/2606.09629

Banik, N., Bovy, J., Bertone, G., Erkal, D., & deBoer, T. J. L. 2021, MNRAS, 502, 2364, doi: 10.1093/mnras/stab210

Baydin, A. G., Pearlmutter, B. A., Radul, A. A., & Siskind, J. M. 2018, Journal of Machine Learning Research, 18, 1. https://arxiv.org/abs/1502.05767

Belokurov, V., Sanders, J. L., Fattahi, A., et al. 2020, MNRAS, 494, 3880, doi: 10.1093/mnras/staa876

Benettin, G., Galgani, L., Giorgilli, A., & Strelcyn, J.-M. 1980a, Meccanica, 15, 9, doi: 10.1007/BF02128236

Benettin, G., Galgani, L., Giorgilli, A., & Strelcyn, J.-M. 1980b, Meccanica, 15, 21, doi: 10.1007/BF02128237

Binney, J. 2010, MNRAS, 401, 2318, doi: 10.1111/j.1365-2966.2009.15845.x

Binney, J. 2012, MNRAS, 426, 1324, doi: 10.1111/j.1365-2966.2012.21757.x

Binney, J., & McMillan, P. 2011, MNRAS, 413, 1889, doi: 10.1111/j.1365-2966.2011.18268.x

Binney, J., & McMillan, P. J. 2016, MNRAS, 456, 1982, doi: 10.1093/mnras/stv2734

Boubert, D., Guillochon, J., Hawkins, K., et al. 2018, MNRAS, 479, 2789, doi: 10.1093/mnras/sty1601

Bovy, J. 2014, ApJ, 795, 95, doi: 10.1088/0004-637X/795/1/95

Bovy, J. 2015, ApJS, 216, 29, doi: 10.1088/0067-0049/216/2/29

Bovy, J., Bahmanyar, A., Fritz, T. K., & Kallivayalil, N. 2016, ApJ, 833, 31, doi: 10.3847/1538-4357/833/1/31

Bovy, J., & Rix, H.-W. 2013, ApJ, 779, 115, doi: 10.1088/0004-637X/779/2/115

Bradbury, J., Frostig, R., Hawkins, P., et al. 2018, JAX: composable transformations of Python+NumPy programs, 0.11.2 http://github.com/jax-ml/jax

Buckley, M. R., Lim, S. H., Putney, E., & Shih, D. 2023, MNRAS, 521, 5100, doi: 10.1093/mnras/stad843

Campagne, J.-E., Lanusse, F., Zuntz, J., et al. 2023, The Open Journal of Astrophysics, 6, 15, doi: 10.21105/astro.2302.05163

Cautun, M., Benítez-Llambay, A., Deason, A. J., et al. 2020, MNRAS, 494, 4291, doi: 10.1093/mnras/staa1017

Chemaly, D., Sola, E., Belokurov, V., et al. 2026, MNRAS, 550, stag1161, doi: 10.1093/mnras/stag1161

Chen, R. T. Q., Rubanova, Y., Bettencourt, J., & Duvenaud, D. 2018, in Advances in Neural Information Processing Systems 31. https://arxiv.org/abs/1806.07366

Chen, Y., Valluri, M., Gnedin, O. Y., & Ash, N. 2025, ApJS, 276, 32, doi: 10.3847/1538-4365/ad9904

Contigiani, O., Rossi, E. M., & Marchetti, T. 2019, MNRAS, 487, 4025, doi: 10.1093/mnras/stz1547

Dehnen, W. 1999, AJ, 118, 1201, doi: 10.1086/301010

Duane, S., Kennedy, A. D., Pendleton, B. J., & Roweth, D. 1987, Physics Letters B, 195, 216, doi: 10.1016/0370-2693(87)91197-X

Euclid Collaboration, Starkman, N., Nibauer, J., et al. 2026, arXiv e-prints, arXiv:2606.21774. https://arxiv.org/abs/2606.21774

Evans, N. W., Häfner, R. M., & de Zeeuw, P. T. 1997, MNRAS, 286, 315, doi: 10.1093/mnras/286.2.315

Fardal, M. A., Huang, S., & Weinberg, M. D. 2015, MNRAS, 452, 301, doi: 10.1093/mnras/stv1198

Forbes, J. C., Bannister, M. T., Lintott, C., et al. 2025, ApJ, 988, 121, doi: 10.3847/1538-4357/adc9ac

Frankel, N., Sanders, J., Ting, Y.-S., & Rix, H.-W. 2020, ApJ, 896, 15, doi: 10.3847/1538-4357/ab910c

Fritz, T. K., Battaglia, G., Pawlowski, M. S., et al. 2018, A&A, 619, A103, doi: 10.1051/0004-6361/201833343

Gaia Collaboration, Prusti, T., de Bruijne, J. H. J., et al. 2016, A&A, 595, A1, doi: 10.1051/0004-6361/201629272

Gaia Collaboration, Brown, A. G. A., Vallenari, A., et al. 2018, A&A, 616, A1, doi: 10.1051/0004-6361/201833051

Gaia Collaboration, Vallenari, A., Brown, A. G. A., et al. 2023, A&A, 674, A1, doi: 10.1051/0004-6361/202243940

Galassi, M., Davies, J., Theiler, J., et al. 2009, GNU Scientific Library Reference Manual, 3rd edn. (Network Theory Ltd.)

Gnedin, O. Y., Gould, A., Miralda-Escudé, J., & Zentner, A. R. 2005, ApJ, 634, 344, doi: 10.1086/496958

Green, G. M., Ting, Y.-S., & Kamdar, H. 2023, ApJ, 942, 26, doi: 10.3847/1538-4357/aca3a7

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357, doi: 10.1038/s41586-020-2649-2

Hernquist, L. 1990, ApJ, 356, 359, doi: 10.1086/168845

Hoffman, M. D., & Gelman, A. 2014, Journal of Machine Learning Research, 15, 1593. https://arxiv.org/abs/1111.4246

Hunt, J. A. S., & Bovy, J. 2018, MNRAS, 477, 3945, doi: 10.1093/mnras/sty921

Hunter, J. D. 2007, Computing in Science & Engineering, 9, 90, doi: 10.1109/MCSE.2007.55

Kalda, T., & Green, G. M. 2025, ApJ, 992, 84, doi: 10.3847/1538-4357/adf8ea

Kidger, P. 2021, PhD thesis, University of Oxford. https://arxiv.org/abs/2202.02435

Kidger, P., & Garcia, C. 2021, arXiv e-prints, arXiv:2111.00254. https://arxiv.org/abs/2111.00254

Kinoshita, H., Yoshida, H., & Nakai, H. 1991, Celestial Mechanics and Dynamical Astronomy, 50, 59, doi: 10.1007/BF00048986
Koposov, S. E., Boubert, D., Li, T. S., et al. 2020, MNRAS, 491, 2465, doi: 10.1093/mnras/stz3081
Lane, J. M. M., Bovy, J., & Mackereth, J. T. 2022, MNRAS, 510, 5119, doi: 10.1093/mnras/stab3755
Lane, J. M. M., Bovy, J., & Mackereth, J. T. 2023, MNRAS, 526, 1209, doi: 10.1093/mnras/stad2834
Li, Y., Modi, C., Jamieson, D., et al. 2024, ApJS, 270, 36, doi: 10.3847/1538-4365/ad0ce7
Lienen, M., & Günnemann, S. 2022, arXiv e-prints, arXiv:2210.12375. https://arxiv.org/abs/2210.12375
Loken, C., Gruner, D., Groer, L., et al. 2010, Journal of Physics Conference Series, 256, 012026, doi: 10.1088/1742-6596/256/1/012026
Lucey, M., Pearson, S., Hunt, J. A. S., et al. 2023, MNRAS, 520, 4779, doi: 10.1093/mnras/stad406
Mackereth, J. T., & Bovy, J. 2018, PASP, 130, 114501, doi: 10.1088/1538-3873/aadcdd
Mackereth, J. T., Bovy, J., Leung, H. W., et al. 2019, MNRAS, 489, 176, doi: 10.1093/mnras/stz1521
Malhan, K., & Ibata, R. A. 2019, MNRAS, 486, 2995, doi: 10.1093/mnras/stz1035
Marchetti, T., Evans, F. A., & Rossi, E. M. 2022, MNRAS, 515, 767, doi: 10.1093/mnras/stac1777
Modi, C., Lanusse, F., & Seljak, U. 2021, Astronomy and Computing, 37, 100505, doi: 10.1016/j.ascom.2021.100505
Necib, L., & Lin, T. 2022, ApJ, 926, 189, doi: 10.3847/1538-4357/ac4244
Nibauer, J., & Bonaca, A. 2025, ApJL, 985, L22, doi: 10.3847/2041-8213/add0a9
Nibauer, J., Bonaca, A., Spergel, D. N., et al. 2025, ApJ, 983, 68, doi: 10.3847/1538-4357/adb8e8
O'Doherty, T. N., Bahramian, A., Miller-Jones, J. C. A., et al. 2023, MNRAS, 521, 2504, doi: 10.1093/mnras/stad680
Pace, A. B., Erkal, D., & Li, T. S. 2022, ApJ, 940, 136, doi: 10.3847/1538-4357/ac997b
Paszke, A., Gross, S., Massa, F., et al. 2019, in Advances in Neural Information Processing Systems 32. https://arxiv.org/abs/1912.01703
Pelupessy, F. I., van Elteren, A., de Vries, N., et al. 2013, A&A, 557, A84, doi: 10.1051/0004-6361/201321252
Ponce, M., van Zon, R., Northrup, S., et al. 2019, in Proceedings of the Practice and Experience in Advanced Research Computing on Rise of the Machines (learning), PEARC '19 (ACM), 1–8, doi: 10.1145/3332186.3332195
Pontzen, A., Roškar, R., Stinson, G., & Woods, R. 2013, pynbody: N-Body/SPH analysis for python,, Astrophysics Source Code Library, record ascl:1305.002
Portegies Zwart, S., & McMillan, S. 2018, Astrophysical Recipes: The art of AMUSE (IOP Publishing), doi: 10.1088/978-0-7503-1320-9
Portegies Zwart, S., McMillan, S., Harfst, S., et al. 2009, NewA, 14, 369, doi: 10.1016/j.newast.2008.10.006
Price-Whelan, A. M. 2017, The Journal of Open Source Software, 2, 388, doi: 10.21105/joss.00388
Price-Whelan, A. M., Hunt, J. A. S., Horta, D., et al. 2025, ApJ, 979, 115, doi: 10.3847/1538-4357/ad969a
Qian, Y., Arshad, Y., & Bovy, J. 2022, MNRAS, 511, 2339, doi: 10.1093/mnras/stac238
Reino, S., Rossi, E. M., Sanderson, R. E., et al. 2021, MNRAS, 502, 4170, doi: 10.1093/mnras/stab304
Sanders, J. L., Bovy, J., & Erkal, D. 2016, MNRAS, 457, 3817, doi: 10.1093/mnras/stw232
Shu, F. 1969, ApJ, 158, 505, doi: 10.1086/150214
Starkman, N., Nibauer, J., Pearson, S., Wu, S., & Necib, L. 2026a, arXiv e-prints, arXiv:2609.40057. https://arxiv.org/abs/2609.40057
Starkman, N., Price-Whelan, A., & Nibauer, J. 2025, Journal of Open Source Software, 10, 7771, doi: 10.21105/joss.07771
Starkman, N., Price-Whelan, A., Nibauer, J., & Chen, Y. 2026b, GalacticDynamics/galax: v0.0.3, v0.0.3 Zenodo, doi: 10.5281/zenodo.21985205
Stefánsson, G., Mahadevan, S., Miguel, Y., et al. 2023, Science, 382, 1031, doi: 10.1126/science.abo0233
Swiggum, C., Alves, J., Benjamin, R., et al. 2024, Nature, 631, 49, doi: 10.1038/s41586-024-07496-9
Szpilfidel, A., Boldrini, P., Bovy, J., & Di Matteo, P. 2026, A&A, 710, A189, doi: 10.1051/0004-6361/202558231
Teuben, P. J. 1995, in Astronomical Society of the Pacific Conference Series, Vol. 77, Astronomical Data Analysis Software and Systems IV, ed. R. A. Shaw, H. E. Payne, & J. J. E. Hayes, 398
The Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167, doi: 10.3847/1538-4357/ac7c74
Tonnesen, S. 2019, ApJ, 874, 161, doi: 10.3847/1538-4357/ab0960
Trick, W. H., Bovy, J., & Rix, H.-W. 2016, ApJ, 830, 97, doi: 10.3847/0004-637X/830/2/97
Vasiliev, E. 2019, MNRAS, 482, 1525, doi: 10.1093/mnras/sty2672
Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, Nature Methods, 17, 261, doi: 10.1038/s41592-019-0686-2

Viterbo, G., & Buck, T. 2025, arXiv e-prints, arXiv:2511.22468. https://arxiv.org/abs/2511.22468

Webb, J. J., Bovy, J., Carlberg, R. G., & Gieles, M. 2019, MNRAS, 488, 5748, doi: 10.1093/mnras/stz2118

Webb, J. J., Price-Jones, N., Bovy, J., et al. 2020, MNRAS, 494, 2268, doi: 10.1093/mnras/staa788

Wu, S., Starkman, N., Pearson, S., et al. 2026, arXiv e-prints, arXiv:2605.07809. https://arxiv.org/abs/2605.07809

Xiang, M., & Rix, H.-W. 2022, Nature, 603, 599, doi: 10.1038/s41586-022-04496-5

Yang, T., Boruah, S. S., & Afshordi, N. 2020, MNRAS, 493, 3061, doi: 10.1093/mnras/staa441

Yoshida, H. 1990, Physics Letters A, 150, 262, doi: 10.1016/0375-9601(90)90092-3

Zucker, C., Goodman, A. A., Alves, J., et al. 2022, Nature, 601, 334, doi: 10.1038/s41586-021-04286-5